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The dimensions of safety culture: A review of quantitative, qualitative and mixed methods for assessing safety culture in hospitals

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The dimensions of safety culture: A review of quantitative, qualitative and mixed methods for assessing safety culture in hospitals

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ABSTRACT

Objectives: The concept of 'safety culture' has received attention over the past two decades in health care because it is thought to form a basis for the safe delivery of high-quality care. However, the study of safety culture and its relationship to patient care have been challenged by variation in definition, dimensionality and assessment. This suggests a lack of consensus in the literature about what comprises safety culture. This review aimed to map the range of methods and tools used to assess safety culture in hospitals, analyse the prevalence of these methods and tools in the published research literature, and examine the dimensions of safety culture captured through these processes.

Methods: The review was conducted using four bibliometric databases (PubMed, CINAHL, Scopus, and Web of Science) with studies from 2008 to May 2020. We included quantitative tools, mixed methods and qualitative approaches to assessing safety culture.

Results: After removing duplicates, the search yielded 1,765 papers, of which 701 met full-text inclusion criteria. While findings revealed a myriad of methods to assess safety culture, most studies exclusively utilised surveys (n=669; 95.4%), with 87 different survey tools identified. Only 31 studies (4.4%) used qualitative or mixed methods. These methods and tools assessed 11 themes related to safety culture, with 'Leadership' being the most common. Qualitative and mixed methods approaches were more likely to identify additional dimensions of safety culture not covered by the 11 themes, including improvisation and contextual pressures.

Conclusions: We assessed the extent to which safety culture dimensions mapped to specific quantitative and qualitative tools and methods of assessing safety culture. No single method or tool appeared to measure all of the dimensions of safety culture. Findings suggest that

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future attempts to assess safety culture in hospitals should consider incorporating qualitative methods into survey studies to evaluate this multi-faceted construct.

KEYWORDS

Safety culture, Patient safety, Hospital, Measurement, Organisational culture

ARTICLE SUMMARY

- Reviewed both qualitative and quantitative methods for assessing safety culture in hospitals.
- Examined the extent diverse methods and different surveys provide coverage of safety culture dimensions.
- Included only published, peer-reviewed and English language empirical studies; potential insights from grey literature not explored.
- Provides guidance for using a comprehensive surveys alongside qualitative methods to assess safety culture in hospital.

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BACKGROUND

Over the last two decades, the notion that hospitals have a "safety culture" that can be measured, understood and improved, has become a pillar of the patient safety movement,[1] and is increasingly embodied in accreditation and regulatory standards.[2 3] Accordingly, the need for robust assessment of safety culture has become evident,[4] and tools have proliferated.[5 6] Nowadays, safety culture assessment is widely used in hospitals for four key reasons, namely, to: (1) diagnose safety culture and identify areas for improvement; (2) evaluate patient safety interventions or programs and track change over time; (3) conduct internal and external benchmarking; and (4) fulfill directives or regulatory requirements.[1 7] Despite expanding interest in studying and addressing safety culture over the last decade and a half, several challenges persist in defining and assessing the dimensions of safety culture.

Defining safety culture

There are many definitions of safety culture, both within and outside of health care. Cooper[8] evocatively summarised the situation well in his description of the field as a "definitional swamp", highlighting 51 separate definitions of safety culture and 30 separate definitions of safety climate. Compounding equivocality, many researchers mark a distinction between safety *climate* and safety *culture*,[7] suggesting the former involves peoples' *perceptions* of their organisations (its procedures, practices, and the kinds of behaviour that are tolerated or rewarded), whereas culture operates on a deeper, more enduring level, representing the behaviours and practices and the underlying, sometimes unconscious beliefs and values within an organisation.[9] Others in the field use culture and climate interchangeably.[10]

A commonly-occurring definition of safety culture is:[11] "The product of individual and group values, attitudes, perceptions, competencies, and patterns of behaviour that determine the commitment to, and the style and proficiency of, an organization's health and safety management" (p. 339).[12] Although this provides some guidance on which constructs to

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consider when assessing safety culture, the specificities of the attitudes, values and norms conducive to safety, as well as how to assess them, is less clear.

Assessment of safety culture and its dimensions

Safety culture is considered multi-faceted, consisting of several interrelated attributes or dimensions, although consensus on what these dimensions are remains elusive.[12 13] For example, while some conceptualise safety culture as exclusively the prioritization of safety by the unit and organisational leadership,[14] others construe it more broadly to include learning, reporting, and blame orientation.[15-17] Recognising and responding to this equivocality, over the years several reviews of safety culture assessment in health care have examined dimensions, highlighting the most frequent dimensions[9 12 18 19] While the themes these reviews have delineated overlap, they differ considerably in scope (i.e., some are broad, while others are specific) and quantity. These similarities and differences are summarised in Table 1, where thematically related dimensions are presented on the same row.

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Table 1. Dimensions derived from reviews of safety culture assessment in health care.

Flin and colleagues[9] Safety management themes	Singla and colleagues[18] Dimensions of safety culture	Halligan and Zecevic[12] Most frequent safety culture dimensions	Alsalem and colleagues[19] Safety climate dimensions
Review focus			
Examined 12 quantitative safety culture survey studies. Identified 10 safety management themes.	Reviewed 13 survey instruments, inductively and deductively categorising each item into 23 different dimensions (under six broad categories).	Reviewed 137 articles on safety culture in health care, of which 113 explored dimensions. Identified six most frequent dimensions.	Reviewed five most common safety culture surveys by item. Identified 12 dimensions.
Dimensions			
Management and supervisors	Management and institutional commitment to safety (Management and supervision)	Leadership commitment to safety	Top management support & institutional commitment to safety
	Institutional responses (Management and supervision)		
	Adequacy of training and supervision (Management and supervision)		Training & continuous education
Reporting/speaking up	Nonpunitive response to error (Management and supervision)	A non-punitive approach to adverse event reporting and analysis	Reporting incidents & "non-punitive" response to error
Risk perception	Risk taking (Risk)		
	Willingness to ask for help (Risk)		
Safety systems	Detection infrastructure (Safety system)		Safety systems: "Policies & procedures, safety planning, hand offs & transitions, staffing, equipment"
	Handoffs and transitions and coordination of care (Safety system)		
	Patient safety planning (Safety system)		

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	Adequacy of staffing (Safety system)		
	Adequacy of equipment, information and processes (Safety system)		
	Reporting infrastructure (Safety system)		
	Adequacy of crisis management (Competence)		
	What should be reported and to whom (Procedure/rules)		
	Compliance with rules and procedures (Procedure/rules)		
Job demands	Work pressure (Work pressure)		Work pressure
Safety attitudes/behaviours	Overall perception of safety	A shared belief in the importance of safety	Safety perceptions & attitudes of staff, risk perceptions
Communication/feedback	Communication openness (Other)	Open communication founded on trust	Communication openness
	Feedback and communication (Other)		Feedback & communication about adverse events
Teamwork	Teamwork (Other)	Teamwork	Teamwork
Personal resources (e.g., stress)	Job satisfaction (Other)		Staff satisfaction
Organisational factors	Organisational learning (Other)	Organisational learning	Organizational learning and continuous improvement
	Beliefs about the causes of errors and adverse events		Beliefs about the causes of errors & adverse events
			Other

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Only one of these reviews included qualitative and mixed method approaches alongside quantitative surveys.[12] Almost a decade ago, Halligan and Zecevic found only 14 of 137 articles utilised qualitative methods (e.g., interviews, focus groups); given the preponderance of quantitative surveys, limited discussion was given to these methods, and it was not clear whether or how they mapped to the dimensions of safety culture they identified or whether these studies uncovered additional insights into safety culture in hospitals.

The widespread, exclusive use of surveys to assess safety culture differentiates health care from other industries where a range of methods are typically used together.[20-22] This is because surveys alone do not expose rich insights into dimensions of culture.[23] Following Schein's model of organisational culture[24] which has been widely adopted in health care,[25] and to understand safety culture,[13 26] surveys capture the first two layers of artifacts and espoused beliefs rather than the deeper underlying shared assumptions at the heart of an organisation's culture. Cultural dimensions are also arguably to some extent contextually dependent, cannot be pre-specified and hence require rich interpretive methods to uncover: "every organization has a unique profile of cultural assumptions that any questionnaire inevitably misses" (p. 615).[27] In-depth, open-ended, qualitative methods are considered better suited to accessing these deeper facets and contextual nuances of culture,[24 28] because their iterative, interactive nature allows for questions, and therefore the conceptualisation of safety culture, to evolve in response to feedback as data are collected.[29]

The present review

There has been limited integration of qualitative research in past reviews of safety culture assessment in health care, with none examining dimensions by type of method. The potential for qualitative methods to uncover additional insights into safety culture dimensions has not been considered.[12] We set out to map the range of methods and tools used to assess safety culture in hospitals, including qualitative, quantitative and mixed approaches. We sought to

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analyse the prevalence of use of these methods and tools and examine the ways safety culture was conceptualised—including its dimensions—through the application of methods or tools.

METHOD

Search strategy and criteria

This systematic review followed the Preferred Reporting Items for Systematic Reviews (PRISMA) guidelines (see Appendix 1 for PRISMA checklist).[30] Four databases were searched: PubMed, CINAHL, Scopus, and Web of Science. The search was from 2008 to May 2020, using a search string based on previous reviews:[12] ('acute care' OR 'hospital') AND ('quality culture' OR 'safety culture' OR 'culture of safety' OR 'safety climate' OR 'safety attitude') AND ('survey' OR 'questionnaire' OR 'tool' OR 'instrument' OR measur* OR assess* OR checklist OR 'check list' OR observ*) AND ('patient safety' OR 'public safety' OR 'workplace safety'). The full search for Scopus is provided in Appendix 2.

To be eligible for inclusion in the review, studies were required to have: (1) been an empirical study published in a peer-reviewed journal; (2) utilised a method(s) or tool(s) for the purposes of assessing, studying or measuring safety culture or climate; (3) collected data in hospital(s); and (4) the study was published in English. Review articles were retained for snowballing and context, but were otherwise excluded from this review.

Study selection

Titles and abstracts were divided and independently reviewed by four of the authors (CP, LE, KC, AO). Fleiss' kappa statistic[31] was used to evaluate inter-rated agreement of 5% of the titles/abstracts by two authors (CP, KC). Snowballing was also used to identify studies from reference lists of included studies and reviews identified.

Examination of methods and tools

Although all types of papers describing methods to assess safety culture were included, data extraction differed depending on method. Among the studies exclusively using quantitative surveys, data extraction focused on details of the tool itself rather than the study, including

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survey origins and details of dimensions and items. For surveys that had been developed by included studies, we noted information on the development of the tool including what, if any, other surveys it had been derived from. For studies that used an already established survey, where required, we found the original source to collect these details.

Due to their bespoke nature, studies that used qualitative or mixed methods to assess safety culture were examined individually. Information on study design (e.g., ethnography), methods of data collection (e.g., interviews, observations), analysis (e.g., thematic analysis), and sometimes results—in the case of iterative data collection and analysis—were examined to extract information on the method of assessment and the conceptualisation of safety culture. This included review of the types of questions asked during interview (or the interview schedule, if provided), and details of any framework or theory used for analysis.

To extract information on dimensions of safety culture, three of the authors (CP, LE, KC) first deductively developed a draft coding framework by examining dimensions and themes identified in previous reviews.[9 12 18 19] As shown in the rows of Table 1, similar dimensions across reviews were organised together, with some refined into broader themes rather than more specific safety categories (e.g., *safety systems* or *resources and constraints*, rather *detection infrastructure, adequacy of staffing*). In a workshop session, the framework was then applied to all items (n=275) from eight of the most common surveys. These were coded individually by each author and then responses were discussed together to refine the coding framework and its definitions, resolve any disagreement, and ensure consistent coding. During this process, the *prioritisation of safety* and the *perceptions of safety* (i.e., how safe an organisation is) were broken up into distinct themes, because though related, there was sufficient evidence of a distinction and that both were important. After establishing that the coding framework could be applied consistently, the remaining surveys and studies were divided up and coded.

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Patient and public involvement

No patients involved.

RESULTS

The search retrieved a total of 4,020 papers, which were downloaded into the reference management software EndNote and duplicates removed, giving 1,761 articles. Four studies identified from the reference lists of papers were also reviewed, giving a total of 1,765 studies. Following title and abstract screening, 1,024 articles were included for full-text review (See Figure 1). Of these, 701 articles met inclusion criteria and are included in this review.

The overwhelming majority of included papers (n=670; 95.6%) exclusively utilised self-report quantitative surveys to assess safety culture. A further 19 papers (2.7%) reported using surveys in combination with other methods, such as interviews, while 12 papers (1.7%) reported solely using qualitative methods.

[INSERT FIGURE 1 HERE]

Quantitative surveys assessing safety culture

In total, 87 surveys were identified, 45 of which were new surveys developed by authors of included papers. The most widely-used were the Hospital Survey on Patient Safety Culture (HSOPS; n=321; 45.8%) by the Agency for Healthcare Research and Quality,[32] and the Safety Attitudes Questionnaire (SAQ; n=207; 29.5%) developed by Sexton and colleagues at the University of Texas.[33] This was followed rather distantly by the Patient Safety Climate in Healthcare Organisations (PSCHO), the Safety Climate Survey (SCSu), the Hospital Safety Climate Scale (HSCS), the Teamwork and Safety Climate Survey (TSCS)[33 34] and the Operating Room Management Attitudes Questionnaire (ORMAQ).[35] Sixty-one surveys were used in only one study. See Table 2 for the most common surveys.

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Table 2. Most frequently used surveys to assess safety culture in hospitals.

Name of survey	Number of studies	%*
Hospital Survey on Patient Safety Culture[32]	321	45.8
Safety Attitudes Questionnaire[33]	207	29.5
Patient Safety Climate in Healthcare Organizations[6]	21	3.0
Safety Climate Survey[36]	9	1.3
Hospital Safety Climate Scale[37]	8	1.1
Teamwork and Safety Climate Survey[33 34]	7	1.0
Operating Room Management Attitudes Questionnaire[35]	5	0.7

*Proportion of papers using surveys across all included studies (n=701).

The 87 surveys encompassed not only entirely new and unique measures, but also surveys that built on or refined existing tools, or that used some of the same questions. For example, the PSCHO was derived from the Veteran Affairs (VA) Palo Alto/Stanford Patient Safety Center of Inquiry (PSCI) survey.[6 38] The Modified Stanford Instrument and Canadian-Patient Safety Climate Survey—both by Ginsburg and colleagues[4 39]—are also based on the PSCI but included some questions from the HSOPS.

Of the 45 new measures, many were based at least in part on established measures of safety culture, including the HSOPS and SAQ. Others focused on particular aspects of safety culture within hospitals, such as the Just Culture Assessment Tool[40] and the Catheter-Related Bloodstream Infection Prevention Practices Survey;[41] or certain staff groups (e.g., Junior Doctor-Patient Safety Attitudes and Climate Questionnaire);[42] were developed for specific settings (e.g., maternal and child health institutions,[43] acute geriatric units)[44] or languages (e.g., Chinese).[45] Distinct from other self-report surveys, the Patients' Perceptions of Safety Culture scale was designed to assess patient, rather than staff, perceptions of hospital safety culture.[46]

Qualitative and mixed methods studies assessing safety culture

The review identified 31 studies (4.4%) that used qualitative or mixed methods to assess safety culture. Interviews were the most common approach (n=23; 3.3%), followed by frameworks and theories (n=11, 1.6%), and focus groups (n=9, 1.3%) (see Table 3). Most studies reported their interview schedule or example questions along with the main topics

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covered. These questions varied in how they were framed and their content (e.g., broad and exploratory: "Can you tell me about what patient safety means to you?" versus more specific and contained: "How safe do you think is the care provided in the ICU?").

Frameworks or theories were used in 11 studies to guide the collection and analysis of qualitative data, and included grounded theory,[47] a lifeworld versus system distinction,[48] and the SAQ's dimensions.[49 50] The Manchester Patient Safety Framework (MaPSaF) was used across four studies, typically in its conventional facilitated workshop,[51-53] or as an adapted version.[54] Similar to surveys, it presupposes a number of dimensions for which staff are required to rate the safety culture in their unit or organisation. However, the response range is not a Likert scale, but classifies cultures along five theorised levels of cultural maturity (i.e., pathological, reactive, bureaucratic, proactive, generative).[55] A quantitative survey tool based on the MaPSaF and using a similar ordinal response range (the Manchester Patient Safety Culture Assessment Tool, MaPSCAT) was also identified by this review.[56]

Table 3. Qualitative methods to assess safety culture in hospitals.

	Number of outputs*	%
Interviews	23	3.3
Framework or theory	11	1.6
Focus groups	9	1.3
Observations	5	0.7
Document analysis	5	0.7
Ethnography	2	0.3
Total	32	4.6

** The numbers of studies in this column exceeds the total number of studies due to studies using multiple methods/tools.*

Observations were used in only five studies, always in conjunction with interviews, focus groups, or surveys. One of these studies involved novel nurse led photo elicitation during walkabouts together with focus groups to study the safety culture related to medication management in a Brazilian intensive care unit.[57] Five studies analysed organisational documents and data as a window into safety culture. For example, safety

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checklists used during management walkarounds were analysed and triangulated with survey responses and focus group data in a study of intensive care unit safety culture following a safety intervention.[58] Finally, two studies utilised a combination of qualitative methods, including observations, informal conversations, interviews, within an ethnographic study design to examine how team communication affected safety culture in surgery,[59] and the developing patient safety culture of medical residents.[48]

More than half of studies using qualitative methods did so in conjunction with surveys to assess safety culture in a mixed methods design (n=19, 59.4%), to elaborate[51 60] or compare[61] with quantitative findings, develop a safety culture survey,[62] or identify factors influencing patient safety culture.[63] Some survey studies also incorporated abbreviated qualitative data by including open response questions.[60 64] However, there was insufficient detail on the questions used and any analysis to warrant these studies being classified as mixed methods. An exception was Boussat and colleagues'[65] mixed method study, which used open-response comments from a HSOPS survey of a French hospital to develop interview probes and included these answers in the analysis.

Themes for safety culture dimensions

Details of all 87 surveys and 31 qualitative and mixed methods studies were examined to identify common themes in safety culture dimensions, as well as new or emerging ones. This process led to the identification of 11 themes and an "other" category. Table 4 displays definitions, example items, and how often themes were present in methods/tools reviewed. Appendix 3 shows the mapping of each method or tool to the themes of safety culture identified.

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Table 4. Safety culture themes

Theme	Definition	Example item	No. studies/surveys used in (%)	
1. Leadership	Leadership and their support and commitment to safety.	My supervisor often discusses safe work practices with me. (HSCS)	Total*	84 (77.1)
			Quantitative^	65 (78.3)
			Qualitative#	4 (33.3)
			Mixed methodsº	15 (78.9)
2. Perceptions of safety	Perceptions of how safe the organisation is.	Please give your unit an overall grade on patient safety. (MSI)	Total	65 (59.6)
			Quantitative	41 (49.4)
			Qualitative	7 (58.3)
			Mixed methods	17 (89.4)
3. Teamwork and collaboration	Working together as a team and coordination of care among staff.	I enjoy working as part of a team. (ORMAQ)	Total	61 (55.9)
			Quantitative	41 (49.4)
			Qualitative	4 (33.3)
			Mixed methods	16 (84.2)
4. Safety systems	Systems, procedures and processes exist that facilitate patient safety (e.g., rewards, reporting systems).	Things "fall between the cracks" when transferring patients from one unit to another. (HSOPS)	Total	58 (53.2)
			Quantitative	40 (48.2)
			Qualitative	4 (33.3)
			Mixed methods	14 (73.7)
5. Prioritisation of safety	Shared belief, behaviours and norms in which staff in the work area prioritise and value safety.	Patient safety is never sacrificed to get more work done. (HSOPS)	Total	58 (53.2)
			Quantitative	41 (49.4)
			Qualitative	2 (16.7)
			Mixed methods	15 (78.9)
6. Resources and constraints	Resources for safety including staffing, equipment, lack of time and training.	My unit provides training on teamwork in order to improve patient care performance and safety. (PSCHO)	Total	57 (52.3)
			Quantitative	42 (50.6)
			Qualitative	1 (8.3)
			Mixed methods	14 (73.7)
7. Reporting and just culture	Willingness to report and a culture that does not assign blame.	Staff feel like their mistakes are held against them. (HSOPS)	Total	53 (48.6)
			Quantitative	36 (43.4)
			Qualitative	2 (16.7)
			Mixed methods	15 (78.9)
8. Openness	Open communication, staff feeling comfortable to express their issues or	In this clinical area, it is difficult to discuss errors. (SAQ)	Total	54 (49.5)
			Quantitative	35 (42.2)

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	concerns and question behaviours.		Qualitative	4 (33.3)
			Mixed methods	15 (78.9)
9. Learning and improvement	A focus on learning from mistakes, responding to, and improving systems.	When errors happen, we discuss how we could have prevented them. (SOS)	Total	51 (46.7)
			Quantitative	34 (40.9)
			Qualitative	3 (25.0)
			Mixed methods	14 (73.7)
10. Awareness of human limits	Awareness of individual ability to be safe and how that can be limited by various factors (e.g., fatigue).	I am less effective at work when fatigued. (SAQ)	Total	24 (22.0)
			Quantitative	16 (19.3)
			Qualitative	3 (25.0)
			Mixed methods	5 (26.3)
11. Wellbeing	Job satisfaction, burnout and other psychosocial factors.	Morale in this clinical area is high. (SAQ)	Total	17 (15.6)
			Quantitative	10 (12.0)
			Qualitative	1 (8.3)
			Mixed methods	6 (31.5)
Other	New, emerging themes and those unable to be classified elsewhere. Includes: flexibility, monitoring, personal values, patient and family involvement	It is important that my competence be acknowledged by others. (ORMAQ)	Total	30 (27.5)
			Quantitative	16 (19.3)
			Qualitative	6 (50.0)
			Mixed methods	8 (42.1)

HSCS: Hospital Safety Culture Survey, HSOPS: Hospital Survey on Patient Safety Culture, MSI: Modified Stanford Instrument, ORMAQ: Operating Room Management Attitudes Questionnaire, PSCHO: Patient Safety Climate in Healthcare Organizations survey, SAQ: Safety Attitude Questionnaire, SOS: Safety Organizing Scale.

* % of total calculated on n = 109. Insufficient detail on five surveys and four surveys were counted in the mixed methods studies because they were single-use and developed by the authors.

^ % of total calculated on n = 83

% of total calculated on n = 12

º % of total calculated on n = 19

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Leadership was the most frequently assessed safety culture theme, present in three-quarters of methods and tools reviewed. Awareness of human limits and wellbeing were the least frequent across the different methods. With the exception of leadership, all themes were covered to a greater extent by mixed methods studies, compared with using qualitative methods or surveys alone. However, qualitative studies had a relatively higher inclusion of "other" dimensions.

Although dimensions classified to the "other" category were too infrequent to constitute their own theme, there were some recurrent concepts. Specifically, across qualitative methods and quantitative surveys, there were dimensions related to patient involvement in safety culture;[43 63 66-68] interpersonal dynamics around trusting colleagues, developing a shared understanding, and valuing and empowering individuals;[40 62] and the need for flexibility and improvisation.[28 69 70] However, extending upon the latter of these themes, qualitative and mixed methods studies also reported a range of complexities,[69] external pressures,[71] contextual dependencies,[59] and the need to manage and prioritise risks in delivering safe care to patients,[47] which were not adequately captured by the narrow scope of established themes, such as "resource and constraints".

DISCUSSION

This review sought to: (1) map the range of methods and specific tools used to assess safety culture in hospitals; (2) analyse the prevalence of these in published research; and (3) examine the dimensions of safety culture captured in these methods and tools. We found that the overwhelming majority of studies exclusively utilised surveys. Few used surveys in combination with other methods or solely qualitative methods. We identified 11 safety culture themes and one residual category to cover the range of safety culture dimensions in methods and tools.

Safety culture surveys

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The most widely-utilised surveys were the HSOPS,[32] and the SAQ,[33] a finding consistent with past reviews on this topic.[12] The SAQ also covered the most safety culture dimensions of any of the reviewed survey, but this does not unequivocally suggest its superiority. Its "Stress recognition" subscale (coded to "Awareness of human limits") has been found to poorly correlate with other dimensions and hence is suggested to not adequately fit into the safety culture construct.[72] Sexton and colleagues more recent survey, Safety, Communication, Operational Reliability, and Engagement (SCORE), which was used in a couple of studies in this review,[73 74] includes burnout and work-life balance but not stress recognition.[75] This highlights that there is still no gold standard tool for measuring safety culture, limiting the capacity for comparison across studies, and the synthesis and meta-analysis of safety culture research, which makes surveys so appealing in the first instance.

Qualitative and mixed methods approaches

The present review found a relative lack of qualitative and mixed methods. Authors of studies that used qualitative or mixed methods, however, cited their value in providing deeper insights into what affects safety culture,[63 76] and what the priorities should be in addressing it. For example, Sine and Northcutt[61] compared findings from patient safety culture dimension survey scores (HSOPS) and the prioritisation of those dimensions by representative focus groups, which enabled understanding of how dimensions of safety culture interact with each other and the larger system of patient safety. Boussat and colleagues'[65] mixed method study, used the HSOPS, open response comments and in-depth interviews, finding the qualitative data complemented and provided more detail to the surveys. Practically-speaking, qualitative methods were suggested to redress a common limitation of safety culture surveys in hospitals: lower than desired response rates.[76] Hence, evidence suggests that qualitative methods may deepen or enrich the exploration of complex issues related to safety culture, identify priority dimensions and provide insight into areas for improvement.

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Dimensions of safety culture

We identified 11 themes related to safety culture assessment, which were broadly comparable to older reviews, suggesting consistency over time in the conceptualisation of core aspects of safety culture, including 'Leadership' which continued to be the most commonly explored. [9 12 19] On the other hand, more person-specific themes such as 'Wellbeing' and 'Awareness of human limits' were less widely represented, although they were present in the popular SAQ, as well as many of the tools and methods adapted from or influenced by it. Unlike previous reviews, our framework marks a distinction between the *perceptions of safety* (e.g., "I would feel safe being treated here as a patient" from the SAQ) and the normalised *prioritisation of safety* in the attitudes and behaviour of staff (e.g., "Patient safety is never sacrificed to get more work done" from the HSOPS). This allowed us to differentiate the HSOPS as providing greater coverage of the prioritisation of safety than the SAQ, a dimensions which some theorists suggest is the core feature of an organisation's safety culture.[14]

We found that mixed methods studies, particularly those that used comprehensive surveys (e.g., SAQ, HSOPS) in conjunction with qualitative methods, covered the most safety culture themes. However, no single method or tool appeared to measure all identified dimensions of safety culture. Qualitative and mixed methods studies were more likely to assess "other" aspects of safety culture, though, which suggests their capacity to derive unexpected, richer or more contextually relevant findings. These other aspects of safety culture covered the involvement of patients in safety culture,[43 63 66-68] including by surveying their perceptions of the construct,[46] interpersonal dynamics[40 62] and the context of how safety is enacted in a pressurised health care environment.[47 59]

Implications

Exclusively using a survey was the most common approach to assess safety culture. The HSOPS and SAQ were the two most widely used; both provide coverage of most of the safety culture dimension. However, both have recently undergone major revisions. The

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HSOPS version 2.0 was released in 2019 and involved deleting, rewording and adding multiple items, with the removal of a dimension for the overall perception of patient safety.[77]The SAQ has arguably been superseded by the SCORE, also developed by Sexton and colleagues, which removed one of the original dimensions and added a number of others with a greater focus on wellbeing.[75] Given the recency of these developments, it is unclear the extent to which these revised surveys will be taken up by hospitals and researchers, though some support exists for making this transition.[77]

In general, safety culture surveys are practical, able to be used by non-experts, and time-efficient for gathering large amounts of data from one or more participant groups in a reliable and reproducible manner.[11 50 78] No doubt part of their appeal, they also support comparison and benchmarking, activities which are omnipresent in health care, although not without critique including that such measurement may lead to *tunnel vision*, focusing on the "dimensions of performance that are included in the measurement system" (p. 570).[79] Surveys have encountered similar critiques from organisational culture theorists, that by reducing culture in any one organisation to a finite number of dimensions, reflected in items on a questionnaire, they may miss the "unique profile of cultural assumptions" (p. 614).[27] Our review highlights that qualitative and mixed methods approaches, on the other hand, were flexible to assessing other aspects of a hospital safety culture, and may uncover new, context specific or unexpected issues. Furthermore, qualitative data can be used to augment surveys,[65] particularly when one has failed to achieve a desired response rate.[76]

All of this points to the need for careful consideration and planning of the methodological approach one takes to an assessment of safety culture. Clearly using a survey is practical and feasible for capturing the perspectives of a large number of participants; however, the tool should be judiciously selected so that it covers, if not the most dimensions of safety culture, then those most important and appropriate for one's purposes, the setting

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and the participants. Ideally, assessment should include the use of qualitative methods in addition to surveys to enable not only greater exploration of specified dimensions, but the potential to identify other relevant aspects of safety culture.

Strengths and limitations

A strength of this study was searching four large academic databases. One limitation was the lack of information provided in the method of included studies to, for example, adequately identify safety culture dimensions, or even if a survey was new or an extension of an existing survey. Numerous studies of safety culture in hospitals provided inappropriate references (e.g., citing a secondary source that had used the SAQ, not calling the tool by its established name). Numerous surveys also have similar survey names and origins. Accurately capturing which surveys were used sometimes required a lengthy process of tracking through multiple sources to determine the tool being referred to. Finally, a workshop and multiple group meetings were conducted to ensure consistent coding of safety culture themes.

CONCLUSION

There are a multitude of methods and tools available to assess safety culture in hospitals, yet no clear agreement on all of its dimensions. Certainly, some surveys and mixed method approaches appear to be more comprehensive in their coverage of safety culture themes. The review highlights the discrepancy in the large volume of studies that exclusively use surveys compared with the much smaller number employing qualitative or mixed methods. Nevertheless, it indicates the value in incorporating qualitative and mixed methods approaches to verify or extend the findings of quantitative surveys and understand some of the contextual aspects of safety culture.

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COMPETING INTERESTS

The authors declare that they have no competing interests.

AUTHOR CONTRIBUTIONS

KC, LAE, AH, and MB had the original idea for the manuscript. KC and LAE led the search and data extraction. CP, AO, and JCL assisted with running the searches, screening papers and data extraction. KC, LAE and CP wrote portions of the manuscript. JB provided critical feedback on the manuscript and wrote the conclusion. All authors critically revised the manuscript.

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	p.1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	p.1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	p.7-8
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p.7-8
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	No protocol exists.
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	p.8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	p.8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	p.8 and Appendix 1.
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p.8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p.8-9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	p.8-9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	No formal risk of bias was appropriate given the focus of the review was to analyse prevalence of methods and tools (not examine study findings).
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	p.8-9

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			Method or tool used to assess safety culture.
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	p.8-9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p.10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Appendix 3. The focus of the review is on methods/tools rather than study characteristics.
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	No formal risk of bias was appropriate given the focus of the review was to analyse prevalence of methods and tools.
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Results summarised p.10-16.
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	No additional analyses.
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	p.16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	p.20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications	p.20

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		for future research.	
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding for a literature search, which guided the development of this review, was awarded to KC, LE, AH, and MB by the Australian Commission on Safety and Quality in Health Care. JB reports funding from the National Health and Medical Research Council: NHMRC Partnership Centre in Health System Sustainability (Grant ID 9100002) and NHMRC Investigator Grant APP1176620.

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Appendix 1 – Search strategy for Scopus

(TITLE-ABS-KEY("acute care" OR hospital) AND TITLE-ABS-KEY("quality culture" OR "safety culture" OR "culture of safety" OR "safety climate" OR "safety attitude") AND TITLE-ABS-KEY(survey OR questionnaire OR tool OR instrument OR measur* OR assess* OR checklist OR "check list" OR observ*) AND TITLE-ABS-KEY("patient safety" OR "public safety" OR "workplace safety")

2008- May 2020 = 1289 citations

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Name of survey, tool or approach	New survey developed by study authors?	Quantitative, Qualitative or Mixed methods	Details	As used in	Leadership and their support for safety	Perceptions of safety	Teamwork and collaboration	Safety systems	Prioritisation of safety	Resources and constraints	Reporting and just culture	Openness	Learning and improvement	Awareness of human limits	Wellbeing	Other	Details of other	Number of dimensions covered by tool
SAQ + interviews analysed using framework anaysis method		Mixed meth	Following survey, semi-structured interviews were co	Abdi Z, Delgoshaei B, Ravaghi H, Abbasi M, Heyrani A	1	1	1	1		1	1	1	1	1	1	1	Recommend Documen	11
Pharmacy Survey on Patient Safety Culture (PSOPSC) - based on HSC		Quantitative	Originally for community pharmacy it has been adapt	Jia PL, Zhang LH, Zhang MM, et al. Safety culture in a	1	1	1	1	1	1	1	1	1	1		1		11
SAQ + interviews		Mixed meth	Operating room (OR) version of the SAQ used first.	Khoshbin A, Lingard L, Wright JG. Evaluation of precas	1	1	1	1	1	1	1	1	1	1	1			11
Patient Safety Behavisoral Intent (PSBI) - developed by W	Yes	Quantitative	constructs based in part on the SAQ: 1. Prior	Wakefield JG, McLaws ML, Whitby M, Patton L, Patie	1	1		1	1	1	1	1	1	1	1	1	Norms	11
SAQ and Safety Climate Scale + interviews		Mixed meth	Interviews followed survey (which included open-end	Allen S, Chiarella M, Homer CS. Lessons learned from	1	1	1	1		1	1	1	1	1	1			10
HSOPSC + analysis of open responses and interviews		Mixed meth	Interviews were based on a thematic guide comprisin	Boussat, B., K. Kamalnavin and P. François (2018). "	1	1	1	1	1	1	1	1	1		1			10
Operating Room Management Attitudes Questionnaire (ORMAQ)		Quantitative	Fifty-six item questionnaire covering eight dimensions	Geraghty, A., Reid, S., & McIlhenny, C. (2014). Exami	1		1	1	1	1	1	1	1	1	1	1	Values	10
new Chinese SAQ - developed by Taiwan Joint Commission		Quantitative	The new Chinese SAQ has 46 items with additional d	Huang, C.-H., H.-H. Wu and Y.-C. Lee (2018). "The p	1	1	1	1	1	1	1	1	1	1	1			10
Manchester Patient Safety Framework (MaPSaF) + SAQ		Mixed meth	Evaluation of an improvement project using baseline	Lachman P, Linkson L, Evans T, Clausen H, Hothi D. I	1			1	1	1	1	1	1	1	1			10
Safety Attitudes Questionnaire - developed by Sexton et al. 2006		Quantitative	A 60 item survey of health professionals attitudes to s	Sexton JB, Sharek PJ, Thomas EJ, et al. Exposure to L	1	1	1	1		1	1	1	1	1	1			10
MaPSaF + HSOPSC + interviews, focus groups + analysis of reports		Mixed meth	Pre-post intervention study with MaPSaF run as work	Simons PAM, Houben R, Vlayen A, et al. Does lean m	1	1	1	1	1	1	1	1	1	1	1	1	Systems	10
Patient Safety Climate in Healthcare Organizations - developed by Sing	Quantitative	Developed from the PSCI, this 38 item survey evalua	Singer S, Lin S, Falwell A, Gaba D, Baker L. Relationsl	1	1	1	1	1	1	1	1	1	1	1	1	Flexibility	10	
Patient and Occupational Safety Culture Survey - developed Yes	Quantitative	Based on both the HSOPSC and SAQ, this survey ass	Wagner, A., M. Michaelis, F. Luntz, A. Wittich, M. Sch	1	1	1	1	1	1	1	1	1	1				10	
HSOPSC including open responses + interviews		Mixed meth	Interview schedule not focused on dimensions of safe	Campione, J. and T. Famolaro (2018). "Promising Prac	1	1	1	1	1	1	1	1	1					9
Focus groups + HSOPSC		Mixed meth	Focus group interviews first conducted, with the inter	Elder NC, Brunus SM, Nagy M, Kudel I, Render ML. I	1	1	1	1	1	1	1	1	1					9
HSOPSC + interviews and framework for analysis		Mixed meth	Expanding on survey results, three safety domains w	Livorsi D, Knobloch MJ, Blue LA, et al. A rapid assess	1	1	1	1	1	1	1	1	1					9
HSOPSC+ focus groups with interactive qualitative analysis		Mixed meth	Following administration of the surve, a focus group	Sine DM, Northcutt N. Interactive qualitative assessmen	1	1	1	1	1	1	1	1	1					9
Hospital Survey on Patient Safety Culture - developed by the Agency f	Quantitative	(dimensions): 1. Management support for patient	Sorra JS, Dyer N. Multilevel psychometric properties of	1	1	1	1	1	1	1	1	1	1				9	
HSOPSC + focus groups, Theoretical Domains Framework		Mixed meth	HSOPSC data used to identify areas to cover in inter	Stewart, D., B. Thomas, K. MacLure, A. Pallivalapila,	1	1	1	1	1	1	1	1	1					9
HSOPSC + interviews and framework for analysis		Mixed meth	Following survey, the interviews were aimed at identi	Wami SD, Demissie AF, Wassie MM, Ahmed AN. Patie	1	1	1	1		1	1	1	1			1	Patient fac	9
Interviews, focus groups + MSI		Mixed meth	Survey conducted prior to intervention, while intervie	Zecevic AA, Li AH, Ngo C, Halligan M, Kothari A. Imp	1	1	1	1	1		1	1	1					9
Junior Doctor-Patient Safety Attitudes and Climate Questio	Yes	Quantitative	The Junior Doctors' Patient Safety Attitudes and Clin	Durani P, Dias J, Singh HP, Taub N. Junior doctors an	1	1		1	1	1	1	1	1	1				8
Safety Climate Survey (SCSu) - based on SAQ - developed by Sexton	Quantitative	A 19-21 item derivative of the SAQ, it assesses safet	Sujeres M, Touranzeau A, Laschinger HKS, Dorian D.	1	1			1	1	1	1	1					8	
Chinese Hospital Survey on Patient Safety Climate - develo	Yes	Quantitative	Based theoretically on Zohar's model of safety cultur	Zhu J, Li L, Zhao H, Han G, Wu AW, Weingart SN. De	1	1	1	1	1	1	1	1	1					8
Safety Climate Survey -developed by Ahmed et al. 2018	Yes	Quantitative	Insufficient detail to classify entirely, but contains the	Ahmed, S. M., N. Fekry and F. A. Abed (2018). "Asse	1	1	1	1	1	1	1	1	1		1			7
The Patient Safety and Infection Prevention Assessment - described in	Quantitative	Survey includes Likert and open response questions a	Braun BI, Harris AD, Richards CL, et al. Does health c	1	1	1	1	1	1	1	1	1					7	
Hazardous Drug Handling Questionnaire - developed by Polovich and	Quantitative	This survey has 65 questions that measure variables	Callahan A, Ames NJ, Manning ML, Touchton-Leonard	1	1	1	1	1	1	1	1	1			1	E.g., My	7	
Modified Stanford Patient Safety Culture Survey Instrument (MSI) - de	Quantitative	Available in a number of iterations (2005, 2006), the	Castel ES, Ginsburg LR, Zaheer S, Tamim H. Understan	1	1	1	1	1	1	1	1	1	1				6	
untitled survey - developed by Johnston et al. 2018 + interv	Yes	Mixed meth	Survey of junior and senior surgeons included the SA	Johnston, M. J., S. Arora, D. King and A. Darzi (2018). "Improving the Quali	1	1	1	1	1		1				1	1	Supervisio	7
Hospital Organizational Culture Questionnaire - developed	Yes	Quantitative	A Japanese language survey with 24 items in eight dir	Kobuse H, Morishima T, Tanaka M, Murakami G, Hirose M, Imanaka Y. Visualizing variat	1	1	1	1	1	1	1	1	1	1				7
Manchester Patient Safety Culture Assessment Tool (MaPSCAT) - dev	Quantitative	Based on the MaPSaF, this 24 question survey uses o	Law MP, Zimmerman R, Baker GR, Smith T. Assessme	1		1	1	1	1	1							7	
untitled survey - developed by Matsubara et al. 2008	Yes	Quantitative	A 50 item, Japanese language safety culture survey a	Matsubara S, Hazhara A, Nobutomo K, Matsubara S,	1		1	1	1		1	1	1			1	Patient	7
Patient Safety Culture Questionnaire - developed by Veterans Affairs	Quantitative	Similar to the HSOPSC, the survey consisted of 68 ar	Mohr, D. C., J. L. Eaton, K. M. McPhaul and M. J. Ho	1	1			1	1	1							7	
Patients' Perceptions of Safety Culture scale (PaPSC) - dev	Yes	Quantitative	Developed from the HSOPSC, SAQ and patient expe	Monaca, C., B. Bestmann, M. Kattein, D. Langner, H.	1	1	1	1	1	1						1	Communic	7
Hospital Safety Climate Scale (HSCS) - developed by Gershon et al. 20	Quantitative	A 20 item survey covering the following six dimensio	Stringer B, Haines AT, Goldsmith CH, Berguer R, Blytt	1		1	1	1	1	1					1	E.g., my	7	
Pharmacy Safety Climate Questionnaire - developed by Ashcroft and P	Quantitative	Thirty-four item survey covering: investigating and le	Walpola RL, Chen TF, Fois RA, Ashcroft DM, Lalor D.	1		1	1	1	1	1		1					7	
Safety Climate and Capability (SCC) - developed by Benn	Yes	Quantitative	Based on established safety culture surveys, this 31 i	Benn J, Burnett S, Parand A, Pinto A, Vincent C. Facto	1	1	1	1	1		1	1	1					6
Victorian Safety Climate Survey (SCS) - developed by Sexton et al. 20	Quantitative	The 74 survey items were adapted from the SAQ and	Bennett PN, Ockerby C, Stinson J, Wilcocks K, Chalm	1		1	1	1	1	1			1				6	
Leading a Culture of Quality in Infection Prevention (LCQ-IP) - develo	Quantitative	The 19 item survey measures four dimensions of an i	Colet, P. C., J. P. Cruz, G. Cacho, H. Al-Qubeilat, S. S.	1		1		1	1	1	1	1					6	
Patient Safety Culture Assessment Scale (PSCAS) - adapted by Chen a	Quantitative	Based on the SAQ for use in China, the PSCAS conta	Hong S, Li Q. The reasons for Chinese nursing staff to	1	1	1			1		1	1	1				6	
Safety Climate Scale - developed by Zohar and Luria 2005		Quantitative	Originally two 27 item scales assessing organisationa	Hughes LC, Chang Y, Mark BA. Quality and strength o	1			1	1	1		1	1					6
Quality and Patient Safety Culture Survey - developed by Ki	Yes	Quantitative	Based on PSCI/PSCHO and HSOPSC, and other inst	Kagan, I., N. Porat and S. Barnoy (2019). "The quality	1		1	1		1	1					1	Perception	6
Patient Safety Culture Scale - developed by Calnan/ Health Region	Quantitative	Management and Leadership (17 items), Employees'	Klein TJ, Wilhesis C, Ghali WA. Predicting patient com	1	1			1	1	1							6	
untitled survey - developed by Liu et al. 2009	Yes	Quantitative	Developed for the Chinese population, this survey inc	Liu Y, Kalisch BJ, Zhang L, Xu J. Perception of safety	1	1	1		1	1	1							6
The National Emergency Department Safety Study Survey	Yes	Quantitative	The survey contains 50 questions about systems in th	Magid DJ, Sullivan AF, Cleary PD, et al. The safety of	1	1	1	1		1								6
untitled survey - developed by Schwonke et al. 2016	Yes	Quantitative	A Portuguese language survey based on the PSCI, it a	Schwonke CRGB, Filho WDL, Lunardi GL, da Silveira	1	1				1	1	1		1				6
untitled survey - developed by Walston et al. 2010	Yes	Quantitative	In part based on Zohar (1980), survey is made up of	Walston SL, Al-Omar BA, Al-Mutari FA. Factors affect	1		1	1	1	1			1					6
Patient Safety Culture Survey - Maternal and Child Health	Yes	Quantitative	This survey focuses on safety culture in the context o	Chang, Y., Hang, L., Qiu, C., Li, X., Yang, L., Wang,	1			1	1	1				1	1	1	Patient En	6
Attitudes to Patient Safety Questionnaire III (APSQ III) (Carruthers et a	Quantitative	Survey measures patient safety attitudes over nine fa	Abu-El-Noor, N. I., M. K. Abu-El-Noor, Y. Z. Abuswda, M. Alfaqawi and B. Böttcher (20	1			1			1	1		1		1	Patient inv	5	
General Safety Climate Survey - developed by Neal et al. 2000	Quantitative	A 25 item survey of safety culture in general (rather t	Amponsah-Tawiah K, Adu MA. Work Pressure and Sa	1				1	1	1							5	
California-Patient Safety Culture Survey (Cal-PSCS) - deve	Yes	Quantitative	A 22 item survey covering four factors: 1. Assistance	Avramchuk, A. S. and S. J. J. McGaire (2018). "Patient	1			1		1	1	1						5
untitled survey - developed by Birkmeyer et al. 2013	Yes	Quantitative	Survey developed from the HSOPSC and SAQ; focus	Birkmeyer NJ, Finks JF, Greenberg CK, et al. Safety cu	1	1	1	1		1								5
Interviews and observation		Quantitative	Interviews and observations used to evaluate training	De Korne DF, Van Wijngaarden JD, Yan Dyck C, Hak	1			1			1		1	1				5
untitled survey - developed by Ferrer et al. 2018	Yes	Quantitative	Developed in part from the HSOPSC, this 35 item m	Ferrer, M., R. Santa, A. Medhekar, E. Calvo and L. P. Molina-Sánchez (2018). "Drivers of	1		1	1	1		1		1	1				5
untitled survey - developed by Gibbs et al. 2018	Yes	Quantitative	Specifically focused on infection control, this survey	Gibbs, K., S. DeMaria, S. McKinsey, A. Fede, A. Harrington, D. Hutchison,	1			1		1			1	1				5
Focus groups and observation		Qualitative	Data collection and data analysis occurred in success	Gimenes FR, Torrieri MC, Gabriel CS, et al. Applying an ecological restoratio	1	1	1			1			1					5
untitled survey - developed by Kirkegaard et al. 2018 + ob	Yes	Mixed meth	Focused on occupational health and safety, and viole	Kirkegaard, M. L., P. Knæs, H. B. Nielsen and A. H. G	1	1			1		1					1	Risk-taking	5
untitled survey - developed by Kudo et al. 2008	Yes	Quantitative	A Japanese survey with 30 items assessing the follow	Kudo Y, Kido S, Tarzuraka Shahzad M, Saegusa Y, Sat	1		1			1	1		1					5
untitled survey - developed by Listyowardojo et al. 2012	Yes	Quantitative	Developed from the SAQ and Jobs Diagnostic Survey	Listyowardojo TA, Nap RE, Johnson A. Variations in h	1		1		1	1					1			5
SAQ + interviews with SAQ as framework for analysis		Mixed meth	Interviews used to probe topic of safety culture deepe	Listyowardojo TA, Yan X, Leyshon S, et al. (2017) A s	1	1	1			1						1	Scope of	5
Interviews, focus groups + Cultural Assessment Survey		Mixed meth	Qualitative data used to inform survey development.	Milne JK, Bendaly N, Bendaly L, Worsley J, Fitzgerald J, Nisker J. A measure	1	1	1		1			1				1	wanting	5
Patient Safety Dialogue - developed Ohrn et al. 2011	Qualitative	Based on the MaPSaF, questions about patient safety	Ohrn A, Rutberg H, Nilsen P. Patient safety dialogue: e	1	1		1				1	1					5	
Just Culture Assessment Tool (JCAT) - developed by Petse	Yes	Quantitative	of safety culture (from Reason's theory: "a	Petschonek S, Burlison J, Cross C, et al. Development of the just culture assessment tool: measuring the pa	1			1			1	1				1	Trust -	5
Veterans Health Administration (VHA) Hospital Survey on	Yes	Quantitative	This survey assesses 5 domains of safety climate: 1. p	Pringle J, Weber RJ, Rice K, Kirisci L, Sirio C. Examin	1		1	1	1							1	Patient saf	5
Hospital Transfusion Service Error Reporting and Safety Culture Surve	Quantitative	transfusion medicine, this survey contains 79 survey	Sorra J, Nieva V, Fastman BR, Kaplan H, Schreiber G.	1	1	1			1	1							5	
Vienna Safety Climate Questionnaire (VSCQ) - developed	Yes	Quantitative	A German-language survey based in large part on oth	Steyrer J, Schiffinger M, Huber C, Valentin A, Strunk G	1	1				1	1	1						5
Interviews		Qualitative	Study sought to evaluate a program. Post intervention	Sustain MA. A novel tool for organisational learning and its impact on safety cu	1	1				1	1							5
Human Factors Attitude Survey (HFAS) - developed by Gregan et al. 2	Quantitative	A 23 item survey based on one in aviation, it assesses	Wu WY, Wu YL, Hou SM, et al. Examining the effects	1		1	1				1	1	1				5	
Safety Climate of Nurses (SCN) - Sarasangi et al. 2015	Quantitative	Focused specifically on nurses, it contains 22 questio	Zarei E, Khakzad N, Reniers G, Akbari R. On the relati	1		1			1	1				1			5	
Medical Students' Patient Safety Questionnaire - developed by Flin et a	Quantitative	Originally developed to assess the students' levels of	Ahmed M, Arora S, Baker P, Hayden J, Vincent C, Sevdalis N. Building capacity and capability for patient safety educat	1					1	1	1						4	
Survey on Patient Safety Climate - developed by Nayeh et al. 2005	Quantitative	The survey covers four dimensions: - managerial safe	Custo, P. T., R. T. Custo and S. Buttigieg (2019). "The	1				1	1	1							4	
untitled survey - developed by Hahn & Murphy, 2008	Yes	Quantitative	A 6 item measure of global work safety climate, ques	Hahn SE, Murphy LR. A short scale for measuring safe	1				1		1		1					4
untitled survey - developed by Lowe 2012	Yes	Quantitative	Survey is comprised of the following six items: Errors	Lowe G. How employee engagement matters for hospital performance. Healt	1			1			1		1					4
Error Orientation Scale and Zohar Safety Climate Scale - re	Quantitative	This survey combines two scales, the Error Orientati	Mark BA, Hughes LC, Belyea M, Bacon CT, Chang Y,	1	1	1	1										4	
Medical Safety Climate Questionnaire - developed by Madigosky et al.	Quantitative	Questionnaire with 28 items assessing knowledge, sk	McCarthy SE, O'Boyle CA, O'Shaughnessy A, Walsh G. Online patient safety education programme for junior doctors: i	1				1		1		1	1				4	
untitled survey - developed by Gimeno et al. 2005	Quantitative	Based in part on the HSCS and developed for Costa	Menendez CC, Gimeno D, Barrientos-Gutiérrez T, Bur	1				1		1	1						4	
VA Employee Safety Perception Survey - developed by the National Sa	Quantitative	The survey consisted of 50 questions that represente	Mohr, D. C., J. L. Eaton, K. M. McPhaul and M. J. Ho	1	1		1										4	
untitled survey - developed by Pazz et al. 2013	Yes	Quantitative	One domain of their survey focused on safety culture	Pazz K, Schur C, Cuzzao L, Lacado J. A national study o	1	1				1			1					4
untitled survey - developed by Simsekler 2019 + interviews	Yes	Mixed meth	Survey based on some dimensions of the MaPSaF (m	Simsekler, M. C. E. (2019). "The link between healthcare risk identification and patient safety culture." International Jou	1			1			1					1	Dimension	4
Interviews and focus groups		Qualitative	Interview questions were broad and exploratory- (a)	Wailleg, J., B. Robinson and M. Coombs (2019). "Surv	1	1										1	Safety as s	4
Safety Climate Questionnaire - developed by Zeidi et al. 2012	Quantitative	This 37 item questionnaire includes 8 aspects of mun	Yari, S., M. H. Naseri, H. Akbari, S. Shahsavari and H.	1	1	1		1									4	
untitled survey - developed by Al Faqeeh et al. 2019	Yes	Quantitative	An English-language survey assessing: safety behavio	Al Faqeeh, F., K. Khalid and A. Osman (2019). "Integ	1	1			1									3
untitled survey - developed by Berman et al. 2018	Yes	Quantitative	A surgery-focused questionnaire, asking surgeons wh	Berman L, Ranapi A, Cuddy D, Sharda M, Otosc	1	1			1									3
untitled survey - developed by Boyer et al. 2012	Yes	Quantitative	A brief 7 item measure of safety climate with the foll	Boyer KK, Gardner JW, Schweckhart S. Process qualit	1	1						1						3
Document analysis + untitled survey - developed by Chapp	Yes	Mixed meth	To assess radiation safety culture, analysis of safety r	Chapple CL, Bradley A, Murray M, et al. Radiation safety culture in the UK a	1				1	1								3
Clinician Safety Culture and Leadership Questionnaire - de	Yes	Quantitative	A 35 item questionnaire covering safety culture, team	Clay-Williams, R., N. J. Taylor, H. P. Ting, T. Winata,	1	1	1											3
Ethnography - Interviews and observation	Qualitative	Interview schedule focused on teamwork and comm	Gillespie BM, Gwinner K, Chaboyer W, Fairweather N. Team communications in surgery -	1							1					developing	3	
Interviews and grounded theory	Qualitative	Interviews conducted in a grounded theory design (st	Graves TS, Fitzgald-Conner D, Wakefield BJ. It's always something: hospital	1								1			1	managing	3	
untitled survey - developed by Ji et al. 2019	Yes	Quantitative	In part developed from the PSCHO, this survey is spe	Ji, C. H., F. H. Xu and S. Pan (2019). "Patient safety culture regarding intrave	1				1	1								3
Interviews and document analysis using neo-institutional theory	Qualitative	Study examined safety culture and how it was influen	Kyratsis, Y., R. Ahmad, M. Iwami, E. Castro-Sanchez, R. Atun and A. H. Ho	1				1								macro (req	3	
Work Safety Scale (WSS) - Hayes et al. 1998	Quantitative	A 50 item survey with the following five dimensions:	McCaughey D, DelliFraine JL, McGhan G, Bruning NS.	1		1			1								3	
Patient Safety Attitude Scale-Nurse Aide (PSAS-NA) - dev	Yes	Quantitative	Developed to assess nurse aides' safety culture in Tai	Peng SJ, Yu ML. Psychometric testing of an instrument measuring nurse aides' patient safety attitudes. J	1	1	1		1									3
untitled survey - developed by Peterson et al. 2016	Yes	Quantitative	A 10 item survey derived from Gershon et al. (2000)	Peterson K, Rogers BME, Brosseau LM, et al. Differen	1		1		1									3
untitled survey - developed by Santa et al., 2018	Yes	Quantitative	Insufficient detail to classify entirely, but contains the	Santa, R., S. Borrero, M. Ferrer and D. Gherisi (2018). "Fostering a healthcare sector quali	1		1		1		1							3
Speaking Up about Patient Safety Questionnaire (SUPS-Q)	Yes	Quantitative	The survey contains: three behavior-related scales (1	Schwappach, D., G. Sendlhofer, L. Häsler, V. Gombotz, K. Leitgeb, M. Hoffi	1					1	1	1						3
Teamwork and Safety Climate Questionnaire (TSCQ) - developed by S	Quantitative	A short, 27 item version of the SAQ that contains onl	Schwartz, M. E., D. E. Welsh, D. E. Paull, R. S. Know	1	1	1											3	
Interviews	Qualitative	An inductively focused interview study, with intervie	Sirriyeh R, Lawton R, Armitage G, Gardner P, Ferguso	1	1										1	safety	3	
Interviews	Quantitative	Two stages were conducted to assess perceived risk a	Scholz SJ, Strand K, Soreide E. Percutaneous dilatational tracheotomy in the IC		1								1	1			3	
Patient Safety Climate Scale - developed by Weng et al. 20	Yes	Quantitative	For the survey, patient safety climate defined as "the	Weng R-H, Chen J-C, Pong L-J, Chen L-M, Lin T-C. T	1			1	1									3
Psychosocial safety climate (PSC) - developed by Hall et al. 2010	Quantitative	Examines psychosocial safety climate, comprised of	Zadow AJ, Dollard MF, McLinton SS, Lawrence P, Tu	1							1					1	Organizati	3
Physical safety climate (PSC) - developed by Idris et al. 2012	Quantitative	Examines physical safety climate, comprised of 12 it	Zadow AJ, Dollard MF, McLinton SS, Lawrence P, Tu	1							1					1	Organizati	3
untitled survey - described by Mercer 2019	Quantitative	A two item measure of safety climate developed by a	Alajbri, D., A. Vaughn, M. Austin, L. White, Z. Li, J. N					1									Psychology	2
Safety Organising Scale (SOS) - developed by Vogus and Sutcliffe 200	Quantitative	The nine item SOS focuses on understanding and uni	Pensschenck S, Burlison J, Edwards R, Hoffman J. Ass	1			1									1	Psychology	2
untitled survey - developed by Hatoun et al. 2016	Yes	Quantitative	Patient safety climate was assessed using the	Hatoun J, Suen W, Liu C, et al. Elucidating Reasons for	1	1												2
untitled survey - developed by Quan et al. 2016	Yes	Quantitative	Assessed by the following four items: Establishment o	Quan M, Wang X, Wu H, et al. Influencing factors on u	1			1										2
Catheter-Related Bloodstream Infection Prevention Practic	Yes	Quantitative	Two statements in this survey are used to assess safe	Ullman AJ, Long DA, Rickard CM. Prevention of centr	1	1												2
Violence Prevention Climate - developed by Kessler et al. 2008	Quantitative	The study used only the the four-item "Practices" sul	Arnetz, J., L. E. Hamblin, S. Sudan and B. Arnetz (201	1														1
Ethnography - interviews, observations, document analysis using Haber	Qualitative	Informal conversations, interview conducted. Docume	Bressers, G., I. Wallenburg, R. Stalmeijer, M. oude Fabrink and K. Lombarts (2020). "Patient safety in medical residency training: Balancing bravery and checklists." Health (United Kingdom).												1	workaroun	1	
Interviews and focus groups	Qualitative	Focus group questions were about patient safety situ	Sabry, H. A., M. A. Soliman, H. M. Bazaraa and A. A. Hegazy (2020). "Improving patient safety at pediatric intensive c	1													1	
Insufficient detail of survey provided	Quantitative	Insufficient detail provided	Bishop AC, Boyle TA. The Role of Safety Culture in Influencing Provider Perceptions of Patient Safety. Journal of Patient Safety. 2016;12(4):204-209.														0	
Insufficient detail of survey provided	Quantitative	Insufficient detail provided	Muralidhar S, Taneja A, Ramesh V. Patient safety culture-perception of health care workers in a tertiary care hospital. The International journal of risk & safety in medicine. 2012;24(4):191-199.														0	
Insufficient detail of survey provided	Quantitative	Insufficient detail provided	Sahebalzamani M, Mohammady M. A study of patient safety management in the framework of clinical governance according to the nurses working in the ICU of the hospitals in the East of Tehran. Iranian journal of nursing and midwifery														0	
Insufficient detail of survey provided	Quantitative	Insufficient detail provided	Shaw KN, Ruddy RM, Olsen CS, et al. Pediatric patient safety in emergency departments: unit characteristics and staff perceptions. Pediatrics. 2009;124(2):485–493.														0	
Insufficient detail of survey provided	Quantitative	Insufficient detail provided	Theeb Moeidh A, Abdulaziz Shah F, Mohammed Al-Matari E. The relationship between prosocial voice and the patient safety culture in the Saudi public hospitals. Asian Social Science. 2015;11(16):267-277.														0	
					84	65	61	58	58	57	53	54	51	24	17	30	0	612

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The dimensions of safety culture: A systematic review of quantitative, qualitative and mixed methods for assessing safety culture in hospitals

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ABSTRACT

Background: The study of safety culture and its relationship to patient care have been challenged by variation in definition, dimensionality and methods of assessment. This systematic review aimed to map methods to assess safety culture in hospitals, analyse the prevalence of these methods in the published research literature, and examine the dimensions of safety culture captured through these processes.

Methods: We included studies reporting on quantitative, qualitative and mixed methods to assess safety culture in hospitals. The review was conducted using four academic databases (PubMed, CINAHL, Scopus, and Web of Science) with studies from January 2008 to May 2020. A formal quality appraisal was not conducted. Study purpose, type of method and safety culture dimensions were extracted from all studies, coded thematically, and summarised narratively and using descriptive statistics where appropriate.

Results: A total of 694 studies were included. A third (n=244, 35.2%) had a descriptive or exploratory purpose, 225 (32.4%) tested relationships among variables, 129 (18.6%) evaluated an intervention, while 13.8% (n=96) had a methodological focus. Most studies exclusively utilised surveys (n=663; 95.5%), with 88 different surveys identified. Only 31 studies (4.5%) used qualitative or mixed methods. Thematic analysis identified 11 themes related to safety culture across the methods, with "Leadership" being the most common. Qualitative and mixed methods approaches were more likely to identify additional dimensions of safety culture not covered by the 11 themes, including improvisation and contextual pressures.

Discussion: We assessed the extent to which safety culture dimensions mapped to specific quantitative and qualitative tools and methods of assessing safety culture. No single method

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or tool appeared to measure all 11 themes of safety culture. Risk of publication bias was high in this review. Future attempts to assess safety culture in hospitals should consider incorporating qualitative methods into survey studies to evaluate this multi-faceted construct.

KEYWORDS

Safety culture, Patient safety, Hospital, Measurement, Organisational culture

ARTICLE SUMMARY

- Reviewed both qualitative and quantitative methods for assessing safety culture in hospitals.
- Examined the extent diverse methods and different surveys provide coverage of safety culture dimensions.
- The literature search was performed in four academic databases.
- Included only published, peer-reviewed and English language empirical studies; potential insights from grey literature not explored.
- Evaluated qualitative and mixed methods separately from quantitative surveys.

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BACKGROUND

Over the last two decades, the notion that hospitals have a "safety culture" that can be measured, understood and improved, has become a pillar of the patient safety movement,[1] and is increasingly embodied in accreditation and regulatory standards.[2 3] Accordingly, the need for robust assessment of safety culture has become evident,[4] and tools have proliferated.[5 6] Nowadays, safety culture assessment is widely used in hospitals for four key reasons, namely, to: (1) diagnose safety culture and identify areas for improvement; (2) evaluate patient safety interventions or programs and track change over time; (3) conduct internal and external benchmarking; and (4) fulfill directives or regulatory requirements.[1 7] Despite expanding interest in studying and addressing safety culture over the last decade and a half, several challenges persist in defining and assessing the dimensions of safety culture.

Defining safety culture

There are many definitions of safety culture, both within and outside of health care. Cooper[8] evocatively summarised the situation well in his description of the field as a "definitional swamp", highlighting 51 separate definitions of safety culture and 30 separate definitions of safety climate. Compounding equivocality, many researchers mark a distinction between safety *climate* and safety *culture*,[7] suggesting the former involves peoples' *perceptions* of their organisations (its procedures, practices, and the kinds of behaviour that are tolerated or rewarded), whereas culture operates on a deeper, more enduring level, representing the behaviours and practices and the underlying, sometimes unconscious beliefs and values within an organisation.[9] Others in the field use culture and climate interchangeably.[10]

A typical definition of safety culture is:[11] "The product of individual and group values, attitudes, perceptions, competencies, and patterns of behaviour that determine the commitment to, and the style and proficiency of, an organization's health and safety management" (p. 339).[12] Although this provides some guidance on which constructs to

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consider when assessing safety culture, the specificities of the attitudes, values and norms conducive to safety, as well as how to assess them, is less clear.

Assessment of safety culture and its dimensions

Safety culture is considered multi-faceted, consisting of several interrelated attributes or dimensions, although consensus on what these dimensions are remains elusive.[12 13] For example, while some conceptualise safety culture as exclusively the prioritization of safety by the unit and organisational leadership,[14] others construe it more broadly to include learning, reporting, and blame orientation.[15-17] Recognising and responding to this equivocality, over the years several reviews of safety culture assessment in health care have examined dimensions, highlighting the most frequent dimensions[9 12 18 19] While the themes these reviews have delineated overlap, they differ considerably in scope (i.e., some are broad, while others are specific) and quantity. These similarities and differences are summarised in Table 1, where thematically related dimensions are presented on the same row.

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Table 1. Dimensions derived from reviews of safety culture assessment in health care.

Flin and colleagues[9] Safety management themes	Singla and colleagues[18] Dimensions of safety culture	Halligan and Zecevic[12] Most frequent safety culture dimensions	Alsalem and colleagues[19] Safety climate dimensions
Review focus			
Examined 12 quantitative safety culture survey studies. Identified 10 safety management themes.	Reviewed 13 survey instruments, inductively and deductively categorising each item into 23 different dimensions (under six broad categories).	Reviewed 137 articles on safety culture in health care, of which 113 explored dimensions. Identified six most frequent dimensions.	Reviewed five most common safety culture surveys by item. Identified 12 dimensions.
Dimensions			
Management and supervisors	Management and institutional commitment to safety (Management and supervision)	Leadership commitment to safety	Top management support & institutional commitment to safety
	Institutional responses (Management and supervision)		
	Adequacy of training and supervision (Management and supervision)		Training & continuous education
Reporting/speaking up	Nonpunitive response to error (Management and supervision)	A non-punitive approach to adverse event reporting and analysis	Reporting incidents & "non-punitive" response to error
Risk perception	Risk taking (Risk)		
	Willingness to ask for help (Risk)		
Safety systems	Detection infrastructure (Safety system)		Safety systems: "Policies & procedures, safety planning, hand offs & transitions, staffing, equipment"
	Handoffs and transitions and coordination of care (Safety system)		
	Patient safety planning (Safety system)		

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	Adequacy of staffing (Safety system)		
	Adequacy of equipment, information and processes (Safety system)		
	Reporting infrastructure (Safety system)		
	Adequacy of crisis management (Competence)		
	What should be reported and to whom (Procedure/rules)		
	Compliance with rules and procedures (Procedure/rules)		
Job demands	Work pressure (Work pressure)		Work pressure
Safety attitudes/behaviours	Overall perception of safety	A shared belief in the importance of safety	Safety perceptions & attitudes of staff, risk perceptions
Communication/feedback	Communication openness (Other)	Open communication founded on trust	Communication openness
	Feedback and communication (Other)		Feedback & communication about adverse events
Teamwork	Teamwork (Other)	Teamwork	Teamwork
Personal resources (e.g., stress)	Job satisfaction (Other)		Staff satisfaction
Organisational factors	Organisational learning (Other)	Organisational learning	Organizational learning and continuous improvement
	Beliefs about the causes of errors and adverse events		Beliefs about the causes of errors & adverse events

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Only one of these reviews included qualitative and mixed method approaches alongside quantitative surveys.[12] A decade ago, Halligan and Zecevic found only 14 of 137 articles utilised qualitative methods (e.g., interviews, focus groups); given the preponderance of quantitative surveys, limited discussion was given to these methods, and it was not clear how they mapped to the dimensions of safety culture they identified or whether these studies uncovered additional insights into safety culture in hospitals.

The widespread, exclusive use of surveys to assess safety culture differentiates health care from other industries where a range of methods are typically used together.[20-22] This is because surveys alone do not expose rich insights into dimensions of culture.[23] Following Schein's model of organisational culture[24] which has been widely adopted in health care,[25] and to understand safety culture,[13 26] surveys capture the first two layers of artifacts and espoused beliefs rather than the deeper underlying shared assumptions at the heart of an organisation's culture. Cultural dimensions are also arguably to some extent contextually dependent, cannot be pre-specified and hence require rich interpretive methods to uncover: "every organization has a unique profile of cultural assumptions that any questionnaire inevitably misses" (p. 615).[27] In-depth, open-ended, qualitative methods are considered better suited to accessing these deeper facets and contextual nuances of culture,[24 28] because their iterative, interactive nature allows for questions, and therefore the conceptualisation of safety culture, to evolve in response to feedback as data are collected.[29]

The present review

There has been limited integration of qualitative research in past reviews of safety culture assessment in health care, with none examining dimensions by type of method. The potential for qualitative methods to uncover additional insights into safety culture dimensions has not been considered.[12] We set out to map the range of methods and tools used to assess safety culture in hospitals, including qualitative, quantitative and mixed approaches. We sought to

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analyse the prevalence of use of these methods and tools, the purpose for using them, and examine the ways safety culture was conceptualised—including its dimensions—through the application of methods or tools.

METHOD

Search strategy and criteria

This systematic review followed the Preferred Reporting Items for Systematic Reviews (PRISMA) guidelines (see Appendix 1 for PRISMA checklist).[30] Four databases were searched: PubMed, CINAHL, Scopus, and Web of Science. The search was from 2008 to May 2020, using a search string based on previous reviews:[12] ('acute care' OR 'hospital') AND ('quality culture' OR 'safety culture' OR 'culture of safety' OR 'safety climate' OR 'safety attitude') AND ('survey' OR 'questionnaire' OR 'tool' OR 'instrument' OR measur* OR assess* OR checklist OR 'check list' OR observ*) AND ('patient safety' OR 'public safety' OR 'workplace safety'). The full search for Scopus is provided in Appendix 2.

To be eligible for inclusion in the review, studies were required to have: (1) been an empirical study published in a peer-reviewed journal; (2) utilised a method(s) or tool(s) for the purposes of assessing, studying or measuring safety culture or climate; (3) collected data in hospital(s); and (4) the study was published in English. Review articles were retained for snowballing and context but were otherwise excluded from this review.

Study selection

Titles and abstracts were divided and independently reviewed by four of the authors (CP, LAE, KC, AO). Fleiss' kappa statistic[31] was used to evaluate inter-rater agreement of 5% of the titles/abstracts by two authors (CP, KC). Snowballing was also used to identify studies from reference lists of included studies and reviews identified.

Examination of methods used to assess safety culture

The study purpose for all included studies was extracted. The primary study aim was coded into one of four categories to determine whether certain methods were selectively used for

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certain purposes. These categories, developed collaboratively by the first and second author, were: (1) Descriptive or exploratory assessment of safety culture including comparisons between groups (e.g., doctors and nurses, countries) and studies establishing a "baseline" or identifying areas for improvement; (2) Testing relationships among components of safety culture, or between safety culture and other measured variables; (3) Methodological focus including validation studies or those furthering the assessment of safety culture; or (4) Evaluation of an intervention, typically a program to improve safety, safety culture or quality. As categories (2) and (3) had a theoretical and methodological focus, respectively, they were deemed purposed for research, whereas categories (1) and (4) were more applied and, therefore, deemed improvement focused.

Other aspects of data extraction differed depending on method. Among the studies exclusively using quantitative surveys, data extraction focused primarily on details of the tool itself rather than the study, including survey origins and details of dimensions and items. For surveys that had been developed by included studies, we noted information on the development of the tool including what, if any, other surveys it had been derived from. For studies that used an already established survey, where required, we found the original source to collect these details.

Due to their bespoke nature, studies that used qualitative or mixed methods to assess safety culture were examined individually. Information on study design (e.g., ethnography), methods of data collection (e.g., interviews, observations), analysis (e.g., thematic analysis), and sometimes results—in the case of iterative data collection and analysis—were inspected to extract information on the method of assessment and the conceptualisation of safety culture. This included review of the types of questions asked during interview (or the interview schedule, if provided), and details of any framework or theory used for analysis.

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To extract information on dimensions of safety culture, three of the authors (CP, LAE, KC) first deductively developed a draft coding framework by examining dimensions and themes identified in previous reviews.[9 12 18 19] As shown in the rows of Table 1, similar dimensions across reviews were organised together, with some refined into broader themes rather than more specific safety categories (e.g., *safety systems* or *resources and constraints*, rather *detection infrastructure, adequacy of staffing*). In a workshop session, the framework was then applied to all items (n=275) from eight of the most common surveys. These were coded individually by each author and then responses were discussed together to refine the coding framework and its definitions, resolve any disagreement, and ensure consistent coding. During this process, the *prioritisation of safety* and the *perceptions of safety* (i.e., how safe an organisation is) were broken up into distinct themes, because though related, there was sufficient evidence of a distinction and that both were important. After establishing that the coding framework could be applied consistently, the remaining surveys and studies were divided up and coded.

Quality appraisal

A formal quality assessment was not conducted due to the wide variation in study design, the large number of included studies, and the focus of this review being on methods rather than the results of the included studies. However, the information provided in the methods sections and reference lists of included studies was assessed as part of the data extraction process.

Patient and public involvement

No patients involved.

Ethics statement

Ethical approval was not required for this systematic review as no human participants were involved.

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RESULTS

The search retrieved a total of 4,020 papers, which were downloaded into the reference management software EndNote and duplicates removed, giving 1,751 articles. Four studies identified from the reference lists of papers were also reviewed, giving a total of 1,755 studies. Following title and abstract screening, 1,014 articles were included for full-text review (See Figure 1). Of these, 694 articles met inclusion criteria and are included in this review.

[INSERT FIGURE 1 HERE]

The overwhelming majority of included papers (n=663; 95.5%) exclusively utilised self-report quantitative surveys to assess safety culture. A further 19 papers (2.7%) were mixed methods studies that reported using surveys in combination with other methods, such as interviews, while 12 papers (1.7%) reported solely using qualitative methods. The largest proportion of studies had a descriptive or exploratory purpose (n=244, 35.2%), followed closely by testing relationships among variables (n=225, 32.4%). Almost one-fifth of studies assessing safety culture evaluated an intervention (n=129, 18.6%), while 14% (n=96) had a methodological focus. Purposes differed slightly according to the type of method utilised (see Figure 2), with qualitative and mixed methods studies more likely to be exploratory, and no qualitative studies having an explicit methodological focus. The full list of all included studies is in Appendix 3.

[INSERT FIGURE 2 HERE]

Quantitative surveys assessing safety culture

In total, 88 surveys were identified, 45 of which were new surveys developed by authors of included papers. The most widely-used surveys were the Hospital Survey on Patient Safety

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Culture (HSOPS; n=312; 45.7%) by the Agency for Healthcare Research and Quality,[32] and the Safety Attitudes Questionnaire (SAQ; n=209; 30.6%) developed by Sexton and colleagues at the University of Texas.[33] This was followed rather distantly by the Patient Safety Climate in Healthcare Organisations (PSCHO), the Safety Climate Survey (SCSu), the Hospital Safety Climate Scale (HSCS), the Teamwork and Safety Climate Survey (TSCS)[33 34] and the Operating Room Management Attitudes Questionnaire (ORMAQ).[35] A large proportion (n=62, 70.4%) of the 88 surveys identified were used only a single time among the 682 survey studies. See Table 2 for the most common surveys.

Table 2. Most frequently used surveys to assess safety culture in hospitals.

Name of survey	Number of studies	%*
Hospital Survey on Patient Safety Culture[32]	312	45.7
Safety Attitudes Questionnaire[33]	209	30.6
Patient Safety Climate in Healthcare Organizations[6]	20	2.9
Safety Climate Survey[36]	9	1.3
Hospital Safety Climate Scale[37]	8	1.2
Teamwork and Safety Climate Survey[33 34]	7	1.0
Operating Room Management Attitudes Questionnaire[35]	5	0.7

*Proportion of papers using surveys across all studies using surveys (n=682).

The 88 surveys encompassed not only entirely new and unique measures, but also surveys that built on or refined existing tools, or that used some of the same questions. For example, the PSCHO was derived from the Veteran Affairs (VA) Palo Alto/Stanford Patient Safety Center of Inquiry (PSCI) survey.[6 38] The Modified Stanford Instrument and Canadian-Patient Safety Climate Survey—both by Ginsburg and colleagues[4 39]—are also based on the PSCI but included some questions from the HSOPS.

Of the 45 new measures, many were based at least in part on established measures of safety culture, including the HSOPS and SAQ. Others focused on particular aspects of safety culture within hospitals, such as the Just Culture Assessment Tool[40] and the Catheter-Related Bloodstream Infection Prevention Practices Survey;[41] or certain staff groups (e.g., Junior Doctor-Patient Safety Attitudes and Climate Questionnaire);[42] were developed for specific settings (e.g., maternal and child health institutions,[43] acute geriatric units)[44] or languages

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(e.g., Chinese).[45] Distinct from other self-report surveys, the Patients' Perceptions of Safety Culture scale was designed to assess patient, rather than staff, perceptions of hospital safety culture.[46]

Qualitative and mixed methods studies assessing safety culture

The review identified 31 studies (4.5%) that used qualitative or mixed methods to assess safety culture. Interviews were the most common approach (n=23; 74.2%), followed by frameworks and theories (n=11, 35.5%), and focus groups (n=9, 29.0%) (see Table 3). Most studies reported their interview schedule or example questions along with the main topics covered. These questions varied in how they were framed and their content (e.g., broad and exploratory: "Can you tell me about what patient safety means to you?" versus more specific and contained: "How safe do you think is the care provided in the ICU?").

Frameworks or theories were used in 11 studies (35.5%) to guide the collection and analysis of qualitative data, and included grounded theory,[47] a lifeworld versus system distinction,[48] and the SAQ's dimensions.[49 50] The Manchester Patient Safety Framework (MaPSaF) was used across four studies, typically in its conventional facilitated workshop,[51-53] or as an adapted version.[54] Similar to surveys, it presupposes a number of dimensions for which staff are required to rate the safety culture in their unit or organisation. However, the response range is not a Likert scale, but classifies cultures along five theorised levels of cultural maturity (i.e., pathological, reactive, bureaucratic, proactive, generative).[55] A quantitative survey tool based on the MaPSaF and using a similar ordinal response range (the Manchester Patient Safety Culture Assessment Tool, MaPSCAT) was also identified by this review.[56]

Table 3. Qualitative methods to assess safety culture in hospitals.

	Number of studies*	%^
Interviews	23	74.2
Framework or theory	11	35.5
Focus groups	9	29.0
Observations	5	16.1

Document analysis	5	16.1
Ethnography	2	6.5

** Frequencies exceed the total number of qualitative/mixed methods studies due to studies using multiple methods/tools.*

^ % based on total number of qualitative and mixed methods studies (n=31).

Observations were used in only five studies (16.1%), always in conjunction with interviews, focus groups, or surveys. One of these studies involved novel nurse led photo elicitation during walkabouts together with focus groups to study the safety culture related to medication management in a Brazilian intensive care unit.[57] Five studies (16.1%) analysed organisational documents and data as a window into safety culture. For example, safety checklists used during management walkarounds were analysed and triangulated with survey responses and focus group data in a study of intensive care unit safety culture following a safety intervention.[58] Finally, two studies (6.5%) utilised a combination of qualitative methods, including observations, informal conversations, interviews, within an ethnographic study design to examine how team communication affected safety culture in surgery,[59] and the developing patient safety culture of medical residents.[48]

More than half of studies using qualitative methods did so in conjunction with surveys to assess safety culture in a mixed methods design (n=19, 61.3%), to elaborate[51 60] or compare[61] with quantitative findings, develop a safety culture survey,[62] or identify factors influencing patient safety culture.[63] Some survey studies also incorporated abbreviated qualitative data by including open response questions.[60 64] However, there was insufficient detail on the questions used and any analysis to warrant these studies being classified as mixed methods. An exception was Boussat and colleagues'[65] mixed method study, which used open-response comments from a HSOPS survey of a French hospital to develop interview probes and included these answers in the analysis.

Themes for safety culture dimensions

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Details of all 88 surveys and 31 qualitative and mixed methods studies were examined to identify common themes in safety culture dimensions, as well as new or emerging ones. This process led to the identification of 11 themes and an "other" category. Table 4 displays definitions, example items, and how often themes were present in methods/tools reviewed. Information on the proportion of different methods covering each theme is charted in Figure 3. Appendix 4 shows the mapping of each method or tool to the themes of safety culture identified.

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Table 4. Safety culture themes

Theme	Definition	Example item	No. studies/surveys used in (%)	
1. Leadership	Leadership and their support and commitment to safety.	My supervisor often discusses safe work practices with me. (HSCS)	Total*	85 (77.3)
			Quantitative^	66 (78.6)
			Qualitative#	4 (33.3)
			Mixed methodsº	15 (78.9)
2. Perceptions of safety	Perceptions of how safe the organisation is.	Please give your unit an overall grade on patient safety. (MSI)	Total*	65 (59.1)
			Quantitative^	41 (48.8)
			Qualitative#	7 (58.3)
			Mixed methodsº	17 (89.4)
3. Teamwork and collaboration	Working together as a team and coordination of care among staff.	I enjoy working as part of a team. (ORMAQ)	Total*	61 (55.5)
			Quantitative^	41 (48.8)
			Qualitative#	4 (33.3)
			Mixed methodsº	16 (84.2)
4. Safety systems	Systems, procedures and processes exist that facilitate patient safety (e.g., rewards, reporting systems).	Things "fall between the cracks" when transferring patients from one unit to another. (HSOPS)	Total*	58 (52.7)
			Quantitative^	40 (47.6)
			Qualitative#	4 (33.3)
			Mixed methodsº	14 (73.7)
5. Prioritisation of safety	Shared belief, behaviours and norms in which staff in the work area prioritise and value safety.	Patient safety is never sacrificed to get more work done. (HSOPS)	Total*	59 (53.6)
			Quantitative^	42 (50.0)
			Qualitative#	2 (16.7)
			Mixed methodsº	15 (78.9)
6. Resources and constraints	Resources for safety including staffing, equipment, lack of time and training.	My unit provides training on teamwork in order to improve patient care performance and safety. (PSCHO)	Total*	58 (52.7)
			Quantitative^	43 (51.2)
			Qualitative#	1 (8.3)
			Mixed methodsº	14 (73.7)
7. Reporting and just culture	Willingness to report and a culture that does not assign blame.	Staff feel like their mistakes are held against them. (HSOPS)	Total*	54 (49.1)
			Quantitative^	37 (44.0)
			Qualitative#	2 (16.7)
			Mixed methodsº	15 (78.9)
8. Openness	Open communication, staff feeling comfortable to express their issues or concerns and question	In this clinical area, it is difficult to discuss errors. (SAQ)	Total*	54 (49.1)
			Quantitative^	35 (41.7)

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	behaviours.		Qualitative#	4 (33.3)
			Mixed methodsº	15 (78.9)
9. Learning and improvement	A focus on learning from mistakes, responding to, and improving systems.	When errors happen, we discuss how we could have prevented them. (SOS)	Total*	51 (46.4)
			Quantitative^	34 (40.5)
			Qualitative#	3 (25.0)
			Mixed methodsº	14 (73.7)
10. Awareness of human limits	Awareness of individual ability to be safe and how that can be limited by various factors (e.g., fatigue).	I am less effective at work when fatigued. (SAQ)	Total*	24 (21.8)
			Quantitative^	16 (19.0)
			Qualitative#	3 (25.0)
			Mixed methodsº	5 (26.3)
11. Wellbeing	Job satisfaction, burnout and other psychosocial factors.	Morale in this clinical area is high. (SAQ)	Total*	17 (15.5)
			Quantitative^	10 (11.9)
			Qualitative#	1 (8.3)
			Mixed methodsº	6 (31.6)
Other	New, emerging themes and those unable to be classified elsewhere. Includes: flexibility, monitoring, personal values, patient and family involvement.	It is important that my competence be acknowledged by others. (ORMAQ)	Total*	30 (27.3)
			Quantitative^	16 (19.0)
			Qualitative#	6 (50.0)
			Mixed methodsº	8 (42.1)

HSCS: Hospital Safety Culture Survey, HSOPS: Hospital Survey on Patient Safety Culture, MSI: Modified Stanford Instrument, ORMAQ: Operating Room Management Attitudes Questionnaire, PSCHO: Patient Safety Climate in Healthcare Organizations survey, SAQ: Safety Attitude Questionnaire, SOS: Safety Organizing Scale.

* % of total calculated on n = 110. Insufficient detail on five surveys and four surveys were counted in the mixed methods studies because they were single-use and developed by the authors.

^ % of total calculated on n = 84

% of total calculated on n = 12

º % of total calculated on n = 19

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[INSERT FIGURE 3 HERE]

"Leadership" was the most frequently assessed safety culture theme, present in three-quarters of methods and tools reviewed. "Awareness of human limits" and "Wellbeing" were the least frequent across the different methods. All themes were covered to a greater extent by mixed methods studies, compared with using qualitative methods or surveys alone. However, qualitative studies had a relatively higher inclusion of "other" dimensions.

Although dimensions classified to the "other" category were too infrequent to constitute their own theme, there were some recurrent concepts. Specifically, across qualitative methods and quantitative surveys, there were dimensions related to patient involvement in safety culture;[43 63 66-68] interpersonal dynamics around trusting colleagues, developing a shared understanding, and valuing and empowering individuals;[40 62] and the need for flexibility and improvisation.[28 69 70] However, extending upon the latter of these themes, qualitative and mixed methods studies also reported a range of complexities,[69] external pressures,[71] contextual dependencies,[59] and the need to manage and prioritise risks in delivering safe care to patients,[47] which were not adequately captured by the narrow scope of established themes, such as "resources and constraints".

DISCUSSION

This review sought to: (1) map the range of methods and specific tools used to assess safety culture in hospitals; (2) analyse the prevalence of these in published research; and (3) examine the dimensions of safety culture captured in these methods and tools. We found that the overwhelming majority of studies exclusively utilised surveys. Few used surveys in combination with other methods or solely qualitative methods. These studies had one of four primary purposes, with quantitative studies more likely to test relationships among variables,

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and qualitative and mixed methods studies typically more focused on describing and exploring safety culture or evaluating some intervention. We identified 11 safety culture themes and one residual category to cover the range of safety culture dimensions in methods and tools.

Safety culture surveys

The most widely-utilised surveys were the HSOPS,[32] and the SAQ,[33] a finding consistent with past reviews on this topic.[12] The SAQ also covered the most safety culture dimensions of any of the reviewed survey, but this does not unequivocally suggest its superiority. Its "Stress recognition" subscale (coded to "Awareness of human limits") has been found to poorly correlate with other dimensions and hence is suggested to not adequately fit into the safety culture construct.[72] Sexton and colleagues more recent survey, Safety, Communication, Operational Reliability, and Engagement (SCORE), which was used in a couple of studies in this review,[73 74] includes burnout and work-life balance but not stress recognition.[75] This highlights that there remains no gold standard tool for measuring safety culture, limiting the capacity for comparison across studies, and the synthesis and meta-analysis of safety culture research, which is a factor that makes surveys appealing in the first instance.

Qualitative and mixed methods approaches

The present review found a relative lack of qualitative and mixed methods. Authors of studies that used qualitative or mixed methods, however, cited their value in providing deeper insights into what affects safety culture,[63 76] and what the priorities should be in addressing it. For example, Sine and Northcutt[61] compared findings from patient safety culture dimension survey scores (HSOPS) and the prioritisation of those dimensions by representative focus groups, which enabled understanding of how dimensions of safety culture interact with each other and the larger system of patient safety. Boussat and colleagues'[65] mixed method study, using the HSOPS, open response comments and in-depth interviews, found the qualitative

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data complemented and provided more detail to the surveys. Practically-speaking, qualitative methods were suggested to redress a common limitation of safety culture surveys in hospitals: lower than desired response rates.[76] Hence, evidence suggests that qualitative methods may deepen or enrich the exploration of complex issues related to safety culture, identify priority dimensions and provide insight into areas for improvement.

Dimensions of safety culture

The 11 themes related to safety culture assessment mapped to older reviews, suggesting consistency over time in the conceptualisation of core aspects of safety culture.[9 12 19] The areas of overlap are displayed in Figure 4. Only a few dimensions of safety culture identified by Singla, et al. [77] could not be categorised elsewhere and so were included under our "other". Nevertheless, these categories were comparable to some of the "other" dimensions we identified including the adequacy of junior-senior doctor supervision,[78] and risk-taking behaviour.[79]

[INSERT FIGURE 4 HERE]

Similar to past reviews,[9 12 19] we found "Leadership" to be the most commonly applied dimension, while person-specific themes such as "Wellbeing" and "Awareness of human limits" were less widely represented, although they were present in the popular SAQ, as well as many of the tools and methods adapted from or influenced by it. Unlike some of the previous reviews, our framework marks a distinction between the "Perceptions of safety" (e.g., "I would feel safe being treated here as a patient" from the SAQ) and the normalised "Prioritisation of safety" in the attitudes and behaviour of staff (e.g., "Patient safety is never sacrificed to get more work done" from the HSOPS). This allowed us to differentiate the HSOPS as providing greater coverage of the prioritisation of safety than the SAQ, a

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dimensions which some theorists suggest is the core feature of an organisation's safety culture.[14]

We found that mixed methods studies, particularly those that used comprehensive surveys (e.g., SAQ, HSOPS) in conjunction with qualitative methods, covered the most safety culture themes. However, no single method or tool appeared to measure all identified dimensions of safety culture. Qualitative and mixed methods studies were more likely to assess "other" aspects of safety culture, though, which suggests their capacity to derive unexpected, richer or more contextually relevant findings. These other aspects of safety culture covered the involvement of patients in safety culture,[43 63 66-68] interpersonal dynamics[40 62] and the context of how safety is enacted in a pressurised health care environment.[47 59] An emerging focus on the patient and carer and their role in safety culture was also affirmed by studies that surveyed these groups about their perceptions.[46 80]

Implications for assessing safety culture for research and improvement

Exclusively using a survey was the most common approach to assess safety culture and a comparatively large proportion of these studies had a research focus (theoretical or methodological). Studies using qualitative methods, while few in our dataset, were more often exploratory or conducting a program evaluation, suggesting the value of these methods in understanding, and improving safety culture. Nevertheless, the volume of studies using surveys for all purposes demonstrates their suitability for quality improvement through to research.

The HSOPS and SAQ were the two most widely used surveys; both provide coverage of most of the safety culture dimension. However, both have recently undergone major revisions. The HSOPS version 2.0 was released in 2019 and involved deleting, rewording and adding multiple items, with the removal of a dimension for the overall perception of patient safety.[81] The SAQ has arguably been superseded by the SCORE, also developed by Sexton

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and colleagues, which removed one of the original dimensions and added a number of others with a greater focus on wellbeing.[75] Given the recency of these developments, it is unclear the extent to which these revised surveys will be taken up by hospitals and researchers, though some support has emerged for making this transition.[81]

In general, safety culture surveys are practical, able to be used by non-experts, and time-efficient for gathering large amounts of data from one or more participant groups in a reliable and reproducible manner, which is important for researchers.[11 50 82] The anonymity usually involved in this form of data collection could also make them appealing for quality improvement, as they facilitate the contributions of staff who may be uncomfortable expressing their views openly. Another aspect of their attraction, surveys support comparison and benchmarking, activities which, are often required by policymakers in health care. These approaches are not without critique including that such measurement may lead to *tunnel vision*, focusing on the "dimensions of performance that are included in the measurement system" (p. 570).[83]

Surveys have encountered similar critiques from organisational culture theorists; by reducing culture in any one organisation to a finite number of dimensions, reflected in items on a questionnaire, they may miss the "unique profile of cultural assumptions" (p. 614).[27] Our review highlights that qualitative and mixed methods approaches, on the other hand, were flexible to assessing other aspects of a hospital safety culture, and may uncover new, context specific or unexpected issues. Not only may such insights lead to future directions in research into safety culture, but also contribute to better understanding of culture on a micro-level and how to improve it. Furthermore, for researchers, qualitative data on safety culture can be used to augment surveys,[65] particularly when one has failed to achieve a desired response rate.[76] However, these methods are typically labour intensive so may not always be suited for time-

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poor quality improvement units within hospitals, in which case a more structured form of qualitative inquiry like the MaPSaF could be useful.[51-53]

Overall, findings point to the need for careful consideration and planning of the methodological approach one takes to an assessment of safety culture. Clearly using a survey is practical and feasible for capturing the perspectives of a large number of participants; however, the tool should be judiciously selected so that it covers, if not the most dimensions of safety culture, then those most important and appropriate for one's purposes, the setting and the participants. Ideally, assessment should include the use of qualitative methods in addition to surveys to enable not only greater exploration of specified dimensions, but the potential to identify other relevant aspects of safety culture.

Strengths and limitations

A strength of this study was searching four large academic databases. However, only searching academic literature leads to a high risk of publication bias. It is plausible that studies in grey literature or unpublished quality improvement projects are more inclined to contain qualitative methods than studies in academic journals where significance testing and large sample sizes are valued.

Another issue was the lack of information provided in the method of included studies to, for example, adequately identify safety culture dimensions, or even if a survey was new or an extension of an existing survey. Numerous studies of safety culture in hospitals provided inappropriate references (e.g., citing a secondary source that had used the SAQ, not calling the tool by its established name). Many surveys also have similar survey names and origins. Accurately capturing which surveys were used sometimes required a lengthy process of tracking through multiple sources to determine the tool being referred to. Finally, a workshop and multiple group meetings were conducted to ensure consistent coding of safety culture themes.

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CONCLUSION

In their comprehensive review a decade ago, Halligan and Zecevic [12] highlighted the need to move toward a common set of definitions and dimensions of safety culture in health care. Alas, our review has shown that there are a multitude of methods and tools available to assess safety culture in hospitals, and still no clear agreement on all of its dimensions. Certainly, some surveys and mixed method approaches appear to be more comprehensive in their coverage of safety culture themes. The review highlights the discrepancy in the large volume of studies that exclusively use surveys compared with the much smaller number employing qualitative or mixed methods. Nevertheless, it indicates the value in incorporating qualitative and mixed methods approaches to verify or extend the findings of quantitative surveys and understand some of the contextual aspects of safety culture.

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DATA SHARING STATEMENT

All data relevant to the study are included in the article or uploaded as supplementary information.

COMPETING INTERESTS

None declared

AUTHOR CONTRIBUTIONS

KC, LAE, AH, and MB had the original idea for the manuscript. KC and LAE led the search and data extraction. CP, AO, and JCL assisted with running the searches, screening papers and data extraction. KC, LAE and CP wrote portions of the manuscript. JB provided critical

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feedback on the manuscript and wrote the conclusion. All authors critically revised the manuscript.

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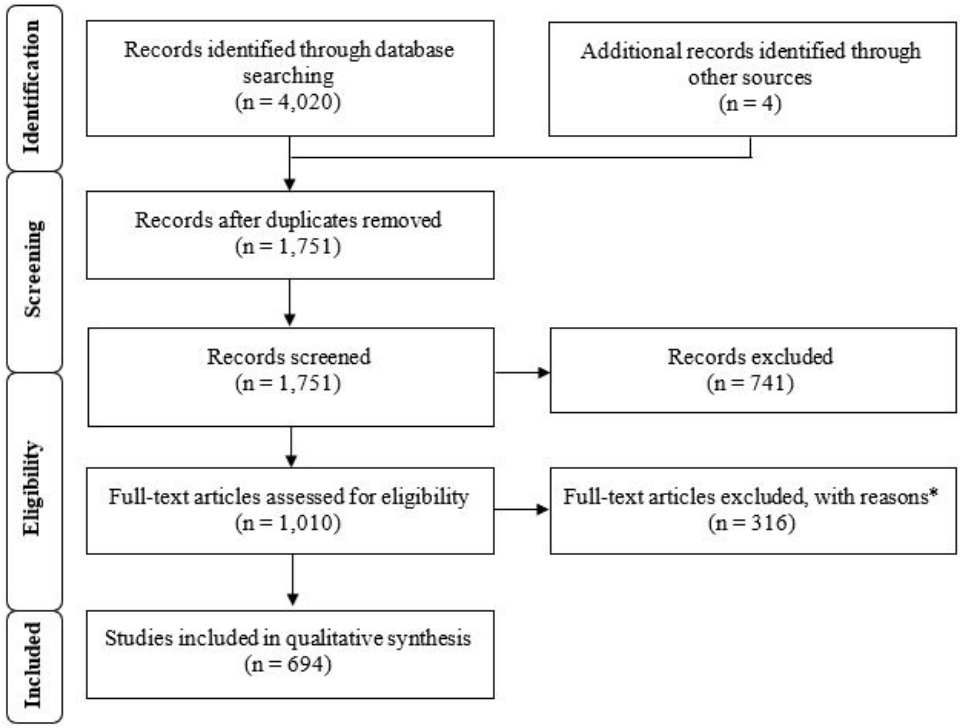
Figure 1: Flow chart of the review of literature to identify methods and tools.

Figure 2. Proportion of included studies for each primary study purpose broken down by method.

Figure 3. Proportion of methods that cover each safety culture dimension broken down by quantitative (n=84), qualitative (n=12) and mixed methods (n=19).

Figure 4. Overlap in dimensions of safety culture identified by this review with past reviews.

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	Reasons for exclusion	Number of studies^
1	Language other than English	87
2	Does not assess *safety culture*	81
3	Setting other than acute care/hospitals	40
4	Unable to access/find full text	66
5	Not an empirical journal article	53

^ The numbers of studies in this column exceeds the total number of exclude studies due to a proportion of studies being excluded for multiple reasons.

Figure 1: Flow chart of the review of literature to identify methods and tools.

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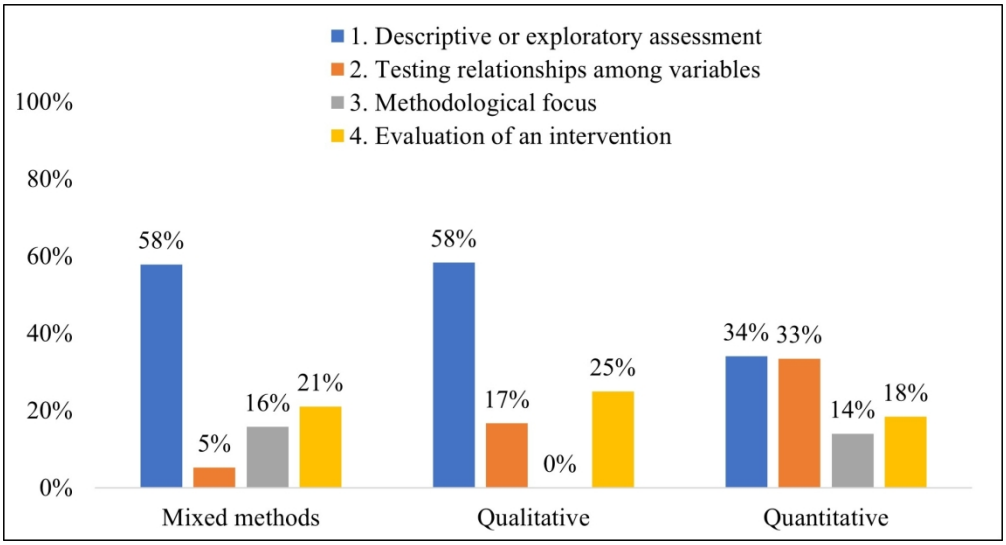


Figure 2. Proportion of included studies for each primary study purpose broken down by method.

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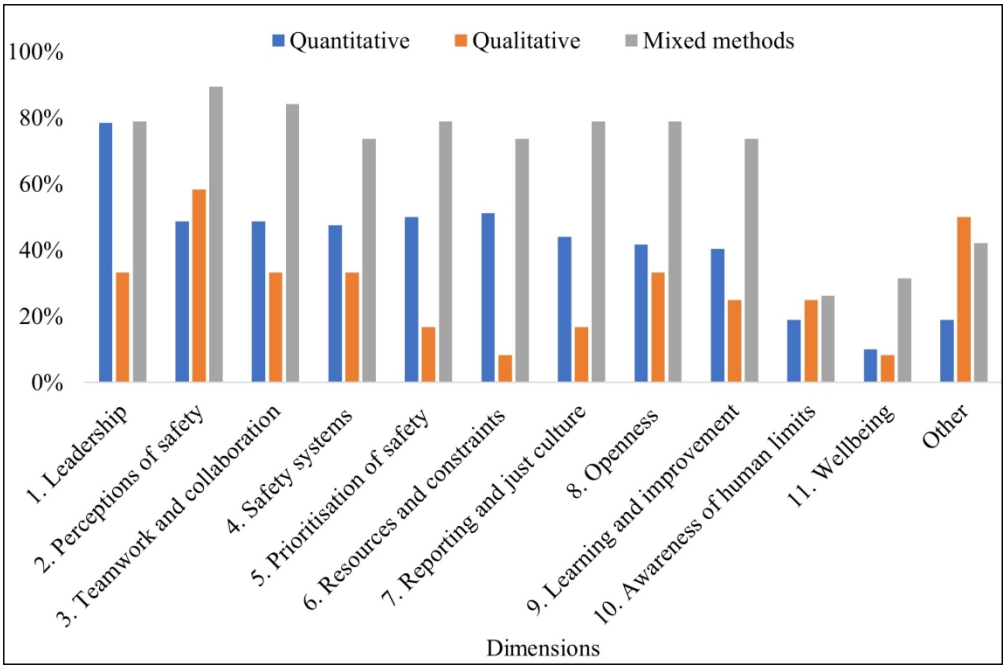


Figure 3. Proportion of methods that cover each safety culture dimension broken down by quantitative (n=84), qualitative (n=12) and mixed methods (n=19).

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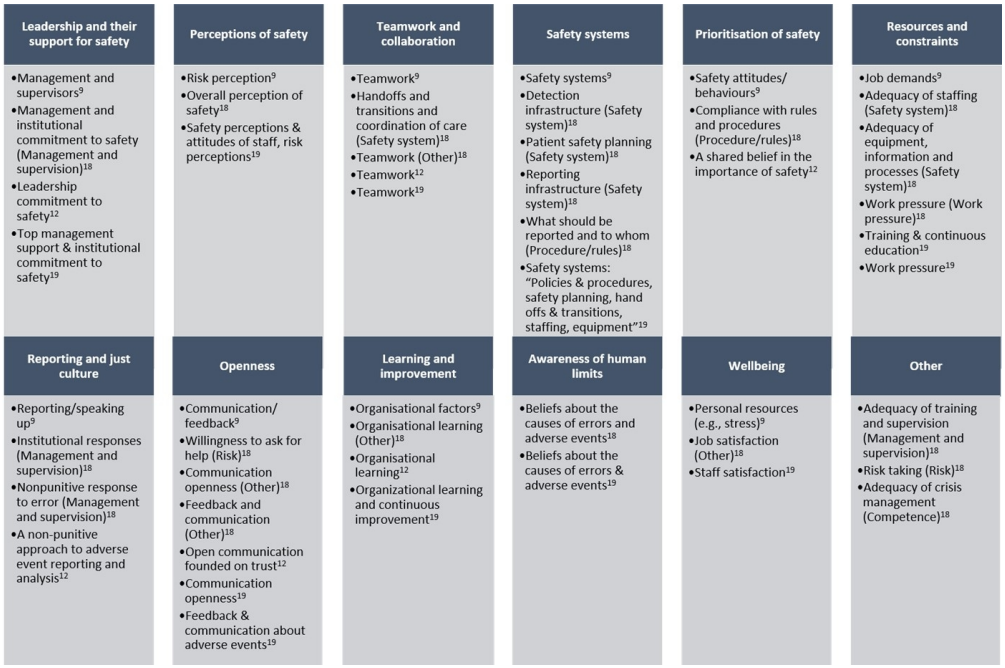


Figure 4. Overlap in dimensions of safety culture identified by this review with past reviews.

118x78mm (300 x 300 DPI)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	p.1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	p.1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	p.7-8
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p.7-8
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	No protocol exists.
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	p.8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	p.8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	p.8 and Appendix 1.
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p.8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p.8-9
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	p.8-9
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	No formal risk of bias was appropriate given the focus of the review was to analyse prevalence of methods and tools (not examine study findings).
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	p.8-9

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			Method or tool used to assess safety culture.
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	p.8-9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p.10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Appendix 3. The focus of the review is on methods/tools rather than study characteristics.
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	No formal risk of bias was appropriate given the focus of the review was to analyse prevalence of methods and tools.
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Results summarised p.10-16.
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	No additional analyses.
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	p.16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	p.20
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications	p.20

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		for future research.	
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding for a literature search, which guided the development of this review, was awarded to KC, LE, AH, and MB by the Australian Commission on Safety and Quality in Health Care. JB reports funding from the National Health and Medical Research Council: NHMRC Partnership Centre in Health System Sustainability (Grant ID 9100002) and NHMRC Investigator Grant APP1176620.

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Appendix 1 – Search strategy for Scopus

(TITLE-ABS-KEY("acute care" OR hospital) AND TITLE-ABS-KEY("quality culture" OR "safety culture" OR "culture of safety" OR "safety climate" OR "safety attitude") AND TITLE-ABS-KEY(survey OR questionnaire OR tool OR instrument OR measur* OR assess* OR checklist OR "check list" OR observ*) AND TITLE-ABS-KEY("patient safety" OR "public safety" OR "workplace safety")

2008- May 2020 = 1289 citations

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Citation	Study year	Study purpose	Coded purpose	Methods	Survey?	Survey name	Interviews?	Focus groups?	Organisational data and	Observation?	Framework?
Aaberg, O. R., R. Ballangrud, S. I. E. Husebø and M. L. Hall-Lord (2019). "An interprofessional team training	2019	The aim of this study was to explore if an interprofessional teamw	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Abbas HAE, Bassiuni NA, Baddar FM. Perception of front-line healthcare providers toward patient safety: a	2008	The current study targeted the perceptions of front-line healthcar	1. Descriptive or exploratory assess	Quantitative	Yes	SCSu					
Abdallah, W., C. Johnson, C. Nitzl and M. A. Mohammed (2019). "Organizational learning and patient safet	2019	The purpose of this paper is to explore the relationship between c	1. Descriptive or exploratory assess	Quantitative	Yes	urvey on Patient Safety Culture					
Abdi Z, Delgoshaei B, Ravaghi H, Abbasi M, Heyrani A. The culture of patient safety in an Iranian intensive	2015	To explore nurses' and physicians' attitudes and perceptions relev	1. Descriptive or exploratory assess	Mixed methods	Yes	SAQ	Yes				
Abdou HA, Sabe KM. A baseline assessment of patient safety culture among nurses at Student University H	2011	to assess patient safety culture among nurses at Student Universit	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Aboshaiqah AE, Baker OG. Assessment of nurses' perceptions of patient safety culture in a Saudi Arabia ho	2013	The purpose of this study was to identify factors that nurses perce	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Abou Hashish EA, El-Bialy GG. Nurses' Perceptions of Safety Climate and Barriers to Report Medication Err	2013	to assess nurses' perceptions of safety climate and barriers to rep	1. Descriptive or exploratory assess	Quantitative	Yes	ate scale (Blegen et al., 2005)					
Aboul-Fotouh AM, Ismail NA, Ez Elarab HS, Wassif GO. Assessment of patient safety culture among healthc	2012	assessed healthcare providers' perceptions of patient safety cultu	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Abrahamson K, Hass Z, Morgan K, Fulton B, Ramanujam R. The Relationship Between Nurse-Reported Safe	2016	The purpose of this study was to better understand the relationsh	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Abrahamson, K., F. Rong, K. Pressler, B. R. Fulton, K. H. Morgan and J. G. Anderson (2018). "Organizational	2018	This article examines the impact of organizational role on percept	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Abstoss KM, Shaw BE, Owens TA, Juno JL, Commiskey EL, Niedner MF. Increasing medication error reportir	2011	This study analyses patterns in reporting rates of medication error	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
AbuAlRub RF, Abu Alhijaa EH. The impact of educational interventions on enhancing perceptions of patien	2014	The purpose of this study was to examine the impact of patient sa	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
AbuAlRub RF, Gharaibeh HF, Bashayreh AEI. The Relationships Between Safety Climate, Teamwork, and Int	2012	to examine the relationships among safety climate, teamwork, an	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Abu-El-Noor NI, Hamdan MA, Abu-El-Noor MK, Radwan A-KS, Alshaer AA. Safety Culture in Neonatal Inten	2017	to assess the safety culture in the neonatal intensive care units (N	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Abu-El-Noor, N. I., M. K. Abu-El-Noor, Y. Z. Abuowda, M. Alfaqawi and B. Böttcher (2019). "Patient safety c	2019	This study aimed to assess attitudes of nurses working in governm	1. Descriptive or exploratory assess	Quantitative	Yes	estionnaire III (APSQ III) (Carruthers et al., 2009)					
Abuosi, A. A., A. Akologo and E. A. Anaba (2020). "Determinants of patient safety culture among healthcar	2020	To find out the determinants of patient safety culture among heal	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Adibi H, Khalesi N, Ravaghi H, Jafari M, Jeddian AR. Development of an effective risk management system i	2012	The aim of this study was to develop and implement a risk manag	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Agnew C, Flin R, Mearns K. Patient safety climate and worker safety behaviours in acute hospitals in Scotla	2013	to obtain a measure of hospital safety climate from a sample of N	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Ahmed M, Arora S, Baker P, Hayden J, Vincent C, Sevdalis N. Building capacity and capability for patient saf	2013	To develop, implement and evaluate a training programme for se	4. Evaluation of an intervention	Quantitative	Yes	nts' Patient Safety Questionnaire					
Ahmed, S. M., N. Fekry and F. A. Abed (2018). "Assessment of safety climate among nurses at selected hos	2018	This study aimed to assess the safety climate among staff nurses a	1. Descriptive or exploratory assess	Quantitative	Yes	ate Survey (Ahmed et al. 2018)					
Al Faqeeh, F., K. Khalid and A. Osman (2019). "Integrating safety attitudes and safety stressors into safety c	2019	How safety climates, safety attitudes, and safety behaviors are rel	2. Testing relationships among varia	Quantitative	Yes	l Faqeeh et al. 2019					
Al Ma'mari, Q., L. A. Sharour and O. A. Omari (2019). "Predictors of perceptions of patient safety culture ar	2019	Predictors of perceptions of patient safety culture and frequency	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Al Nadabi, W., M. Faisal and M. A. Mohammed (2019). "Patient safety culture in Oman: A national study."	2019	This study aims to measure patient safety culture across maternit	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Al Salem, G., P. Bowie and J. Morrison (2019). "Hospital Survey on Patient Safety Culture: psychometric ev	2019	This study aims to investigate the psychometric properties of the	3. Methodological focus	Quantitative	Yes	HSOPS					
Alahmadi HA. Assessment of patient safety culture in Saudi Arabian hospitals. Quality & safety in health ca	2010	The purpose of this study to evaluate the extent to which the cult	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Al-Ahmadi TA. Measuring Patient Safety Culture in Riyadh's Hospitals: A Comparison between Public and P	2009	The aim of this research was to explore the perceptions of Riyadh	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Al-Awa B, Al Mazrooa A, Rayes O, et al. Benchmarking the post-accreditation patient safety culture at King	2012	The objective of this study was to perform an unbiased assessmer	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Alayed AS, Loof H, Johansson UB. Saudi Arabian ICU safety culture and nurses' attitudes. International jour	2014	The purpose of this paper is to examine nurses' attitudes towards	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Al-Babtain B, Aljadhey H, Mahmoud MA, Alaqeel S, Ahmed Y. Culture of safety among nurses in a tertiary t	2016	To assess the culture of safety among nurses in a tertiary teaching	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Alemdar, D. K. and G. Yilmaz (2020). "Evaluation of the Relationship Between the Patient Safety Culture an	2020	This study was conducted descriptively so as to determine the rela	2. Testing relationships among varia	Quantitative	Yes	re Scale (PSCS) by Turkmen et al. 2011					
Alenezi, A., R. P. M. Pandaan, J. U. Almazan, I. N. Pandaan, F. S. Casison and J. P. Cruz (2019). "Clinical prac	2019	To assess the perceptions of clinical practitioners regarding the dir	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alex Kim, R. J., Z. H. Chin, P. Sharlyn, B. Priscilla and S. Josephine (2019). "Hospital survey on patient safety	2019	The purpose of this study is to assess the views and perceptions o	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alharbi, W., J. Cleland and Z. Morrison (2018). "Assessment of Patient Safety Culture in an Adult Oncology	2018	We sought to evaluate patient safety culture across diferent healt	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Ali, H., S. Z. Ibrahem, B. Al Mudaf, T. Al Fadalah, D. Jamal and F. El-Jardali (2018). "Baseline assessment of p	2018	The article analysis patient safety culture in public hospitals in Kuv	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Aljabri, D., A. Vaughn, M. Austin, L. White, Z. Li, J. Naessens and A. Spaulding (2020). "An Investigation of H	2020	The purpose of this study was to evaluate the number of work inju	2. Testing relationships among varia	Quantitative	Yes	Mercer 2019					
Allard J, Bleakley A, Hobbs A, Coombes L. Pre-surgery briefings and safety climate in the operating theatre.	2011	To investigate whether exposure to pre-surgery briefings is relate	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Allen S, Chiarella M, Homer CS. Lessons learned from measuring safety culture: an Australian case study. N	2010	this paper reports a case study examining the safety culture in one	1. Descriptive or exploratory assess	Mixed methods	Yes	Multiple - SAQ, SCS	Yes		Yes		
Al-Mandhari A, Al-Zakwani I, Al-Kindi M, Tawilah J, Dorvlo AS, Al-Adawi S. Patient safety culture assessmen	2014	To illustrate the patient safety culture in Oman as gleaned via 12 i	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Al-Mugheed, K. and N. Bayraktar (2020). "Patient safety attitudes among critical care nurses: A case study	2020	this study aimed to examine patient safety attitudes among critica	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Almutairi AF, Gardner G, McCarthy A. Perceptions of clinical safety climate of the multicultural nursing wo	2013	The purpose of this study is to explore the safety climate percepti	1. Descriptive or exploratory assess	Quantitative	Yes	SCSu					
Al-Nawafleh A, Abu-Helalah MA, Hill V, Masoud MI, Al-Mahasneh HA, Salti ETAI. Patient Safety Culture in J	2016	he purpose of this study is to validate an Arabic version of the Hos	3. Methodological focus	Quantitative	Yes	HSOPS					
Alonazi NA, Alonazi AA, Saeed E, Mohamed S. The perception of safety culture among nurses in a tertiary h	2016	the aim of this study is to explore perceptions of patient safety am	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alqattan, H., J. Cleland and Z. Morrison (2018). "An evaluation of patient safety culture in a secondary care	2018	This study carried out an initial evaluation of the patient safety cu	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alquwez, N., J. P. Cruz, A. M. Almoghairi, R. S. Al-otaibi, K. O. Almutairi, J. G. Alicante and P. C. Colet (2018)	2018	To assess the present patient safety culture of three general hosp	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alrowely, Z. and O. Ghazi Baker (2019). "Assessing Building Blocks for Patient Safety Culture-a Quantitative	2019	The study analyzes staffs' perception of a safety culture and their	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alshammari, F., E. Pasay-an, M. Alboliteeh, M. H. Alshammari, T. Susanto, S. Villareal, M. C. L. Indonto and	2019	The present study investigates the perceptions of healthcare prof	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alshammari, M. H. (2019). "Patient safety culture of nurses in public and private hospitals in northwestern	2019	This study determined the patient safety cultures of nurses in pub	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Alves DF, Guirardello EB. Safety climate, emotional exhaustion and job satisfaction among Brazilian paedia	2016	To assess the correlation and predictive effect of emotional exhau	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Alves Gomides, M. D., A. M. de Souza Fontes, A. O. Soares Monteiro Silveira and G. Sadoyama (2019). "Pat	2019	The objective of this study was to evaluate the perception regardi	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Alves, D. F. B., E. Lorenzini and A. C. B. Kolankiewicz (2020). "Patient safety climate in a Brazilian general ho	2020	To measure patient safety climate and the associated f	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Alzahrani, N., R. Jones and M. E. Abdel-Latif (2018). "Attitudes of doctors and nurses toward patient safety	2018	This study aims to investigate doctors' and nurses' attitudes towa	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Alzahrani, N., R. Jones and M. E. Abdel-Latif (2019). "Safety Attitudes among Doctors and Nurses in an Eme	2019	To investigate doctors' and nurses' attitudes towards patient safe	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Ameryoun, A., A. H. Pakpour, M. Nikoobakht, M. Saffari, M. Yaseri, K. G. N. O'Garo and H. G. Koenig (2019)	2019	This study was conducted to determine whether a 1-day educatio	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Amiri, M., Z. Khademian and R. Nikandish (2018). "The effect of nurse empowerment educational program	2018	The present study aimed to determine the effect of empowering n	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Ammouri AA, Tailakh AK, Muliira JK, Geethakrishnan R, Al Kindi SN. Patient safety culture among nurses. In	2015	The purpose of this study was to investigate nurses' perceptions a	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Amponsah-Tawaih K, Adu MA. Work Pressure and Safety Behaviors among Health Workers in Ghana: The (	2016	the aim of the study was to examine how pressure influence safet	2. Testing relationships among varia	Quantitative	Yes	ey (Griffin & Neal, 2000; Neal & Griffin, 2006)					
Anbari, Z., R. Jadidi, J. Nazari, S. Amini, S. Khosravi, M. S. Koushki and T. S. J. Arani (2019). "Workforce Perc	2019	This study aimed to assesses patient safety culture in Iran teachin	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					

Ansari, S. P., M. E. Rayfield, V. A. Wallis, J. E. Jardine, E. P. Morris and E. Prosser-Snelling (2019). "A Safety P	2019	The aim of the study was to evaluate the impact of maternity-orie	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Antino, M., M. Sanchez-Manzanares and A. Ortega (2020). "The hospital survey on patient safety culture in	2020	This study investigates the psychometric properties of the Hospita	3. Methodological focus	Quantitative	Yes	HSOPS					
Armellino D, Quinn Griffin MT, Fitzpatrick JJ. Structural empowerment and patient safety culture among re	2010	The aim of the present study was to examine the relationship betv	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Armstrong K, Laschinger H, Wong C. Workplace Empowerment and Magnet Hospital Characteristics as Pre	2009	This study tested a theoretical model derived from Kanter's theor	2. Testing relationships among varia	Quantitative	Yes	SCM					
Arnetz, J., L. E. Hamblin, S. Sudan and B. Arnetz (2018). "Organizational Determinants of Workplace Violen	2018	To identify organizational factors contributing to workplace violer	2. Testing relationships among varia	Quantitative	Yes	ntion Climate (Kessler et al., 2008)					
Arrieta, A., G. SuÁRez, G. Hakim and G. Suárez (2018). "Assessment of patient safety culture in private and	2018	To assess the patient safety culture in Peruvian hospitals from the	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Auer C, Schwendimann R, Koch R, De Geest S, Ausserhofer D. How hospital leaders contribute to patient sa	2014	The aim of this study was to explore the associations between hos	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Ausserhofer D, Schubert M, Desmedt M, Blegen MA, De Geest S, Schwendimann R. The association of patie	2013	The purpose of this study was to explore the relationship betweer	2. Testing relationships among varia	Quantitative	Yes	SOS					
Ausserhofer D, Schubert M, Engberg S, Blegen M, De G, Schwendimann R. Nurse-reported patient safety c	2012	We aimed (1) to describe the nurse-reported engagement in safet	1. Descriptive or exploratory assess	Quantitative	Yes	SOS					
Avramchuk, A. S. and S. J. J. McGuire (2018). "Patient Safety Climate: A Study of Southern California Health	2018	We developed and tested a short survey instrument intended as a	3. Methodological focus	Quantitative	Yes	nt Safety Culture Survey (Cal-PSCS)					
Azimi L, Bahadori M. The effect of safety culture education on improvement of managers' attitudes towar	2012	to measure attitudes of managers towards patients' safety before	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Babic, B., A. A. Volpe, S. Merola, E. Mauer, Y. Cozacov, C. Y. Ko, F. Michelassi and P. Saldinger (2018). "Sust	2018	Patient care perception was assessed using Safety Attitudes Ques	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Bagnasco A, Tibaldi L, Chirone P, et al. Patient safety culture: an Italian experience. Journal of clinical nursi	2011	To understand what level of awareness health professionals work	3. Methodological focus	Quantitative	Yes	HSOPS					
Bahar, S. and E. Önler (2020). "Turkish surgical nurses' attitudes related to patient safety: A questionnaire	2020	This research aimed to evaluate the patient safety attitudes of Tur	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Bahrami MA, Chalak M, Montazeralfaraj R, Dehghani Tafti A. Iranian nurses' perception of patient safety c	2014	This study was aimed to survey patient safety culture in 2 Iranian	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Ballangrud R, Hedelin B, Hall-Lord ML. Nurses' perceptions of patient safety climate in intensive care units	2012	To investigate registered nurses' perceptions of the patient safety	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Barbosa MH, Floriano DR, Oliveira KFd, Nascimento KGd, Ferreira LA. PATIENT SAFETY CLIMATE AT A PRIVA	2016	Evaluate the patient safety climate from the perspective of the he	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Barbosa MH, Mendes Sousa E, Marques dos Santos Felix M, Faria Oliveira K, Barichello E. Patient safety cli	2015	The study's objective was to assess the safety climate from the pe	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Bartoníčková, D., D. Kalánková, Z. Mikšová, K. Žiaková and L. Mazalová (2019). "Patient safety culture from	2019	to demonstrate a broader context in the assessment of patient sa	3. Methodological focus	Quantitative	Yes	HSOPS					
Bartoníčková, D., D. Kalánková, Z. Mikšová, R. Kurucová, Š. Tomová and K. Žiaková (2019). "THE PSYCHOM	2019	To analyze the psychometric properties of the Hospital Survey on	3. Methodological focus	Quantitative	Yes	HSOPS					
Basson, T., A. Montoya, J. Neily, L. Harmon and B. V. Watts (2018). "Improving Patient Safety Culture: A Re	2018	The aim of the study was to improve the safety culture at a Veter	4. Evaluation of an intervention	Quantitative	Yes	ultiple - TSCS, HSOPS					
Basuni EM, Bayoumi MM. Improvement critical care patient safety: using nursing staff development strate	2015	aimed at using staff development strategies through implementin	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Beck, M. K., M. M. Loro, V. A. Casali Bandeira, C. R. Schmidt, S. D. Pai, L. R. Bandeira and A. C. Bernat Kolan	2018	This study aimed to measure the safety culture of the patient from	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Benn J, Burnett S, Parand A, Pinto A, Vincent C. Factors predicting change in hospital safety climate and ca	2012	(1) To analyse change in a survey measure of organisational patier	4. Evaluation of an intervention	Quantitative	Yes	limate and Capability (SCC)					
Bennett PN, Ockerby C, Stinson J, Willcocks K, Chalmers C. Measuring hospital falls prevention safety clima	2014	The aim of this study was to measure the falls prevention safety c	1. Descriptive or exploratory assess	Quantitative	Yes	Safety Climate Survey (SCS)					
Benning A, Ghaleb M, Suokas A, et al. Large scale organisational intervention to improve patient safety in f	2011	To conduct an independent evaluation of the first phase of the He	4. Evaluation of an intervention	Quantitative	Yes	NHS Staff Survey					
Benzer JK, Meterko M, Singer SJ. The patient safety climate in healthcare organizations (PSCHO) survey: Sh	2017	to develop a 15- to 20-item safety climate survey based on the Pa	3. Methodological focus	Quantitative	Yes	PSCHO					
Berman, L., S. Rangel, A. Goldin, D. Skarda, M. Ottosen, D. Bliss, P. Valusek, M. Fallat, K. Tsao, A. Q. For the	2018	The purpose of this study was to characterize the attitudes and p	1. Descriptive or exploratory assess	Quantitative	Yes	Berman et al. 2018					
Berry JC, Davis JT, Bartman T, et al. Improved Safety Culture and Teamwork Climate Are Associated With D	2016	to document such an association across an entire hospital system	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Bethune R, Canter R, Abrams P. What do surgical trainees think about patient safety culture, and is this dif	2012	This study surveyed both consultant and trainee views on perceiv	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Bilal, H. and H. Y. Sari (2020). "Relationship between burnout and patient safety attitudes in pediatric nurs	2020	to determine the relationship between the patient safety attitude	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Birk K, Padjen L, Markič M. Adverse event reporting in Slovenia – The influence of safety culture, supervisc	2016	The aim of this survey was to determine whether safety culture, s	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Birkmeyer NJ, Finks JF, Greenberg CK, et al. Safety culture and complications after bariatric surgery. Ann Su	2013	To assess relationships between safety culture and complications	2. Testing relationships among varia	Quantitative	Yes	irkmeyer et al. 2013					
Bishop AC, Boyle TA. The Role of Safety Culture in Influencing Provider Perceptions of Patient Safety. Journ	2016	To determine how provider perceptions of safety culture influenc	2. Testing relationships among varia	Quantitative	Yes	irkmeyer et al. 2013					
Black AA, Brauer SG, Bell RA, Economidis AJ, Haines TP. Insights into the climate of safety towards the prev	2011	To explore the climate of safety towards falls prevention among fa	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO					
Blegen, M. A., Gearhart, S., O'Brien, R., Sehgal, N. L., & Alldredge, B. K. (2009). AHRQ's hospital survey on p	2009	This project analyzed the psychometric properties of the Agency f	3. Methodological focus	Quantitative	Yes	HSOPS					
Blegen MA, Sehgal NL, Alldredge BK, Gearhart S, Auerbach AA, Wachter RM. Republished paper: Improving	2010	The goal of this project was to improve unit-based safety culture t	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Boan DM, Nadzam D, Clapp JR, Jr. The Impact of Variance in Perception of the Organization on Capacity to	2012	analyzed the degree of agreement in perceptions of these four gr	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Bodur S, Filiz E. Validity and reliability of Turkish version of "Hospital Survey on Patient Safety Culture" and	2010	The aim of this study is threefold: to determine the validity and re	3. Methodological focus	Quantitative	Yes	HSOPS					
Bognár A, Barach P, Johnson JK, et al. Errors and the Burden of Errors: Attitudes, Perceptions, and the Cult	2008	We assessed the perceptions and attitudes of surgical teams relat	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Borji M, Molavi S, Salimi E, Bastami Y. Studying Patient Safety Culture from the Viewpoint of Nurse in educ	2016	this article aims at studying the condition of patient safety culture	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Boughaba, A., S. Aberkane, Y.-O. Fourar and M. Djebabra (2019). "Study of safety culture in healthcare inst	2019	to measure safety culture dimensions in order to improve and pro	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Boussat, B., K. Kamalanavin and P. François (2018). "The contribution of open comments to understanding	2018	to analyze the qualitative data of the comments section of a Hosp	3. Methodological focus	Mixed methods	Yes	HSOPS	Yes				
Boyer KK, Gardner JW, Schweikhart S. Process quality improvement: An examination of general vs. outcom	2012	We empirically test a conceptual framework of safety culture that	2. Testing relationships among varia	Quantitative	Yes	Boyer et al. 2012					
Brabcova I, Bartlova S, Tothova V, Prokesova R. Selected aspects of "safety culture" in hospitals of the Czec	2015	GOAL OF THE SURVEY was to assess the safety culture with regarc	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Braddock CH, 3rd, Szaflarski N, Forsey L, Abel L, Hernandez-Boussard T, Morton J. The TRANSFORM Patien	2015	We aimed to evaluate the effects of a multifaceted, microsystem-	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Brandis, S., S. Schleimer and J. Rice (2019). ""Pixie dust"." Journal of Health Organization & Management 3	2019	to explore the concept of reflexivity and develop a model for mag	2. Testing relationships among varia	Quantitative	Yes	SOS					
Brasaitė I, Kaunonen M, Martinkėnas A, Mockienė V, Suominen T. Health care professionals' skills regardin	2016	to describe health care professionals' skills regarding patient safet	1. Descriptive or exploratory assess	Quantitative	Yes	lls and Knowledge (PS-ASK, Schnall et al., 2008)					
Brasaite I, Kaunonen M, Martinkenas A, Suominen T. Health care professionals' attitudes regarding patient	2016	to describe health care professionals' attitudes regarding patient s	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Braun BI, Harris AD, Richards CL, et al. Does health care role and experience influence perception of safety	2013	examined the association between professional role and health ca	2. Testing relationships among varia	Quantitative	Yes	and Infection Prevention Assessment					
Brborovic H, Brborovic O, Brumen V, Pavlekovic G, Mustajbegovic J. Are nurse presenteeism and patient sa	2014	The aim of this cross-sectional study was to investigate whether n	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Brborovic H, Sklebar I, Brborovic O, Brumen V, Mustajbegovic J. Development of a Croatian version of the (	2014	Our aim was to determine if all 12 dimensions of the US HSOPSC v	3. Methodological focus	Quantitative	Yes	HSOPS					
Brborović, O., H. Brborović, I. A. Nola and M. Milošević (2019). "Culture of blame—an ongoing burden for c	2019	Our aim was to examine the prevalence and the impact of culture	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Bredesen IM, Bjoro K, Gunningberg L, Hofoss D. Patient and organisational variables associated with press	2015	To investigate the association of ward-level differences in the odd	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Bressers, G., I. Wallenburg, R. Stalmeijer, M. oude Egbrink and K. Lombarts (2020). "Patient safety in medic	2020	This article draws on Habermas' distinction between 'lifeworld' ar	1. Descriptive or exploratory assess	Qualitative			Yes		Yes	Yes	Yes
Brilli RJ, McClead RE, Jr., Crandall WV, et al. A comprehensive patient safety program can significantly redu	2013	To evaluate the effectiveness of a hospital-wide initiative to impro	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Brown DS, Wolosin R. Safety culture relationships with hospital nursing sensitive metrics. Journal for healt	2013	explored linkages between staff perceptions of safety culture (SC)	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Budin WC, Gennaro S, O'Connor C, Contratti F. Sustainability of Improvements in Perinatal Teamwork and	2014	to describe changes in perinatal nurse (n = 70) and physician (n = 4	4. Evaluation of an intervention	Quantitative	Yes	nd Safety Climate Survey (TSCS)					
Bump GM, Calabria J, Gosman G, et al. Evaluating the Clinical Learning Environment: Resident and Fellow f	2015	We sought to address the dearth of metrics that assess trainee pa	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					

Bump GM, Coots N, Liberi CA, et al. Comparing Trainee and Staff Perceptions of Patient Safety Culture. Aca	2017	the authors set out to identify complementary measures of traine	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Burstrom L, Letterstal A, Engstrom ML, Berglund A, Enlund M. The patient safety culture as perceived by st	2014	The aim of the present study was to describe the patient safety cu	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Cabral RA, Eggenberger T, Keller K, Gallison BS, Newman D. Use of a Surgical Safety Checklist to Improve T	2016	To improve surgical team communication, a team at Broward Hea	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Callahan A, Ames NJ, Manning ML, Touchton-Leonard K, Yang L, Wallen R. Factors Influencing Nurses' Use	2016	To identify factors associated with oncology nurses' use of hazard	2. Testing relationships among varia	Quantitative	Yes	ng Questionnaire (Polovich & Clark, 2012)						
Calvache, J. A., E. Benavides, S. Echeverry, F. Agredo, R. J. Stolker and M. Klimek (2020). "Psychometric Pro	2020	developed a Latin American Spanish version of the HSPSC for use	3. Methodological focus	Quantitative	Yes	HSOPS						
Camargo CA, Jr., Tsai CL, Sullivan AF, et al. Safety climate and medical errors in 62 US emergency departme	2012	We describe the incidence and types of medical errors in emerger	3. Methodological focus	Quantitative	Yes	ency Department Safety Study Survey						
Campbell EG, Singer S, Kitch BT, Iezzoni LI, Meyer GS. Patient safety climate in hospitals: act locally on varia	2010	to assess the extent and nature of variation in safety climate acro	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Campbell, D. and K. Dontje (2019). "Implementing Bedside Handoff in the Emergency Department: A Pract	2019	Implementing Bedside Handoff in the Emergency Department: A P	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Campione, J. and T. Famolaro (2018). "Promising Practices for Improving Hospital Patient Safety Culture." J	2018	to gain knowledge about promising best practices used by hospita	1. Descriptive or exploratory assess	Mixed methods	Yes	HSOPS	Yes					
Carney BT, Mills PD, Bagian JP, Weeks WB. Sex differences in operating room care giver perceptions of pat	2010	Sex differences in operating room care giver perceptions of patier	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Carrothers KM, Barr J, Spurlock B, Ridgely MS, Damberg CL, Ely EW. Contextual issues influencing impleme	2013	to identify which contextual factors facilitate/hinder the impleme	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Carvalho PA, Donato Goettems LB, Gomes Maia Pires MR, Cunha de Oliveira ML. Safety culture in the oper	2015	to evaluate the perception of healthcare professionals about the s	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Castel ES, Ginsburg LR, Zaheer S, Tamim H. Understanding nurses' and physicians' fear of repercussions for	2015	examines the influence of clinician demographics (age, gender, an	2. Testing relationships among varia	Quantitative	Yes	MSI						
Castle N, Wagner LM, Perera S, Ferguson JC, Handler SM. Comparing the Safety Culture of Nursing Homes	2011	to compare the patient safety culture between nationally represe	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Chaboyer W, Chamberlain D, Hewson-Conroy K, et al. CNE article: safety culture in Australian intensive car	2013	To establish a baseline understanding of the safety culture in Aust	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Chagolla, B., D. Bingham, B. Wilson and B. Scheich (2018). "Perceptions of Safety Improvement Among Clir	2018	To measure whether implementation of a comprehensive, 18-mo	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Chakravarty A, Sahu A, Biswas M, Chatterjee K, Rath S. A study of assessment of patient safety climate in t	2015	A study of assessment of patient safety climate in tertiary care ho	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Chang, O. H., B. Levy, H. Lytle, R. Pope, H. Phiri, T. Gellhaus, C. Eckhardt and J. Sclafani (2019). "Implement	2019	To evaluate maternal mortality and changes in the culture of safet	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Chapple CL, Bradley A, Murray M, et al. Radiation safety culture in the UK medical sector: A top to bottom	2017	The medical sector provides unique challenges in this regard, and	1. Descriptive or exploratory assess	Mixed methods	Yes				Yes			
Chegini, Z., A. Janati, M. Afkhami, M. Behjat and S. M. Shariful Islam (2020). "A comparative study on patie	2020	To appraise patient safety culture in emergency departments of Ir	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Chen IC, Li HH. Measuring patient safety culture in Taiwan using the Hospital Survey on Patient Safety Cult	2010	to assess the culture of patient safety in Taiwan and attempt to pr	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Chen IC, Ng HF, Li HH. A multilevel model of patient safety culture: cross-level relationship between organi	2012	The main objective of this study was to investigate the cross-level	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Cheng, C. Y., Y. C. Lee, C. H. Huang and H. H. Wu (2019). "Assessing the medical staff's overall satisfaction c	2020	This study adds an additional question to the existing Chinese vers	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Cheng, H., H. Yang, Y. Ding and B. Wang (2020). "Nurses' mental health and patient safety: An extension of	2020	We employed the job demands–resources model to examine the	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Cheng, H.-C., A. M.-F. Yen and Y.-H. Lee (2019). "Factors affecting patient safety culture among dental hea	2019	our objectives were to measure the level of patient safety culture	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Chi, C. Y., C. H. Huang, Y. C. Lee, C. F. Wu and H. H. Wu (2020). "Understanding critical demographic variab	2020	to identify critical demographic variables that would significant in	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Chi, C.-Y., C.-H. Huang, Y.-C. Lee and H.-H. Wu (2019). "Critical Demographic Variables on Affecting Patient	2019	This study intends to identify critical demographic variables that w	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Cho, S. M. and J. Choi (2018). "Patient Safety Culture Associated With Patient Safety Competencies Among	2018	To investigate the relationships between registered nurses' (RNs')	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Chu-Weininger MYL, Wueste L, Lucke JF, Weavind L, Mazabob J, Thomas EJ. The impact of a tele-ICU on pr	2010	To measure teamwork and safety climate in three intensive care u	4. Evaluation of an intervention	Quantitative	Yes	nd Safety Climate Survey (TSCS)						
Clark OL, Zickar MJ, Jex SM. Role Definition as a Moderator of the Relationship Between Safety Climate an	2014	This field study investigated the moderating influence of role defi	2. Testing relationships among varia	Quantitative	Yes	HSCS						
Clay-Williams, R., N. J. Taylor, H. P. Ting, G. Arnolda, T. Winata and J. Braithwaite (2020). "Do quality mana	2020	This study aimed to explore the associations between the organiz	2. Testing relationships among varia	Quantitative	Yes	ulture and Leadership Questionnaire						
Clay-Williams, R., N. J. Taylor, H. P. Ting, T. Winata, G. Arnolda and J. Braithwaite (2020). "The clinician safe	2020	This study aimed to refine and validate a tool to measure safety cl	3. Methodological focus	Quantitative	Yes	ulture and Leadership Questionnaire						
Clay-Williams, R., N. Taylor, H. P. Ting, T. Winata, G. Arnolda, E. Austin and J. Braithwaite (2020). "The relat	2020	We aimed to examine whether Emergency Department (ED) quali	2. Testing relationships among varia	Quantitative	Yes	ulture and Leadership Questionnaire						
Colet, P. C., J. P. Cruz, G. Cacho, H. Al-Qubeilat, S. S. Soriano and C. P. Cruz (2018). "Perceived Infection Pre	2018	This study investigated the staff nurses' perception of the infectio	2. Testing relationships among varia	Quantitative	Yes	on Prevention (LCQ-IP) (Pogorzelska-Maziarz et al., 2015)						
Collier SL, Fitzpatrick JJ, Siedlecki SL, Dolansky MA. Employee Engagement and a Culture of Safety in the In	2016	to explore the relationship between employee engagement and c	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Cooper JB, Blum RH, Carroll JS, et al. Differences in safety climate among hospital anesthesia departments	2008	The introduction of an insurance premium incentive for simulation	4. Evaluation of an intervention	Quantitative	Yes	ans Administration Palo Alto Patient Safety Center of Inquiry						
Corso LD. Mediation effects of safety climate and safety motivation on the relation between organizationa	2008	The aim of this study was to test whether the effects of organizati	2. Testing relationships among varia	Quantitative	Yes	ey (Griffin & Neal, 2000; Neal & Griffin, 2006)						
Cox ED, Carayon P, Hansen KW, et al. Parent perceptions of children's hospital safety climate. BMJ Qual Sa	2013	to develop and evaluate a parent-reported version of the Agency	3. Methodological focus	Quantitative	Yes	HSOPS						
Cruz, E. D. A., D. J. M. Da Rocha, A. B. Mauricio, F. S. Ulbrich, J. Batista and E. C. S. Maziero (2018). "Safety c	2018	assess the safety culture among health professionals of a teaching	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Custo, P. T., R. T. Custo and S. Buttigieg (2019). "The Relationship Between Safety Climate and Performanc	2019	this study tests a theoretical framework addressing relationships a	2. Testing relationships among varia	Quantitative	Yes	ate (Naveh et al., 2005, Katz-Navon et al., 2005)						
da Costa, D. B., D. Ramos, C. S. Gabriel and A. Bernardes (2018). "Patient safety culture: Evaluation by nurs	2018	to evaluate the patient safety culture of nursing teams in the hosp	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
da Silva, G. M., M. V. M. de Lima, M. C. Araripe, S. P. Vasconcelos, S. P. Opitz and G. Z. Laporta (2018). "Eva	2018	To analyze the Patient Safety Culture (PSC) from the perspective o	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Danielsson, M., P. Nilsen, H. Rutberg and K. Arestedt (2019). "A National Study of Patient Safety Culture in	2019	to investigate the patient safety culture in all Swedish hospitals ar	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Daugherty Biddison EL, Paine L, Murakami P, Herzke C, Weaver SJ. Associations between safety culture and	2016	to assess the relationship between safety culture and employee e	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Davies J, Pucher PH, Ibrahim H, Stubbs B. Impact of the introduction of electronic prescribing on staff perc	2017	to assess the impact of the implementation of an EP system on sa	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Davis KK, Harris KG, Mahishi V, Bartholomew EG, Kenward K. Perceptions of Culture of Safety in Hemodialy	2016	Perceptions of Culture of Safety in Hemodialysis Centers.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Davoodi R, Mohammadzadeh Shabestari M, Takbiri A, et al. Patient Safety Culture Based on Medical Staff A	2013	We performed a cross-sectional, analytic-descriptive study in 25 h	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
de Carvalho REFL, Cassiani SHB. Cross-cultural adaptation of the safety attitudes questionnaire - short forn	2012	to perform a cross-cultural adaptation of the Safety Attitudes Que	3. Methodological focus	Quantitative	Yes	SAQ						
De Korne DF, Van Wijngaarden JD, Van Dyck C, Hiddema UF, Klazinga NS. Evaluation of aviation-based safe	2014	to evaluate the implementation of a broad-scale team resource m	4. Evaluation of an intervention	Qualitative			Yes				Yes	
de Mello JF, Barbosa SFF. Patient safety culture in intensive care: Nursing contributions. Texto e Contexto	2013	The aim of this quantitative, comparative, cross-sectional, survey	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
de Souza, L. S., F. L. R. Rocha and L. d. l. Mazzo (2018). "Organizational climate and the occurrence of accid	2018	to evaluate the organizational climate of a public hospital in the S	2. Testing relationships among varia	Quantitative	Yes	SAQ						
De Souza, V. S., J. L. C. De Oliveira, M. A. R. Costa, G. De Vicente, R. R. Mendonça and L. M. Matsuda (2019	2019	To verify the association between safety climate and the workload	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Deilkas E, Hofoss D. Patient safety culture lives in departments and wards: multilevel partitioning of varian	2010	to document 1) that patient safety culture scores vary considerab	3. Methodological focus	Quantitative	Yes	SAQ						
Deilkås ET, Hofoss D. Psychometric properties of the Norwegian version of the Safety Attitudes Questionna	2008	We have translated the SAQ to Norwegian and validated the trans	3. Methodological focus	Quantitative	Yes	SAQ						
Devriendt E, Van den Heede K, Coussement J, et al. Content validity and internal consistency of the Dutch	2012	The purpose of this study is to explore the face- and content valid	3. Methodological focus	Quantitative	Yes	SAQ						
Dodek PM, Wong H, Heyland DK, et al. The relationship between organizational culture and family satisfac	2012	investigated the relationship between measures of organizational	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Dodek PM, Wong H, Jaswal D, et al. Organizational and safety culture in Canadian intensive care units: Rela	2012	The objectives of this study are to describe organizational and safe	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Donahue M, Miller M, Smith L, Dykes P, Fitzpatrick JJ. A leadership initiative to improve communication an	2011	A leadership initiative to improve communication and enhance sa	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Donnelly LF, Dickerson JM, Goodfriend MA, Muething SE. Improving patient safety: effects of a safety prog	2009	The purpose of this study was to evaluate the effects of a program	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						

Doram K, Chadwick W, Bokovoy J, Profit J, Sexton JD, Sexton JB. Got spirit? The spiritual climate scale, psyc	2017	Our objective was to describe a novel survey scale for evaluating s	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Dorigan, G. H. and E. B. Guirardello (2018). "Effect of the practice environment of nurses on job outcomes	2018	to propose and analyze a theoretical model to measure the effect	2. Testing relationships among varia	Quantitative	Yes	SAQ					
dos Santos Alves DF, da Silva D, Guirardello EdB. Nursing practice environment, job outcomes and safety c	2017	To assess correlations between the characteristics of the nursing p	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Durani P, Dias J, Singh HP, Taub N. Junior doctors and patient safety: evaluating knowledge, attitudes and	2013	The Junior Doctor-Patient Safety Attitudes and Climate Questionn	1. Descriptive or exploratory assess	Quantitative	Yes	afety Attitudes and Climate Questionnaire,					
Durgun, H. and H. Kaya (2018). "The attitudes of emergency department nurses towards patient safety." Ir	2018	This research was planned to identify the attitudes of emergency	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Dwyer, L., A. Smith, R. McDermott, C. Breatnach, A. El-Khuffash and J. D. Corcoran (2018). "Staff Attitudes	2018	There is little published research evaluating attitudes towards pat	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Edwards PJ, Scott T, Richardson P, et al. Using staff perceptions on patient safety as a tool for improving sa	2008	This article presents a case study highlighting a pediatric health ca	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Edwards, M. T. (2018). "An Assessment of the Impact of Just Culture on Quality and Safety in US Hospitals.	2018	In pursuit of high reliability, numerous organizations have promot	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Ege, Y., D. Kilic and E. Yildiz (2019). "Patient Safety Culture: Evaluating Turkish Nurses' Attitudes and Knowl	2019	The research was planned with the purpose to determine the per	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Eiras M, Escoval A, Grillo IM, Silva-Fortes C. The hospital survey on patient safety culture in Portuguese hos	2014	The study aims to determine the AHRQ Hospital Survey on Patient	3. Methodological focus	Quantitative	Yes	HSOPS					
Elder NC, Brungs SM, Nagy M, Kudel I, Render ML. Intensive care unit nurses' perceptions of safety after a	2008	To explore perceptions of patient safety among nursing staff in IC	1. Descriptive or exploratory assess	Mixed methods	Yes	HSOPS		Yes	Yes		
El-Jardali F, Dimassi H, Jamal D, Jaafar M, Hemadeh N. Predictors and outcomes of patient safety culture in	2010	assessed the culture of safety in Lebanese hospitals.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
El-Jardali F, Jaafar M, Dimassi H, Jamal D, Hamdan R. The current state of patient safety culture in Lebanes	2010	The objective of this study is to conduct a baseline assessment of	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
El-Jardali F, Sheikh F, Garcia NA, Jamal D, Abdo A. Patient safety culture in a large teaching hospital in Riya	2014	this paper details the findings of a baseline assessment of the pati	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
El-Sherbiny, N. A., E. H. Ibrahim and W. Y. Abdel-Wahed (2020). "Assessment of patient safety culture amo	2020	to assess patient safety culture among paramedical health employ	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Elsous A, Akbari Sari A, ALjeesh Y, Radwan M. Nursing perceptions of patient safety climate in the Gaza Str	2017	This study was undertaken to assess the perception of nurses abo	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Elsous A, Akbarisari A, Rashidian A, Aljeesh Y, Radwan M, Zaydeh HA. Psychometric Properties of an Arabic	2017	This study was done to determine the reliability and validity of the	3. Methodological focus	Quantitative	Yes	SAQ					
Emanuela Fontenele Lima de Carvalho R, De Bortoli Cassiani SH. Cross-cultural adaptation of the Safety Att	2012	The objective of this study was to perform a cross-cultural adapta	3. Methodological focus	Quantitative	Yes	SAQ					
Erestam S, Haglind E, Bock D, Andersson AE, Angenete E. Changes in safety climate and teamwork in the o	2017	to evaluate the safety climate in a Swedish operating room setting	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Etchegaray JM, Thomas EJ. Comparing two safety culture surveys: safety attitudes questionnaire and hosp	2012	to examine the reliability and predictive validity of two patient saf	3. Methodological focus	Quantitative	Yes	ultiple - SAQ, HSOPS					
Etchegaray JM, Thomas EJ. Engaging Employees: The Importance of High-Performance Work Systems for P	2015	To develop and test survey items that measure high-performance	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Fan CJ, Pawlik TM, Daniels T, et al. Association of Safety Culture with Surgical Site Infection Outcomes. Jou	2016	to evaluate the association between safety culture and surgical sit	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Farag A, Blegen M, Gedney-Lose A, Lose D, Perkhounkova Y. Voluntary Medication Error Reporting by ED N	2017	The purpose of this study was to examine the relationship among	2. Testing relationships among varia	Quantitative	Yes	own - improper citation					
Farup PG. Are measurements of patient safety culture and adverse events valid and reliable? Results from	2015	explored associations between the patient safety culture and adve	3. Methodological focus	Quantitative	Yes	HSOPS					
Fassarella, C. S., F. G. Camerini, D. M. Henrique, L. F. Almeida and M. D. C. B. Figueiredo (2018). "Evaluatio	2018	To conduct a benchmarking comparison of the composites of pati	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Fassarella, C. S., L. D. da Silva, F. G. Camerini and M. d. C. Aguiar Barbieri Figueiredo (2019). "Nurse safety c	2019	To evaluate nurse safety culture in a teaching hospital, as well as t	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Feldman, S. S., S. Buchalter, D. Zink, D. J. Slovensky and L. W. Hayes (2019). "Training leaders for a culture c	2019	The purpose of this paper is to understand the degree to which a	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Feng X, Bobay K, Krejci JW, McCormick BL. Factors associated with nurses' perceptions of patient safety cu	2012	To explore nurses' perceptions of patient safety culture and factor	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Feng XQ, Acord L, Cheng YJ, Zeng JH, Song JP. The relationship between management safety commitm	2011	to explore the relationship between management safety commitm	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Fermo VC, Radunz V, da Rosa LM, Marinho MM. Patient safety culture in a bone marrow transplantation u	2015	evaluate the patient safety culture in the perspective of health pr	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Fermo VC, Radunz V, Rosa LM, Marinho MM. Professional attitudes toward patient safety culture in a bon	2016	To identify the attitude of health professionals toward the patient	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Ferrer, M., R. Santa, A. Medhekar, E. Calvo and L. P. Molina-Sánchez (2018). "Drivers of hospital safety cult	2018	The objective of this study is to identify the elements that contrib	2. Testing relationships among varia	Quantitative	Yes	Ferrer et al. 2018					
Fontenele Lima de Carvalho RE, Arruda LP, Pinheiro do Nascimento NK, Sampaio RL, Nunes Cavalcante MU	2017	to assess the culture of safety in three public hospitals.	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Ford EW, Silvera GA, Kazley AS, Diana ML, Huerta TR. Assessing the relationship between patient safety cu	2016	the purpose of this paper is to explore the relationship between h	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
France DJ, Greevy RA, Jr., Liu X, et al. Measuring and Comparing Safety Climate in Intensive Care Units. Me	2010	To measure safety climate in intensive care units (ICU) owned by a	3. Methodological focus	Quantitative	Yes	SAQ					
Frankel A, Grillo SP, Pittman M, et al. Revealing and resolving patient safety defects: the impact of leadersh	2008	To evaluate the impact of rigorous WalkRounds on frontline careg	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Friganovic, A., S. Rezic, B. Kurtovic, S. Vidmanic, R. Zelenikova, C. Rotim, V. Konjevoda, B. Rezek and S. Pisk	2020	To examine Croatian nurses' perception of implicit nursing care ra	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Fujita S, Seto K, Ito S, Wu Y, Huang CC, Hasegawa T. The characteristics of patient safety culture in Japan, T	2013	The aim of this study is to investigate the characteristics of PSC in	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Fujita S, Seto K, Kitazawa T, Matsumoto K, Hasegawa T. Characteristics of unit-level patient safety culture i	2014	to clarify the characteristics of unit-level PSC in hospitals in Japan.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Fujita, S., Y. Wu, S. Iida, Y. Nagai, Y. Shimamori and T. Hasegawa (2019). "Patient safety management syste	2019	This study aimed to identify hospital-level elements related to pat	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Gaber MA. Relationship among nurses' safety compliance, organizational safety climate, worker's variable	2013	to explore relationship among nurses' safety compliance, organiz	2. Testing relationships among varia	Quantitative	Yes	HSCS					
Gabrani A, Hoxha A, Simaku A, Gabrani JC. Application of the Safety Attitudes Questionnaire (SAQ) in Alba	2015	To establish the reliability and validity of the translated version of	3. Methodological focus	Quantitative	Yes	SAQ					
Gadjradj, P. S. and B. S. Harhangi (2019). "Safety Culture and Attitudes Among Spine Professionals: Results	2019	The aim of the current study is to assess attitudes toward safety c	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Galvao, T. F., M. C. Couteiro Lopes, C. C. Carrilho Oliva, M. E. de Almeida Araujo and M. T. Silva (2018). "Pa	2018	to assess patient safety culture in a university hospital.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Gambashidze, N., A. Hammer, N. Ernstmann and T. Manser (2020). "Psychometric properties of the Georg	2020	To study the psychometric properties of the Georgian version of t	3. Methodological focus	Quantitative	Yes	SAQ					
Gampetro, P. J., J. P. Segvich, N. Jordan, B. Velsor-Friedrich and L. Burkhart (2019). "Perceptions of Pediatr	2019	Strategies that identify gaps in hospital's safety culture within the	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Garcia, C. d. L., I. M. Pinheiro Bezerra, J. L. L. Souza Ramos, J. E. T. Mendes Rossette do Valle, M. L. Bezerra de	2019	To analyze the relationship between patient safety culture and bu	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Garrouste-Orgeas M, Perrin M, Soufir L, et al. The Iatroref study: medical errors are associated with sympt	2015	evaluated whether burnout, symptoms of depression and safety c	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Gehring K, Mascherek AC, Bezzola P, Schwappach DL. Safety climate in Swiss hospital units: Swiss version c	2015	we translated the Safety Climate Survey into German and French.	3. Methodological focus	Quantitative	Yes	SCSu					
Geraghty, A., Reid, S., & McIlhenny, C. (2014). Examining patient safety attitudes among urology trainees. A	2014	Examining patient safety attitudes among urology trainees	1. Descriptive or exploratory assess	Quantitative	Yes	ORMAQ					
Gibbs, K., S. DeMaria, S. McKinsey, A. Fede, A. Harrington, D. Hutchison, C. Torchen, A. Levine and A. Goldb	2018	To describe the successful implementation of an in situ simulation	4. Evaluation of an intervention	Quantitative	Yes	Gibbs et al. 2018					
Gillespie BM, Gwinner K, Chaboyer W, Fairweather N. Team communications in surgery - creating a culture	2013	to describe team communication and the ways it fostered or threa	2. Testing relationships among varia	Qualitative			Yes			Yes	
Gillespie, B. M., E. Harbeck, E. Kang, C. Steel, N. Fairweather and W. Chaboyer (2018). "Changes in surgical	2018	The aim of the present study was to describe process changes in s	4. Evaluation of an intervention	Quantitative	Yes	nd Safety Climate Survey (TSCS)					
Gimenes FR, Torrieri MC, Gabriel CS, et al. Applying an ecological restoration approach to study patient saf	2016	To gain an understanding of medication safety culture and other c	1. Descriptive or exploratory assess	Qualitative				Yes		Yes	
Ginsburg L, Gilin D, Tregunno D, Norton PG, Flemons W, Fleming M. Advancing measurement of patient sa	2009	To examine the psychometric and unit of analysis/strength of cul	3. Methodological focus	Quantitative	Yes	MSI					
Ginsburg LR, Tregunno D, Norton PG, Mitchell JI, Howley H. 'Not another safety culture survey': using the C	2014	This study examines the revised Canadian PSCS (Can-PSCS) for use	3. Methodological focus	Quantitative	Yes	CAN PSCS					
Gleeson, L. L., L. Tobin, G. L. O'Brien, E. K. Crowley, A. Delaney, D. O'Mahony and S. Byrne (2020). "Safety c	2020	The aim of this study is to describe the safety culture of a universi	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Godinho Rigobello MC, Fontenele Lima de Carvalho RE, De Bortoli Cassiani SH, Galon T, Capucho HC, de De	2012	To evaluate the perception of the safety climate of nursing profes	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Goldstein DH, Nyce JM, Van Den Kerkhof EG. Safety Perceptions of Health Care Leaders in 2 Canadian Acad	2017	The purpose of this study was to examine the safety climate from	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO					

Golle, L., D. Ciotti, G. E. Gehrke Herr, F. Aozane, C. R. Schmidt and A. C. Bernat Kolankiewicz (2018). "Cultur	2018	To evaluate the perception of active nurses in a private hospital ar	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Goras C, Wallentin FY, Nilsson U, Ehrenberg A. Swedish translation and psychometric testing of the safety /	2013	The aim of the present study was to establish the validity and relia	3. Methodological focus	Quantitative	Yes	SAQ					
Gordon BM, Lam TS, Bahjri K, Hashmi A, Kuhn MA. Utility of Preprocedure Checklists in the Congenital Car	2014	to create, implement, and evaluate a preprocedural meeting in th	3. Methodological focus	Quantitative	Yes	SAQ					
Gore DC, Powell JM, Baer JG, et al. Crew resource management improved perception of patient safety in t	2010	To improve safety in the operating theater, a company of aviation	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Granel, N., J. M. Manresa-Domínguez, A. Barth, K. Papp and M. D. Bernabeu-Tamayo (2019). "Patient safet	2019	The purpose of this paper is to develop a cross-cultural HSOPSC fo	3. Methodological focus	Quantitative	Yes	HSOPS					
Groves PS, Finfgeld-Connett D, Wakefield BJ. It's always something: hospital nurses managing risk. Clinical	2014	to describe the process by which hospital staff nurses keep patien	1. Descriptive or exploratory assess	Qualitative			Yes				
Gu X, Itoh K. A cross-national study on healthcare safety climate and staff attitudes to disclosing adverse e	2010	The present paper reports comparative results of safety climate in	1. Descriptive or exploratory assess	Quantitative	Yes	ORMAQ					
Gu X, Itoh K. A pilot study on safety climate in Chinese hospital. J Patient Saf. 2011;7(4):204-212. OBJECTIV	2011	The present paper has 2 primary objectives as a pilot study on hea	3. Methodological focus	Quantitative	Yes	ORMAQ					
Gunes UY, Gurlek O, Sonmez M. A survey of the patient safety culture of hospital nurses in Turkey. Collegia	2016	to explore and describe nurses' perceptions of patient safety cultu	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Gupta B, Guleria K, Arora R. Patient Safety in Obstetrics and Gynecology Departments of two Teaching Hos	2016	The objective of this study was to assess and compare patient safe	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Gurková, E., R. Zeleníková, A. Friganovic, I. Uchmanowicz, D. Jarošová, E. Papastavrou and K. Žiaková (2019	2019	The aim of the study was to investigate nurses' perceptions of the	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Habahbeh, A. A. and M. A. Alkhalaileh (2020). "Effect of an educational programme on the attitudes towar	2020	The purpose of this study was to evaluate the effect of a surgical s	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Habibi M, Fesharaki MG, Samadinia H, Mohamadian M, Anvari S. Patient Safety Culture and Factors that In	2017	The present study aimed to examine patient safety culture and fa	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Haerkens MH, van Leeuwen W, Sexton JB, Pickkers P, van der Hoeven JG. Validation of the Dutch language	2016	The objective of this study was to assess the psychometric proper	3. Methodological focus	Quantitative	Yes	SAQ					
Haerkens MHTM, Kox M, Noe PM, van der Hoeven JG, Pickkers P. Crew Resource Management in the traur	2017	We determined the effects of CRM implementation in the trauma	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Hahn SE, Murphy LR. A short scale for measuring safety climate. Safety Science. 2008;46(7):1047-1066. A G	2008	A 6-item measure that assesses global work safety climate was va	3. Methodological focus	Quantitative	Yes	ahn & Murphy, 2008					
Halbesleben JR, Wakefield BJ, Wakefield DS, Cooper LB. Nurse burnout and patient safety outcomes: nurse	2008	the relationship between nurse burnout and patient safety indica	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Haller G, Garnerin P, Morales M-A, et al. Effect of crew resource management training in a multidisciplinar	2008	To assess the effect of a Crew Resource Management (CRM) inter	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Hamaideh SH. Mental health nurses' perceptions of patient safety culture in psychiatric settings. Internatio	2016	The aims of this study were to assess the perception of mental he	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hamdan M, Saleem AA. Assessment of patient safety culture in Palestinian public hospitals. Int J Qual Heal	2013	To assess the prevalent patient safety culture in Palestinian public	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Hamdan M. Measuring safety culture in Palestinian neonatal intensive care units using the Safety Attitude	2013	This study aimed to measure safety culture, examine variations ar	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Hamdan, M. and A. A. o. Saleem (2018). "Changes in Patient Safety Culture in Palestinian Public Hospitals:	2018	To assess the changes in the patient safety culture between 2011	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Hammer A, Ernstmann N, Ommen O, et al. Psychometric properties of the Hospital Survey on Patient Safet	2011	examines whether the Hospital Survey on Patient Safety Culture fi	3. Methodological focus	Quantitative	Yes	HSOPS					
Hanifi, N., Z. Yazdanshenas, M. Namadian and N. Motamed (2018). "The Effect of Patient Safety Education	2018	to determine the effect of the patient safety education program o	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Hansen LO, Williams MV, Singer SJ. Perceptions of hospital safety climate and incidence of readmission. He	2011	To define the relationship between hospital patient safety climate	2. Testing relationships among varia	Quantitative	Yes	PSCHO					
Hanskamp-Sebregts M., M. Zegers, G. P. Westert, W. Boeijen, S. Teerenstra, P. J. van Gurp and H. Wollersh	2019	To evaluate the effectiveness of internal auditing in hospital care f	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Harsul, W., A. M. Irwan and E. L. Sjattar (2019). "The relationship between nurse self-efficacy and the cultu	2019	The aim of this study was to analyze the relationship of self-effica	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hartmann CW, Meterko M, Rosen AK, et al. Relationship of hospital organizational culture to patient safet	2009	assessed the relationship between hospitals' organizational cultur	2. Testing relationships among varia	Quantitative	Yes	PSCHO					
Hartmann CW, Rosen AK, Meterko M, et al. An overview of patient safety climate in the VA. Health Serv Re	2008	To assess variation in safety climate across VA hospitals nationally	2. Testing relationships among varia	Quantitative	Yes	PSCHO					
Hatam N, Keshtkar V, Forouzan F, Peivand B. Patient safety culture status in teaching hospitals: A case of s	2012	Patient safety culture status in teaching hospitals: A case of shiraz	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Hatoun J, Suen W, Liu C, et al. Elucidating Reasons for Resident Underutilization of Electronic Adverse Even	2016	Residents at an academic medical center were surveyed about AE	2. Testing relationships among varia	Quantitative	Yes	Hatoun et al. 2016					
Haugen AS, Softeland E, Eide GE, et al. Impact of the World Health Organization's Surgical Safety Checklist	2013	to study the checklist effects on safety culture perceptions in oper	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Haugen AS, Softeland E, Eide GE, Nortvedt MW, Aase K, Harthug S. Patient safety in surgical environments	2010	examined the psychometric properties of a Norwegian translation	3. Methodological focus	Quantitative	Yes	HSOPS					
Hayashi, R., S. Fujita, S. Iida, Y. Nagai, Y. Shimamori and T. Hasegawa (2020). "Relationship of patient safet	2020	This study aimed to examine the relationship between patient saf	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hayes LW, Dobyns EL, DiGiovine B, et al. A multicenter collaborative approach to reducing pediatric codes	2012	The Child Health Corporation of America formed a multicenter col	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Haynes AB, Weiser TG, Berry WR, et al. Changes in safety attitude and relationship to decreased postopera	2011	To assess the relationship between changes in clinician attitude an	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Hedskold M, Pukk-Harenstam K, Berg E, et al. Psychometric properties of the Hospital Survey on Patient Sa	2013	describes this Swedish version and an assessment of its psychome	3. Methodological focus	Quantitative	Yes	HSOPS					
Hefner JL, Hilligoss B, Knupp A, et al. Cultural Transformation After Implementation of Crew Resource Man	2016	examined the impact of a systematic CRM implementation across	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Hellings J, Schrooten W, Klazinga NS, Vleugels A. Improving patient safety culture. International journal of	2010	to describe a patient safety culture improvement approach in five	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Hemmat F, Atashzadeh-Shoorideh F, Mehrabi T, Zayeri F. A survey of nurses' awareness of patient safety c	2015	describes the nurses' awareness of patient safety culture in neona	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Herrington, C. R. and M. W. Hand (2019). "Impact of Nurse Peer Review on a Culture of Safety." Journal of	2019	A pre- and postdesign using the Agency for Healthcare Research a	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Hessels AJ, Genovese-Schek V, Agarwal M, Wurmser T, Larson EL. Relationship between patient safety clim	2016	This study aim was to explore the relationships among safety clim	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hessels, A. J., M. Paliwal, S. H. Weaver, D. Siddiqui and T. A. Wurmser (2019). "Impact of Patient Safety Cul	2019	The purpose of this study was to describe relationships among PS	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hill MR, Roberts MJ, Alderson ML, Gale TC. Safety culture and the 5 steps to safer surgery: an intervention	2015	assessed the safety culture in the elective orthopaedic theatres of	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Hinde T, Gale T, Anderson I, Roberts M, Sice P. A study to assess the influence of interprofessional point of	2016	to assess the impact of interprofessional point of care simulation	4. Evaluation of an intervention	Quantitative	Yes	nd Safety Climate Survey (TSCS)					
Hong S, Li Q. The reasons for Chinese nursing staff to report adverse events: a questionnaire survey. J Nurs	2017	To investigate the impact of nurses' perception of patient safety c	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Hook JN, Boan D, Davis DE, Aten JD, Ruiz JM, Maryon T. Cultural Humility and Hospital Safety Culture. Jour	2016	explore the extent to which cultural humility, which involves open	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Hori Okuyama, J. H., T. F. Galvao, M. T. L. Crozatti and M. T. Silva (2019). "Health professionals' perception	2019	To assess patient safety culture in a university hospital.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Horwitz SK, Horwitz IB. The effects of organizational commitment and structural empowerment on patient	2017	The purpose of this paper is to investigate the relationship betwe	2. Testing relationships among varia	Quantitative	Yes	PSCHO					
Hsu, Y.-J. and J. A. Marsteller (2020). "Who Applies an Intervention to Influence Cultural Attributes in a Qu	2020	This study aimed to identify factors related to teams' adoption of	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Huang DT, Clermont G, Kong L, et al. Intensive care unit safety culture and outcomes: a US multicenter stu	2010	to determine if intensive care unit (ICU) safety culture is independ	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Huang, C.-H., H.-H. Wu and Y.-C. Lee (2018). "The perceptions of patient safety culture: A difference betwe	2018	this study aims to respectively investigate the perceptions of patie	1. Descriptive or exploratory assess	Quantitative	Yes	new Chinese SAQ					
Huang, C.-H., H.-H. Wu, C. Y. Chou, H. Dai and Y.-C. Lee (2018). "The Perceptions of Physicians and Nurses I	2018	We aimed to examine physicians' and nurses' perceptions of patie	1. Descriptive or exploratory assess	Quantitative	Yes	new Chinese SAQ					
Huang, H.-T., C.-H. Tsai and C.-F. Wang (2019). "A Model for Promoting Occupational Safety and Health in	2019	This study used an integrative approach as a starting point and co	2. Testing relationships among varia	Quantitative	Yes	ey (Griffin & Neal, 2000; Neal & Griffin, 2006)					
Hughes LC, Chang Y, Mark BA. Quality and strength of patient safety climate on medical-surgical units. Hea	2009	Study purposes were to describe quality and strength of the patie	1. Descriptive or exploratory assess	Quantitative	Yes	(1980) safety climate scale					
Hutchinson A, Young TA, Cooper KL, et al. Trends in healthcare incident reporting and relationship to safet	2009	This study analyses patterns in reporting of patient safety incident	2. Testing relationships among varia	Quantitative	Yes	NHS Staff Survey					
Hwang JI, Hwang EJ. Individual and work environment characteristics associated with error occurrences in	2011	To determine individual and work environment characteristics ass	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Hwang JI. What are hospital nurses' strengths and weaknesses in patient safety competence? Findings fro	2015	To examine hospital nurses' patient safety competencies and the	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Hwang, J.-I., S. W. Kim and H. J. Chin (2019). "Patient Participation in Patient Safety and Its Relationships w	2019	This study's aim was to examine degrees of patient participation i	2. Testing relationships among varia	Quantitative	Yes	SCM					

Ito S, Seto K, Kigawa M, Fujita S, Hasegawa T, Hasegawa T. Development and applicability of Hospital Surve	2011	Although several conceptual frameworks have been proposed in t	3. Methodological focus	Quantitative	Yes	HSOPS					
Jafarpanah, M. and B. Rezaei (2020). "Association between organizational citizenship behavior and patient	2020	to explore the ways to improve patient safety culture. Organizatio	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Janssen SJ, Teunis T, Guitton TG, Ring D, Herndon JH. Orthopaedic Surgeons' View on Strategies for Improv	2015	This study assessed the experienced safety culture and preferred	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO					
Jasti H, Sheth H, Verrico M, et al. Assessing patient safety culture of internal medicine house staff in an aca	2009	to assess hospital PSC from the perspective of internal medicine h	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Je SM, Kim HJ, You JS, et al. Assessing safety attitudes among healthcare providers after a hospital-wide hi	2014	to evaluate the impact of a hospital-wide quality improvement pr	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Jeong, G. H. J., W. C. Lee, H. H. Liao, F. Y. Chu, T. J. Chen and P. C. Wang (2019). "The hospital patient safet	2019	Our study sought to improve scoring methodology and develop a	3. Methodological focus	Quantitative	Yes	SAQ					
Ji, C. H., F. H. Xu and S. Pan (2019). "Patient safety culture regarding intravenous therapy in guangzhou, Ch	2019	To investigate the patient safety culture regarding intravenous the	1. Descriptive or exploratory assess	Quantitative	Yes	Ji et al. 2019					
Jia PL, Zhang LH, Zhang MM, et al. Safety culture in a pharmacy setting using a pharmacy survey on patient	2014	To explore the attitudes and perceptions of patient safety culture	3. Methodological focus	Quantitative	Yes	HSOPS					
Jiang, K., L. Tian, C. Yan, Y. Li, H. Fang, P. Sun, P. Li, H. Jia, Y. Wang, Z. Kang, Y. Cui, H. Liu, S. Zhao, G. Anasta	2019	This study aims to investigate patient safety culture in secondary	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Johnston, M. J., S. Arora, D. King and A. Darzi (2018). "Improving the Quality of Ward-based Surgical Care v	2018	This study aimed to explore the impact of a human factors interve	4. Evaluation of an intervention	Mixed methods	Yes	ohnston et al. 201	Yes				
Jones KJ, Skinner AM, High R, Reiter-Palmon R. A theory-driven, longitudinal evaluation of the impact of te	2013	evaluated the impact of a year-long team training programme on	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Jones, K. J., J. Crowe, J. A. Allen, A. M. Skinner, R. High, V. Kennel and R. Reiter-Palmon (2019). "The impact	2019	The purpose of this study is to determine associations between cc	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Kaafarani HM, Itani KM, Rosen AK, Zhao S, Hartmann CW, Gaba DM. How does patient safety culture in the	2009	How does patient safety culture in the operating room and post-a	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO					
Kagan I, Barnoy S. Factors associated with reporting of medication errors by Israeli nurses. Journal of nursi	2008	This study investigated medication error reporting among Israeli r	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Kagan, I., N. Porat and S. Barnoy (2019). "The quality and safety culture in general hospitals: Patients', phy	2019	To explore the disparities between patients' and health care work	1. Descriptive or exploratory assess	Quantitative	Yes	Patient Safety Culture Survey					
Kahwati, L. C., A. V. Sorensen, S. Teixeira-Poit, S. Jacobs, S. A. Sommerness, K. K. Miller, E. Pleasants, H. M.	2019	The objective of this study was to describe the SPPC implementati	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Kawamoto AM, Campos de Oliveira JL, Tonini NS, Nicola AL. Leadership and patient safety culture: percept	2016	Analyze the perceptions of health professionals as for patient safe	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Kawano T, Taniwaki M, Ogata K, Sakamoto M, Yokoyama M. Improvement of teamwork and safety climate	2014	we investigated whether implementation of the SSCL improved th	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Kaya S, Barsbay S, Karabulut E. The Turkish version of the safety attitudes questionnaire: psychometric pro	2010	To test the psychometric soundness of the Safety Attitudes Quest	3. Methodological focus	Quantitative	Yes	SAQ					
Khammar, A., M. Poursadeghiyan, H. Marioryad, R. N. Amjad, M. Alimohammadi and M. Khandan (2019). "	2019	The present research was conducted to study the safety climate o	1. Descriptive or exploratory assess	Quantitative	Yes	Kudo et al. 2008					
Khandan M, Koohpaei A, Babaki F. The Effect of Individual and Organizational Variables on Patients' Safety	2016	This study aimed to assess PSC and its association with demograp	1. Descriptive or exploratory assess	Quantitative	Yes	SCSu					
Khater WA, Akhu-Zaheya LM, Al-Mahasneh SI, Khater R. Nurses' perceptions of patient safety culture in Jo	2015	The study aimed to assess patient safety culture in Jordanian hosp	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Kho ME, Perri D, McDonald E, et al. The climate of patient safety in a Canadian intensive care unit. J Crit Ca	2009	The climate of patient safety in a Canadian intensive care unit.	1. Descriptive or exploratory assess	Quantitative	Yes	SCSu					
Khoshakhlagh, A. H., E. Khatooni, I. Akbarzadeh, S. Yazdanirad and A. Sheidaei (2019). "Analysis of affectin	2019	this study aimed to determine and compare the views of healthca	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Khoshbin A, Lingard L, Wright JG. Evaluation of preoperative and perioperative operating room briefings a	2009	to evaluate 2 OR briefing safety initiatives, "07:35 huddles" (preop	4. Evaluation of an intervention	Mixed methods	Yes	SAQ	Yes				
Kim, K. (2020). "Exploring the influence of workplace violence and bystander behaviour on patient safety i	2020	To explore the influences of clinical nurses' workplace violence an	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Kim, K.-J., M. S. Yoo and E. J. Seo (2018). "Exploring the Influence of Nursing Work Environment and Patien	2018	This study aimed to explore the influence of nurse work environm	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Kim, S.-H. and S.-J. Weng (2018). "Incorporating work experience of medical staff into patient safety climate management: a multi-group analysis."	2018	This study aims to provide insights on how to incorporate the wor	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Kirkegaard, M. L., P. Kines, H. B. Nielsen and A. H. Garde (2018). "Occupational safety across jobs and shift	2018	explores whether occupational safety and health (OSH) risk perce	1. Descriptive or exploratory assess	Mixed methods	Yes	rkegaard et al., 2018				Yes	
Kline TJ, Willness C, Ghali WA. Predicting patient complaints in hospital settings. Quality & safety in health	2008	To evaluate the complexity of the patient case and patient safety	2. Testing relationships among varia	Quantitative	Yes	MSI					
Klingner J, Moscovice I, Tupper J, Coburn A, Wakefield M. Implementing patient safety initiatives in rural h	2009	This article describes the Tennessee Rural Hospital Patient Safety	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Kobuse H, Morishima T, Tanaka M, Murakami G, Hirose M, Imanaka Y. Visualizing variations in organizatio	2014	To develop a reliable and valid questionnaire that can distinguish	3. Methodological focus	Quantitative	Yes	Kobuse et al. 2014					
Kristensen S, Badsberg JH, Rischel V, Anhoj J, Mainz J, Bartels P. The patient safety climate in Danish hospit	2015	to describe and analyse the patient safety climate in 15 Danish ho	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Kristensen S, Christensen KB, Jaquet A, et al. Strengthening leadership as a catalyst for enhanced patient s	2016	to investigate staff's perceptions of patient safety culture in a Dan	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Kristensen S, Hammer A, Bartels P, et al. Quality management and perceptions of teamwork and safety cli	2015	This study aimed to investigate the associations of quality manage	2. Testing relationships among varia	Quantitative	Yes	nd Safety Climate Survey (TSCS)					
Kristensen S, Sabroe S, Bartels P, Mainz J, Christensen KB. Adaption and validation of the Safety Attitudes (	2015	The purpose of this study was to adapt the SAQ for use in Danish l	3. Methodological focus	Quantitative	Yes	SAQ					
Kristensen S, Tugvustein N, Zachariassen H, Sabroe S, Bartels P, Mainz J. The virgin land of quality manager	2016	to investigate the patient safety culture in the National Hospital o	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Kudo Y, Kido S, Taruzuka Shahzad M, Saegusa Y, Satoh T, Aizawa Y. Safety climate and motivation toward p	2009	to examine the associations between the dimensions of safety clir	2. Testing relationships among varia	Quantitative	Yes	Kudo et al. 2008					
Kudo Y, Satoh T, Kido S, et al. A pilot study testing the dimensions of safety climate among Japanese nurse	2008	To investigate the dimensions of safety climate among Japanese n	1. Descriptive or exploratory assess	Quantitative	Yes	Kudo et al. 2008					
Kudo Y, Satoh T, Kido S, et al. A pilot study testing the dimensions of safey climate among Japanese nurses	2008	To investigate the dimensions of safety climate among Japanese n	1. Descriptive or exploratory assess	Quantitative	Yes	Kudo et al. 2008					
Kullberg A, Bergenmar M, Sharp L. Changed nursing scheduling for improved safety culture and working co	2016	To evaluate fixed scheduling compared with self-scheduling for nu	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Kumbi, M., A. Hussen, A. Lette, S. Nuriye and G. Morka (2020). "Patient Safety Culture and Associated Fact	2020	Therefore, this study aims to assess patient safety culture and ass	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Kundig F, Staines A, Kinge T, Perneger TV. Numbering questionnaires had no impact on the response rate a	2011	we evaluated whether using nonnumbered vs. numbered questio	3. Methodological focus	Quantitative	Yes	HSOPS					
Kuntz, L., N. Scholten, H. Wilhelm, M. Wittland and H. A. Hillen (2018). "The benefits of agreeing on what r	2018	This empirical study aims to explain differences in the units' safety	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Kuosmanen A, Tiihonen J, Repo-Tiihonen E, Eronen M, Turunen H. Patient safety culture in two Finnish sta	2013	to evaluate patient safety culture, identify areas for improvement	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Kuosmanen, A., J. Tiihonen, E. Repo-Tiihonen, M. Eronen and H. Turunen (2019). "Changes in patient safet	2019	To evaluate how a 3-year patient safety intervention, more specif	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Kuy S, Romero RAL. Improving staff perception of a safety climate with crew resource management trainin	2017	First, to describe implementation of CRM in a Veterans Affair (VA)	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Kvist T, Mantynen R, Turunen H, et al. How magnetic are Finnish hospitals measured by transformational l	2013	This report presents baseline measurements for a longitudinal stu	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Kyratsis, Y., R. Ahmad, M. Iwami, E. Castro-Sanchez, R. Atun and A. H. Holmes (2019). "A multilevel neo-ins	2019	we examined how inter-organisational influences shaped safety p	2. Testing relationships among varia	Qualitative			Yes		Yes		
Lachman P, Linkson L, Evans T, Clausen H, Hothi D. Developing person-centred analysis of harm in a paedia	2015	we developed and tested a tool specifically designed for patients a	4. Evaluation of an intervention	Mixed methods	Yes	SAQ					Yes
Lafleur, A., A. Harvey and C. Simard (2018). "Adjusting to duty hour reforms: residents' perception of the s	2018	We measured residents' perception of the patient safety climate d	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Lake, E. T., K. E. Roberts, P. D. Agosto, E. Ely, A. P. Bettencourt, E. S. Schierholz, W. D. Frankenberger, G. Ca	2018	to describe the current state of patient safety in pediatric acute ca	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Lambrou P, Papastavrou E, Merkouris A, Middleton N. Professional environment and patient safety in eme	2015	to examine nurses' and physicians' perceptions of professional en	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Landgren R, Alawadi Z, Douma C, Thomas EJ, Etchegaray J. Barriers of Pediatric Residents to Speaking Up A	2016	We examined the reasons reported by pediatric residents for not	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Law MP, Zimmerman R, Baker GR, Smith T. Assessment of safety culture maturity in a hospital setting. Hea	2010	to examine the levels of safety culture maturity in four programs a	1. Descriptive or exploratory assess	Quantitative	Yes	MapSCAT					
Lawton R, O'Hara JK, Sheard L, et al. Can staff and patient perspectives on hospital safety predict harm-fre	2015	to investigate whether the safety information provided by patient	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Layne, D. M., L. S. Nemeth, M. Mueller and M. Martin (2019). "Negative Behaviors among Healthcare Prof	2019	Our objective was to examine the presence of negative behaviors	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Lee E. Safety climate and attitude toward medication error reporting after hospital accreditation in South I	2016	This study compared registered nurses' perceptions of safety clim	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Lee SH, Phan PH, Dorman T, Weaver SJ, Pronovost PJ. Handoffs, safety culture, and practices: evidence fro	2016	to analyze how different elements of patient safety culture are as	2. Testing relationships among varia	Quantitative	Yes	HSOPS					

Lee WC, Wung HY, Liao HH, et al. Hospital safety culture in Taiwan: a nationwide survey using Chinese vers	2010	to verify a safety culture survey instrument in Chinese and to asse	3. Methodological focus	Quantitative	Yes	SAQ					
Lee YC, Wu HH, Hsieh WL, Weng SJ, Hsieh LP, Huang CH. Applying importance-performance analysis to pat	2015	to improve and to develop better strategies regarding patient safe	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Lee, S. E., C. Vincent, V. S. Dahinten, L. D. Scott, C. G. Park and K. Dunn Lopez (2018). "Effects of Individual	2018	This study aimed to investigate effects of individual nurse and hos	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Lee, W.-C., S.-H. Han and H.-J. Jeong (2020). "Ensuring Taiwan patient safety culture survey is free from dif	2020	The aim of this study was to examine whether the succinct versio	3. Methodological focus	Quantitative	Yes	SAQ					
Lee, Y.-C., C.-H. Huang, C.-F. Wu, H.-W. Hsueh and H.-H. Wu (2019). "A Longitudinal Study of Identifying Cr	2019	This study uses longitudinal data from a regional teaching hospita	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Lee, Y.-C., P.-S. Zeng, C.-H. Huang, C.-F. Wu, C.-C. Yang and H.-H. Wu (2019). "Causal Relationships of Patie	2019	This study uses the decision-making trial and evaluation laborator	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Lefebvre, G., L. Honey, K. Hines, A. Keough, C. Roye, S. Bellemare, T. D. Piscione, A. Falconer, L. Shepherd, S	2020	This study implemented a quality improvement program based or	4. Evaluation of an intervention	Quantitative	Yes	SCORE					
Legg JS, Aaron L, Dempsey MC. Patient safety perceptions among vascular interventional technologists. Jou	2013	To describe perceptions of patient safety culture (PSC) among US	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Legg JS, Dempsey MC, Aaron L. Patient Safety Perceptions Among U.S. Radiation Therapists. Radiation The	2013	To determine the radiation therapy profession's overall values, be	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Li A-T. Teamwork climate and patient safety attitudes: associations among nurses and comparison with ph	2013	Teamwork climate and patient safety attitudes: associations amor	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Li, Y., Y. Zha, Y. Ha, M. Jiao, H. Ma, B. Teng, K. Yang, T. Sun, Q. Wu and H. Qiao (2018). "Perceptions of patie	2018	The aim of the study is to explore the perception of healthcare em	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Liang HY, Tang FI, Wang TF, Lin KC, Yu S. Nurse characteristics, leadership, safety climate, emotional labou	2016	The aim of this study was to propose a theoretical model and app	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Lim RB, Ng BB, Ng KM. Evaluation of the patient safety Leadership Walkabout programme of a hospital in S	2014	we carried out an evaluation of its effectiveness in improving the	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Lima de Oliveira, I. C., M. L. S. Nunes Cavalcante, S. F. Aires, R. J. Moreira de Freitas, B. V. da Silva, D. M. Fre	2018	To evaluate the culture of patient safety in a mental health service	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Lin, D. M., K. A. Carson, L. H. Lubomski, E. C. Wick and J. C. Pham (2018). "Statewide Collaborative to Reduc	2018	This study's purpose was to evaluate the effectiveness of the AHR	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Ling L, Gomersall CD, Samy W, et al. The Effect of a Freely Available Flipped Classroom Course on Health Ca	2016	to assess the impact of a standardized patient safety course on he	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Listyowardojo TA, Nap RE, Johnson A. Variations in hospital worker perceptions of safety culture. Int J Qual	2012	To compare the attitudes toward and perceptions of institutional	1. Descriptive or exploratory assess	Quantitative	Yes	yowardojo et al. 2012					
Listyowardojo TA, Yan X, Leyshon S, et al. (2017) A safety culture assessment by mixed methods at a public	2017	A safety culture assessment by mixed methods at a public matern	1. Descriptive or exploratory assess	Mixed methods	Yes	SAQ	Yes				
Liu Y, Kalisch BJ, Zhang L, Xu J. Perception of safety culture by nurses in hospitals in china. Journal of nursin	2009	The safety culture in hospitals in China was investigated using a se	1. Descriptive or exploratory assess	Quantitative	Yes	Liu et al. 2009					
Livorsi D, Knobloch MJ, Blue LA, et al. A rapid assessment of barriers and facilitators to safety culture in an	2016	to identify opportunities for safety culture improvement in a 30-b	1. Descriptive or exploratory assess	Mixed methods	Yes	HSOPS	Yes				
Lopez KD, Gerling GJ, Cary MP, Kanak MF. Cognitive work analysis to evaluate the problem of patient falls	2010	To identify factors in the nursing work domain that contribute to t	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Lopez-Jeng, C. and S. D. Eberth (2019). "Improving Hospital Safety Culture for Falls Prevention Through Int	2019	to develop a health education program tailored to their internal a	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Lowe G. How employee engagement matters for hospital performance. Healthcare quarterly (Toronto, On	2012	examines how job, work environment, management and organiza	2. Testing relationships among varia	Quantitative	Yes	Lowe, 2012					
Luiz RB, Simoes AL, Barichello E, Barbosa MH. Factors associated with the patient safety climate at a teach	2015	to investigate the association between the scores of the patient sa	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Lyu H, Wick EC, Housman M, Freischlag JA, Makary MA. Patient satisfaction as a possible indicator of quali	2013	To determine whether patient satisfaction is independent from su	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Macedo TR, Rocha PK, Tomazoni A, Souza S, Anders JC, Davis K. The culture of patient safety from the pers	2016	To identify the patient safety culture in pediatric emergencies fro	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Magid DJ, Sullivan AF, Cleary PD, et al. The safety of emergency care systems: Results of a survey of clinicia	2009	The purpose of this study is to assess the degree to which EDs are	3. Methodological focus	Quantitative	Yes	ency Department Safety Study Survey					
Mahfoozpour S, Mojdehkar R. ATTITUDES OF HEALTH CAREGIVERS ON TEAMWORK AND SAFETY CLIMATE	2010	To study the attitudes of health caregivers on teamwork & safety	1. Descriptive or exploratory assess	Quantitative	Yes	nd Safety Climate Survey (TSCS)					
Mahjoub, M., S. Ben Fredj, N. Bouafia, R. Bouriga, N. Ben Jalleb and M. Njah (2018). "Promoting safety cult	2018	to assess and analyze the healthcare professional-patient relation	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Mahrous, M. S. (2018). "Patient safety culture as a quality indicator for a safe health system: Experience fr	2018	This study aimed to evaluate the extent to which the culture suppl	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Mallouli M, Tlili MA, Aouicha W, et al. Assessing patient safety culture in Tunisian operating rooms: A mult	2017	To assess the patient safety culture (PSC) in operating rooms (ORs	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Manapragada, A., V. Bruk-Lee, A. H. Thompson and L. M. Heron (2019). "When safety climate is not enoug	2019	To examine the association between components of safety climat	2. Testing relationships among varia	Quantitative	Yes	ey (Griffin & Neal, 2000; Neal & Griffin, 2006)					
Mantynen R, Vehvilainen-Julkunen K, Partanen P, Turunen H, Miettinen M, Kvist T. Changes in Transforma	2014	This paper describes the changes in transformational leadership a	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Mardon RE, Khanna K, Sorra J, Dyer N, Famolaro T. Exploring relationships between hospital patient safety	2010	examined relationships between the Agency for Healthcare Resea	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Marinho MM, Radünz V, Barbosa SdFF. Assessment of safety culture by surgical unit nursing teams. Texto	2014	to evaluate the patient safety culture from the perspective of the	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Mark BA, Hughes LC, Belyea M, Bacon CT, Chang Y, Jones CA. Exploring organizational context and structur	2008	To examine relationships among organizational context (characte	2. Testing relationships among varia	Quantitative	Yes	(1980) safety climate scale					
Marsteller JA, Wen M, Hsu YJ, et al. Safety Culture in Cardiac Surgical Teams: Data From Five Programs and	2015	We measured and described safety culture in five cardiovascular s	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Martinez W, Etchegaray JM, Thomas EJ, et al. 'Speaking up' about patient safety concerns and unprofessio	2015	To develop and test the psychometric properties of two new surv	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Martowirono K, Wagner C, Bijnen AB. Surgical residents' perceptions of patient safety climate in Dutch tea	2014	This study aims to take a first step in validating the SCS for use am	3. Methodological focus	Quantitative	Yes	SCSu					
Marzolf S, Zekarias B, Tedla K, et al. Continuing professional education in Eritrea taught by local obstetrics	2015	evaluated a course taught by Obstetrics/Gynaecology residents or	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Mascherek AC, Bezzola P, Gehring K, Schwappach DLB. Effect of a two-year national quality improvement p	2016	The aims of the implementation program were to implement com	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Matsubara S, Hagihara A, Nobutomo K, Matsubara S, Hagihara A, Nobutomo K. Development of a patient s	2008	To develop a Japanese scale to measure patient safety climate an	3. Methodological focus	Quantitative	Yes	atsubara et al. 2008					
Mayeng LM, Wolvaardt JE. Patient safety culture in a district hospital in South Africa: An issue of quality. C	2015	The study identified and analysed the factors that influence the pa	1. Descriptive or exploratory assess	Quantitative	Yes	MapSCAT					
McCarthy SE, O'Boyle CA, O'Shaughnessy A, Walsh G. Online patient safety education programme for junio	2016	1) to determine whether the programme improved junior doctors	4. Evaluation of an intervention	Quantitative	Yes	climate" (Madigosky et al. 2006)					
McCaughey D, DelliFraine JL, McGhan G, Bruning NS. The negative effects of workplace injury and illness o	2013	The purpose of this study is to examine whether workplace injurie	2. Testing relationships among varia	Quantitative	Yes	scale (WSS, Hayes et al., 1998)					
McEachan RR, Lawton RJ, O'Hara JK, et al. Developing a reliable and valid patient measure of safety in hos	2014	The aim of the current study is to test the reliability and validity of	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
McFadden KL, Henagan SC, Gowen CR, III. The patient safety chain: Transformational leadership's effect on	2009	to investigate the existence of a patient safety chain for hospitals.	2. Testing relationships among varia	Quantitative	Yes	SAQ					
McFadden KL, Stock GN, Gowen CR, 3rd. Leadership, safety climate, and continuous quality improvement:	2014	We present a research model that shows how transformational le	2. Testing relationships among varia	Quantitative	Yes	SCSu					
McLaughlin N, Winograd D, Chung HR, Van de Wiele B, Martin NA. Impact of the time-out process on safet	2014	This study assessed the impact of the current protocol for the tim	2. Testing relationships among varia	Quantitative	Yes	ultiple - SAQ, HSOPS					
McLeroy, R. D., J. Ingersoll, P. Nielsen and J. Pamplin (2020). "Implementation of Tele-Critical Care at Gene	2020	Implementation of Tele-Critical Care at General Leonard Wood Ar	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
McMullan SP, Thomas-Hawkins C, Shirey MR. Certified Registered Nurse Anesthetist Perceptions of Factor	2017	investigates relationships among CRNA organizational structures (	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Meddings J, Reichert H, Greene MT, et al. Evaluation of the association between Hospital Survey on Patien	2017	Examine the association between hospital units' results for the Ho	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Meeks DW, Lally KP, Carrick MM, et al. Compliance with guidelines to prevent surgical site infections: As si	2011	to assess predictive factors and compliance with surgical site infec	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Menendez CC, Gimeno D, Barrientos-Gutiérrez T, Burau KD, Felknor SA. Safety climate and verbal abuse ar	2012	We examined whether low level of safety factors were associated	2. Testing relationships among varia	Quantitative	Yes	lenendez et al. 2012					
Meurling L, Hedman L, Sandahl C, Fellander-Tsai L, Wallin C-J. Systematic simulation-based team training in	2013	To examine the relationship between simulation-based team train	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Mielke, J., S. De Geest, S. Beckmann, L. Leppla, X. Luta, R.-A. Guerbaai, S. Hunziker and R. Schwendimann (2	2019	The objectives of this cross-sectional study were to translate the B	2. Testing relationships among varia	Quantitative	Yes	ultiple - SAQ, HSOPS					
Mihdawi, M., R. Al-Amer, R. Darwish, S. Randall and T. Afaneh (2020). "The Influence of Nursing Work Envi	2020	This study aimed to examine the relationship between patient saf	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Mikušová V, Rusnáková V, Naďová K, Boroňová J, Beťková M. Patient safety assessment in Slovak hospitals	2012	to find out how healthcare workers perceive patient safety in thei	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Milne JK, Bendaly N, Bendaly L, Worsley J, Fitzgerald J, Nisker J. A measurement tool to assess culture char	2010	We sought to develop a cultural assessment survey (CAS) to asses	3. Methodological focus	Mixed methods	Yes	ural Assessment Su	Yes	Yes			

Moeidh AT, Shah VA, Yusof RZ, Al-Matari EM, Paules A. The relationship between prosocial voice and the p	2015	This study is focused on achieving two objectives; first, examining	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Moghri J, Akbari Sari A, Yousefi M, et al. Is Scores Derived from the Most Internationally Applied Patient Sa	2013	to assess the questionnaire against this bias.	3. Methodological focus	Quantitative	Yes	HSOPS						
Moghri J, Arab M, Saari AA, et al. The Psychometric Properties of the Farsi Version of "Hospital Survey on P	2012	The aim of this study was to assess the validity and reliability of Fa	3. Methodological focus	Quantitative	Yes	HSOPS						
United States	2018	We examined relationships between employee safety climate and	2. Testing relationships among varia	Quantitative	Yes	urvey; Patient Safety Culture Questionnaire (Veterans Affairs)						
Mohseny, M., M. Montazeri, P. Farshidmehr, H. Z. Mahmoudabadi, E. Rahimpour, N. Zamanpour and K. Ha	2019	The main purpose of the present study is to investigate the insight	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Monaca, C., B. Bestmann, M. Kattein, D. Langner, H. Müller and T. Manser (2020). "Assessing Patients' Per	2020	We aimed to develop a measure explicitly focusing on patients' pe	3. Methodological focus	Quantitative	Yes	ions of Safety Culture scale (PaPSC)						
Monsees, E., J. Goldman, A. Vogelsmeier and L. Popejoy (2020). "Nurses as antimicrobial stewards: Recogn	2020	to determine bedside nurses' recognition and performance confid	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Morgan PJ, Cunningham L, Mitra S, et al. Surgical safety checklist: implementation in an ambulatory surgic	2013	The threefold purpose of this study was to add ambulatory-specif	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Mousinho Tavares, A. P., E. C. Carvalho Moura, F. V. Silva Dantas Avelino, V. C. Aguiar Lopes and L. T. Nogu	2018	to evaluate the patient safety culture from the perspective of the	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Moussavi F, Moghri J, Gholizadeh Y, et al. Assessment of patient safety culture among personnel in the hos	2013	The aim of this study was to assess the patient safety culture at Is	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Muething SE, Goudie A, Schoettker PJ, et al. Quality improvement initiative to reduce serious safety events	2012	We conducted a quality improvement initiative to implement cult	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Muralidhar S, Taneja A, Ramesh V. Patient safety culture-perception of health care workers in a tertiary ca	2012	This study attempts to gain knowledge on the practices and attitu	1. Descriptive or exploratory assess	Quantitative	Yes	uralidhar et al., 2012						
Murphy, T., A. Laptook and J. Bender (2018). "Sustained Improvement in Neonatal Intensive Care Unit Safe	2018	This study aimed to measure safety attitudes in the neonatal inter	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Najjar S, Hamdan M, Baillien E, et al. The Arabic version of the hospital survey on patient safety culture: a	2013	The aim of this study is to investigate the psychometric properties	3. Methodological focus	Quantitative	Yes	HSOPS						
Najjar, S., E. Baillien, K. Vanhaecht, M. Hamdan, M. Euwema, A. Vleugels, W. Sermeus, W. Schrooten, J. He	2018	To investigate the relationships between patient safety culture (PS	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Najjar, S., N. Nafouri, K. Vanhaecht and M. Euwema (2018). "Improving patient safety in Palestinian hospit	2018	The aim of this study was to assess patient safety and explore rela	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Nekoei-Moghadam, M., M. Raadabadi and M. Heidarijamebozorgi (2019). "Patient safety culture in univer	2019	The aim of the study was to determine and to compare patient sa	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Nelson S, Stone PW, Jordan S, et al. Patient safety climate: Variation in perceptions by infection prevention	2011	This study explores perceptions of infection preventionists (IPs) an	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Nguyen G, Gambashidze N, Ilyas SA, Pascu D. Validation of the safety attitudes questionnaire (short form 2	2015	the study is aimed at cross-culturally validating the questionnaire	3. Methodological focus	Quantitative	Yes	SAQ						
Nie Y, Mao X, Cui H, He S, Li J, Zhang M. Hospital survey on patient safety culture in China. BMC Health Ser	2013	This study explores the attitudes and perceptions of patient safety	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Nilsson, U., C. Goras, F. Y. Wallentin, A. Ehrenberg and M. Unbeck (2018). "The Swedish Safety Attitudes Q	2018	To validate the Swedish Safety Attitudes Questionnaire-operating	3. Methodological focus	Quantitative	Yes	SAQ						
Noord, I. V.-v., M. Smits, N. C. Zwijnenberg, P. Spreeuwenberg, C. Wagner and I. Verbeek-van Noord (2019	2019	We examined whether a nation-wide patient safety programme (P	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Notaro, K. A. M., A. R. Corrêa, A. Tomazoni, P. K. Rocha and B. F. Manzo (2019). "Safety culture of multidis	2019	analyze the safety culture of multidisciplinary teams from three ne	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Noureldin, M. and M. A. Noureldin (2020). "Reporting frequency of three near-miss error types among hos	2020	The study objectives were to: 1) explore hospital pharmacists' rep	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Occelli P, Quenon JL, Hubert B, et al. Development of a safety culture: initial measurements at six hospitals	2011	Development of a safety culture: initial measurements at six hosp	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Occelli P, Quenon JL, Kret M, et al. Validation of the French version of the Hospital Survey on Patient Safet	2013	To assess the psychometric properties of the French version of the	3. Methodological focus	Quantitative	Yes	HSOPS						
Occelli, P., J.-L. Quenon, M. Kret, S. Domecq, A. Denis, F. Delaperche, O. Claverie, B. Castets-Fontaine, R. Au	2019	To assess the impact of a vignette-based analysis of adverse event	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
O'Connor S, Carlson E. Safety Culture and Senior Leadership Behavior: Using Negative Safety Ratings to Alig	2016	This report describes how staff-designed behavior changes among	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Odell, D. D., C. M. Quinn, R. S. Matulewicz, J. Johnson, K. E. Engelhardt, J. J. Stulberg, A. D. Yang, J. L. Holl a	2019	We examine the safety culture in a statewide hospital quality imp	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Ohrn A, Rutberg H, Nilsen P. Patient safety dialogue: evaluation of an intervention aimed at achieving an in	2011	This paper evaluates the results and changes after 5 years of the P	4. Evaluation of an intervention	Qualitative								Yes
Okafor, C. H., A. C. Ugwu and I. E. Okon (2018). "Effects of Patient Safety Culture on Patient Satisfaction W	2018	To assess the effect of patient safety culture on patient satisfactio	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Okuyama, J. H. H., T. F. Galvão, M. T. L. Crozatti and M. T. Silva (2019). "Health professionals' perception of	2019	To assess patient safety culture in a university hospital.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
O'Leary KJ, Haviley C, Slade ME, Shah HM, Lee J, Williams MV. Improving teamwork: impact of structured i	2011	to assess the impact of an intervention, Structured Inter-Disciplina	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
O'Leary KJ, Wayne DB, Haviley C, Slade ME, Lee J, Williams MV. Improving teamwork: impact of structured	2010	to assess the impact of an intervention, structured inter-disciplina	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Olsen E, Aase K. A comparative study of safety climate differences in healthcare and the petroleum industr	2010	The aim of this article is to compare safety climate in healthcare a	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Olsen E. Exploring the possibility of a common structural model measuring associations between safety cli	2010	to explore the possibility of identifying general safety climate con	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Olson DP, Fields BG, Windish DM. Geographic Localization of Housestaff Inpatients Improves Patient-Provi	2015	assesses whether geographic localization of housestaff patients co	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Olsson C, Forsberg A, Bjersa K. Safety climate and readiness for implementation of evidence and person ce	2016	to explore the safety climate and readiness to implement evidenc	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Omidi, L., R. Akbari, E. Hadavandi and E. Zarei (2018). "An intelligent algorithm for assessing patient safety	2018	The present study aimed to measure PSC and healthcare professio	3. Methodological focus	Quantitative	Yes	HSOPS						
Önler, E. and N. Akyolcu (2019). "Evaluation of operating room staff's attitudes related to patient safety: A	2019	This research was performed in order to evaluate the safety attitu	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Ortiz de Elguea, J., A. Orkaizagirre-Gomara, M. Sanchez De Miguen, F. Urcola-Pardo, C. German-Bes and I.	2019	The aim was to adapt and validate a new version of the Hospital S	3. Methodological focus	Quantitative	Yes	HSOPS						
Ozer, S., H. Sarsilmaz Kankaya, H. Aktas Toptas and F. S. Aykar (2019). "Attitudes Toward Patient Safety an	2019	The goal of this descriptive, cross-sectional study was to evaluate	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Paez K, Schur C, Zhao L, Lucado J. A national study of nurse leadership and supports for quality improveme	2013	This study assessed the perceptions and actions of rural hospital n	1. Descriptive or exploratory assess	Quantitative	Yes	Paez et al. 2013						
Paine LA, Rosenstein BJ, Sexton JB, Kent P, Holzmueller CG, Pronovost PJ. Assessing and improving safety c	2010	To describe the authors' hospital-wide efforts to improve safety cl	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Palmieri, P. A., J. M. Leyva-Moral, D. E. Camacho-Rodriguez, N. Granel-Gimenez, E. W. Ford, K. M. Mathies	2020	The purpose of this study was to develop a target-language HSOP	3. Methodological focus	Quantitative	Yes	HSOPS						
Palomar M, Alvarez-Lerma F, Riera A, et al. Impact of a national multimodal intervention to prevent cathet	2013	to assess its effectiveness after contextual adaptation at large-sca	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Paradiso, L. (2019). "Just culture: It's more than policy." Nursing Management 50(6): 38-45	2019	The article discusses a study at an urban teaching hospital in New	2. Testing relationships among varia	Quantitative	Yes	ure Assessment Tool (JCAT)						
Parry G, Horowitz L, Goldmann D. Patient safety attitudes of paediatric trainee physicians. Quality & safety	2009	To measure the patient safety attitudes of trainee physicians at an	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Patel S, Wu AW. Safety Culture in Indian Hospitals: A Cultural Adaptation of the Safety Attitudes Questionn	2016	to translate and culturally adapt the Safety Attitudes Questionnair	3. Methodological focus	Quantitative	Yes	SAQ						
Patterson MD, Geis GL, LeMaster T, Wears RL. Impact of multidisciplinary simulation-based training on pat	2013	To improve patient safety in a paediatric ED by implementing a m	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Patterson ME, Bogart MS, Starr KR. Associations between perceived crisis mode work climate and poor inf	2015	Associations between perceived crisis mode work climate and poo	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Patterson ME, Pace HA, Fincham JE. Associations between communication climate and the frequency of m	2013	to identify the extent to which perceived communication climate	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Patterson ME, Pace HA. A Cross-sectional Analysis Investigating Organizational Factors That Influence Near	2016	to determine the extent to which punitive work climate, inadequa	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Patterson, P. D., Huang, D. T., Fairbanks, R. J., & Wang, H. E. (2010). The emergency medical services safety	2010	The emergency medical services safety attitudes questionnaire	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Pavan, N. F. P., A. L. P. Magalhães, D. F. Poncio, R. A. Ascari, P. D. Zanini, N. d. S. Knihs and O. M. Silva (201	2019	To evaluate the patient safety culture in the perioperative renal tr	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Perneger TV, Staines A, Kundig F. Internal consistency, factor structure and construct validity of the French	2014	To assess the psychometric properties of the French-language ver	3. Methodological focus	Quantitative	Yes	HSOPS						
Perng SJ, Yu ML. Psychometric testing of an instrument measuring nurse aides' patient safety attitudes. J N	2013	To develop and test the patient safety attitude scale-nurse aide (P	3. Methodological focus	Quantitative	Yes	t safety attitude scale-nurse aides (PSAS-NA)						
Peterson K, Rogers BME, Brosseau LM, et al. Differences in Hospital Managers', Unit Managers', and Health	2016	This article compares hospital managers' (HM), unit managers' (UM	1. Descriptive or exploratory assess	Quantitative	Yes	Peterson et al. 2016						
Petschonek S, Burlison J, Cross C, et al. Development of the just culture assessment tool: measuring the pe	2013	to develop a measure of individual perceptions of just culture for	3. Methodological focus	Quantitative	Yes	lture Assessment Tool (JCAT), HSOPS						

Pettker CM, Thung SF, Norwitz ER, et al. Impact of a comprehensive patient safety strategy on obstetric ad	2009	We implemented a comprehensive strategy to track and reduce a	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Pettker CM, Thung SF, Raab CA, et al. A Comprehensive Obstetrics Patient Safety Program Improves Safety	2011	The purpose of this study was to determine the effect of an obste	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Pfeiffer Y, Manser T. Development of the German version of the Hospital Survey on Patient Safety Culture:	2010	We developed a German version of the HSOPSC and investigated	3. Methodological focus	Quantitative	Yes	HSOPS					
Pinheiro JPA, Uva AdS. Safety climate in the operating room: Translation, validation and application of the	2016	This study presents the validation of the Operating Room Version	3. Methodological focus	Quantitative	Yes	SAQ					
Pinheiro MP, Junior OCS. Evaluation of the patient safety culture within the hospital organization of a univ	2017	addressing the assessment of nurse caretakers about the patient s	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Poley MJ, van der Starre C, van den Bos A, van Dijk M, Tibboel D. Patient safety culture in a Dutch pediatric	2011	to measure the safety climate in our unit, compare it with benchm	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Pousette A, Larsman P, Eklof M, Torner M. The relationship between patient safety climate and occupation	2017	The aim of this study was to investigate the relationship between	2. Testing relationships among varia	Quantitative	Yes	tional Safety Climate Questionnaire, HSOPS					
Prati G, Pietrantoni L. Attitudes to teamwork and safety among Italian surgeons and operating room nurse	2014	The aim of this study was to assess attitudes about teamwork and	1. Descriptive or exploratory assess	Quantitative	Yes	ORMAQ					
Pringle J, Weber RJ, Rice K, Kirisci L, Sirio C. Examination of how a survey can spur culture changes using a	2009	The objective of this study was to investigate safety climates with	1. Descriptive or exploratory assess	Quantitative	Yes	n(VHA) hospital survey on patient safety culture					
Profit J, Etchegaray J, Petersen LA, et al. Neonatal intensive care unit safety culture varies widely. Archives	2012	To describe NICU care giver assessments of safety culture, explore	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Profit J, Etchegaray J, Petersen LA, et al. The Safety Attitudes Questionnaire as a tool for benchmarking saf	2012	To determine whether the dimensions of safety culture measured	3. Methodological focus	Quantitative	Yes	SAQ					
Profit J, Lee HC, Sharek PJ, et al. Comparing NICU teamwork and safety climate across two commonly used	2016	to assess variation in safety and teamwork climate and in the neo	3. Methodological focus	Quantitative	Yes	ultiple - SAQ, HSOPS					
Profit J, Sharek PJ, Amspoker AB, et al. Burnout in the NICU setting and its relation to safety culture. BMJ Q	2014	1) Test the psychometric properties of a brief four-item burnout s	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Putnam LR, Levy SM, Kellagher CM, et al. Surgical resident education in patient safety: where can we impr	2015	We hypothesized that surgical residents perceive the perioperativ	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Putnam LR, Pham DH, Ostovar-Kermani TG, et al. How Should Surgical Residents Be Educated About Patier	2016	We hypothesized that a dedicated, adjunctive resident safety wor	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Qin C, Xie J, Jiang J, Zhen F, Ding S. Reporting among nurses and its correlation with hospital safety culture	2015	the aims of this study were to analyze the behaviors of nurses in C	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Quan M, Wang X, Wu H, et al. Influencing factors on use of standard precautions against occupational exp	2015	To investigate how specific factors, including knowledge, attitude,	2. Testing relationships among varia	Quantitative	Yes	Quan et al. 2015					
Quillivan RR, Burlison JD, Browne EK, Scott SD, Hoffman JM. Patient Safety Culture and the Second Victim E	2016	to assess the influence of patient safety culture on second victim-	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Raeissi, P., N. Reisi and A. A. Nasiripour (2018). "Assessment of Patient Safety Culture in Iranian Academic	2018	The goal of this study was to establish a baseline for PSC in Iranian	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Raftopoulos V, Pavlakis A. Safety climate in 5 intensive care units: a nationwide hospital survey using the G	2013	The explicit need to focus on quality of care underpins the aim of	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Raftopoulos V, Savva N, Papadopoulou M. Safety culture in the maternity units: a census survey using the	2011	to initially evaluate the safety culture and teamwork climate in the	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Rajalatchumi, A., T. S. Ravikumar, K. Muruganandham, M. Thulasingam, K. Selvaraj, M. M. Reddy and B. Jay	2018	The aim of the study was to assess the perceptions on patient safe	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Ramos, R. R. and C. C. Calidgid (2018). "Patient safety culture among nurses at a tertiary government hosp	2018	To assess the patient safety culture (PSC) among nurses at a gover	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Randmaa M, Mårtensson G, Swenne CL, Engström M. SBAR improves communication and safety climate a	2014	We aimed to examine staff members' perceptions of communicat	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Rasmussen K, Pedersen AH, Pape L, Mikkelsen KL, Madsen MD, Nielsen KJ. Work environment influences a	2014	The purpose of this study was to investigate the occurrence of adv	2. Testing relationships among varia	Quantitative	Yes	SAQ					
azzani, B., F. Atashzadeh-Shoorideh, T. Jamshidi, M. Barkhordari-Sharifabad, Z. Lotfi and V. Skerrett (2020)	2020	investigates the effect of education around ethical principles on n	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Real K, Kim S, Conigliaro J. Using a validated health promotion tool to improve patient safety and increase	2013	This study employed the risk perception attitude (RPA) framework	2. Testing relationships among varia	Quantitative	Yes	ahn & Murphy, 2008					
Reis CT, Laguardia J, Vasconcelos AG, Martins M. Reliability and validity of the Brazilian version of the Hosp	2016	This study explored the reliability and validity of the Brazilian vers	3. Methodological focus	Quantitative	Yes	HSOPS					
Reis, C. T., J. Laguardia, C. G. De Barros, P. B. De Araujo Andreoli and M. Martins (2019). "Reliability and va	2019	This study aimed to reassess the psychometric properties of the H	3. Methodological focus	Quantitative	Yes	HSOPS					
Relihan E, Glynn S, Daly D, Silke B, Ryder S. Measuring and benchmarking safety culture: application of the	2009	To assess the safety culture in an acute medical admissions unit (A	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Reszel, J., D. Weiss, A. E. Sprague, D. B. Fell, S. Dunn, M. C. Walker, D. Sidney, M. Taljaard and W. E. Peters	2019	The purpose of our study was to assess the effect of MOREOB on	4. Evaluation of an intervention	Quantitative	Yes	ural Assessment Survey					
Rezaei, S. and S. Salehi (2018). "A Study of the Relationship between Emotional Intelligence and Patient Sa	2018	the aim of this study was to investigate the relationship between	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Ribeliene, J., A. Blazeviciene, R. J. Nadisauskiene, R. Tameliene, A. Kudreviciene, I. Nedzelskiene and J. Mac	2019	The aim of the study was to evaluate the opinion of nurses and m	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Richter JP, McAlearney AS, Pennell ML. Evaluating the effect of safety culture on error reporting: a compar	2015	The study aims were to identify the organizational factors with the	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Richter JP, McAlearney AS, Pennell ML. The influence of organizational factors on patient safety: Examinin	2016	Our aims were (a) to determine whether perceptions of organizat	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Richter JP, McAlearney AS. Targeted implementation of the Comprehensive Unit-Based Safety Program the	2016	The purpose of this study is to identify the significant factors of sa	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Ricklin, M. E., F. Hess and W. E. Hautz (2019). "Patient safety culture in a university hospital emergency de	2019	the aim of the present study was to assess patient safety culture i	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Rigobello MCG, Carvalho R, Guerreiro JM, Motta APG, Atila E, Gimenes FRE. The perception of the patient	2017	The aim of this study was to assess the patient safety climate from	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Robida A. Hospital Survey on Patient Safety Culture in Slovenia: a psychometric evaluation. Int J Qual Healt	2013	To study the psychometric properties of a translated version of th	3. Methodological focus	Quantitative	Yes	HSOPS					
Roqueta Egea F, Tomas Vecina S, Chanovas Borras MR. Patient safety culture in 30 Spanish hospital emerg	2011	1) To ascertain the level of patient safety culture in hospital emerg	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Rosen AK, Singer S, Shibei Z, Shokeen P, Meterko M, Gaba D. Hospital safety climate and safety outcomes:	2010	explored the potential relationship between safety climate and Ve	2. Testing relationships among varia	Quantitative	Yes	PSCHO					
Røyset, B., B. A. Talseth-Palmer, S. Lydersen and P. G. Farup (2019). "Effects of a fall prevention program in	2019	This study assessed the effects of a fall prevention program on the	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Rozenbojm MD, Nichol K, Spielmann S, Holness DL. Hospital unit safety climate: Relationship with nurses' a	2015	Hospital unit safety climate: Relationship with nurses' adherence t	2. Testing relationships among varia	Quantitative	Yes	HSCS					
Sabry, H. A., M. A. Soliman, H. M. Bazaraa and A. A. Hegazy (2020). "Improving patient safety at pediatric i	2020	The researchers attempted to assess healthcare providers' perspe	1. Descriptive or exploratory assess	Qualitative			Yes	Yes			
Sahebalzamani M, Mohammady M. A study of patient safety management in the framework of clinical gov	2014	to evaluate patient safety management in the framework of clinic	1. Descriptive or exploratory assess	Quantitative	Yes	amani & Mohammady, 2014					
Saladino, L., Pickett, L. C., Frush, K., Mall, A., & Champagne, M. T. (2013). Evaluation of a nurse-led safety p	2013	Evaluation of a nurse-led safety program in a critical care unit.	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Saleh AM, Darawad MW, Al-Hussami M. The perception of hospital safety culture and selected outcomes a	2015	to examine nurses' perceptions of the hospital safety culture in Jo	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Salem, M., J. Labib, A. Mahmoud and S. Shalaby (2019). "Nurses' Perceptions of Patient Safety Culture in Ir	2019	to assess the patient safety culture dimensions from the nurses' p	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Sammer, C., L. D. Hauck, C. Jones, J. Zaiback-Aldinger, M. Li and D. Classen (2020). "Examining the Relation	2020	The objective of this study was to examine the relationship betwe	2. Testing relationships among varia	Quantitative	Yes	ultiple - HSOPS, SAQ					
Samsuri SE, Pei Lin L, Fahrni ML. Safety culture perceptions of pharmacists in Malaysian hospitals and heal	2015	To assess the safety attitudes of pharmacists, provide a profile of	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Santa, R., S. Borrero, M. Ferrer and D. Gherissi (2018). "Fostering a healthcare sector quality and safety cul	2018	The purpose of this paper is to identify the most important QSC de	1. Descriptive or exploratory assess	Quantitative	Yes	Santa et al., 2018					
Santana HT, Rodrigues MCS, Do Socorro Nantua Evangelista M. Surgical teams' attitudes and opinions tow	2016	to assess the attitudes and opinions towards surgical safety amon	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Santiago TH, Turrini RN. Organizational culture and climate for patient safety in Intensive Care Units. Revis	2015	To assess the perception of health professionals about patient saf	3. Methodological focus	Quantitative	Yes	ultiple - SAQ, HSOPS					
Sarac C, Flin R, Mearns K, Jackson J. Hospital survey on patient safety culture: psychometric analysis on a S	2011	To investigate the psychometric properties of the Hospital Survey	3. Methodological focus	Quantitative	Yes	HSOPS					
Scherer D, Fitzpatrick JJ. Perceptions of patient safety culture among physicians and RNs in the perioperati	2008	Perceptions of patient safety culture among physicians and RNs in	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Schubert M, Ausserhofer D, Desmedt M, et al. Levels and correlates of implicit rationing of nursing care in	2013	(1) To describe the levels of implicit rationing of nursing care in Sw	2. Testing relationships among varia	Quantitative	Yes	SOS					
Schwappach, D. and A. Richard (2018). "Speak up-related climate and its association with healthcare work	2018	To determine frequencies of healthcare workers (HCWs) speak up	2. Testing relationships among varia	Quantitative	Yes	Patient Safety Questionnaire (SUPS-Q)					
Schwappach, D., G. Sendlhofer, L. Hässler, V. Gombotz, K. Leitgeb, M. Hoffmann, L. Jantscher and G. Brunne	2018	To analyze speaking up behavior and safety climate with a validate	1. Descriptive or exploratory assess	Quantitative	Yes	Patient Safety Questionnaire (SUPS-Q)					
Schwappach, D., G. Sendlhofer, L. P. Kamolz, W. Köle and G. Brunner (2019). "Speaking up culture of medi	2019	to assess students' speaking up behaviors and perceived culture in	1. Descriptive or exploratory assess	Quantitative	Yes	Patient Safety Questionnaire (SUPS-Q)					

Schwartz, M. E., D. E. Welsh, D. E. Paull, R. S. Knowles, L. D. DeLeeuw, R. R. Hemphill, K. E. Essen and G. L. S	2018	This article focuses on the improvement of communication, team	4. Evaluation of an intervention	Quantitative	Yes	nd Safety Climate Survey (TSCS)						
Schwendimann R, Milne J, Frush K, Ausserhofer D, Frankel A, Sexton JB. A closer look at associations betwe	2013	evaluated the association between WRs and caregiver assessment	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Schwendimann R, Zimmermann N, Kung K, Ausserhofer D, Sexton B. Variation in safety culture dimensions	2013	to explore the variability in safety culture dimensions within and b	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Schwonke CRGB, Filho WDL, Lunardi GL, da Silveira RS, Rodrigues MCS, Guerreiro MO. Cultura de segurida	2016	to characterize the safety culture of the hospital institutions in the	1. Descriptive or exploratory assess	Quantitative	Yes	chwonke et al. 2016						
Sculli GL, Fore AM, West P, Neily J, Mills PD, Paull DE. Nursing crew resource management: a follow-up rep	2013	highlights significant cultural and clinical outcomes from the prog	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Sexton JB, Berenholtz SM, Goeschel CA, et al. Assessing and improving safety climate in a large cohort of in	2011	To evaluate the impact of a comprehensive unit-based safety prog	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Sexton JB, Sharek PJ, Thomas EJ, et al. Exposure to Leadership WalkRounds in neonatal intensive care units	2014	The relationship between WR and caregiver assessments of patier	2. Testing relationships among varia	Quantitative	Yes	ultiple - SAQ, HSOPS						
Sexton, J. B., K. C. Adair, M. W. Leonard, T. C. Frankel, J. Proulx, S. R. Watson, B. Magnus, B. Bogan, M. Jam	2018	This cross-sectional survey study evaluated associations between	2. Testing relationships among varia	Quantitative	Yes	SCORE						
Shahian, D. M., X. Liu, L. P. Rossi, E. A. Mort and S.-L. T. Normand (2018). "Safety Culture and Mortality afte	2018	To investigate the association between hospital safety culture and	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Sharif MR, Motaghi M, Farrokhian A. Assessment of patient safety culture in viewpoints of Kashan hospital	2016	the status of patient safety culture in Kashsn hospitals was examir	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Sharp, L., K. Rannus, A. Olofsson, D. Kelly, W. H. Oldenmenger and E. R. g. The (2019). "Patient safety cultu	2019	To explore the differences in perceived patient safety culture in ca	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Shaw KN, Ruddy RM, Olsen CS, et al. Pediatric patient safety in emergency departments: unit characteristi	2009	The goals were (1) to describe emergency department (ED) charac	1. Descriptive or exploratory assess	Quantitative	Yes	wn - no citation provided						
Sheth S, McCarthy E, Kipps AK, et al. Changes in Efficiency and Safety Culture After Integration of an I-PASS	2016	to determine the impact of a multidisciplinary standardized hando	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Shie HG, Lee WC, Hsiao HF, Lin HL, Yang LL, Jung F. Patient safety attitudes among respiratory therapists in	2011	to assess RTs' safety attitudes in several hospital settings.	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Shoushtarian M, Barnett M, McMahon F, Ferris J. Impact of introducing practical obstetric multi-professior	2014	To assess the introduction of Practical Obstetric Multi-professiona	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Shu Q, Cai M, Tao HB, et al. What Does a Hospital Survey on Patient Safety Reveal About Patient Safety Cu	2015	The objective of this study was to examine the strengths and weal	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Siassakos D, Draycott TJ, Crofts JF, Hunt LP, Winter C, Fox R. More to teamwork than knowledge, skill and a	2010	To assess whether team performance in simulated eclampsia is re	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Siassakos D, Fox R, Hunt L, et al. Attitudes toward safety and teamwork in a maternity unit with embedded	2011	The objective of this study was to identify any residual challenges	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Silkens, M. E. W. M., O. A. Arah, C. Wagner, A. J. J. A. Scherpbier, M. J. Heineman and K. M. J. M. H. Lombar	2018	examined the extent to which residents' self-reported patient safe	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Silva EMB, Pedrosa DLL, Leça APCd, Silva DM. Percepção dos profissionais de saúde sobre a cultura de segur	2016	To assess the health professionals' perceptions of the pediatric pa	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Silva ND, Barbosa AP, Padilha KG, Malik AM. Patient safety in organizational culture as perceived by leader	2016	To identify the perceptions of leaderships toward patient safety c	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Silva-Batalha EMSd, Melleiro MM. Patient safety culture in a teaching hospital: differences in perception e	2015	to evaluate the perception of nursing professionals regarding the	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Simons PAM, Houben R, Vlayen A, et al. Does lean management improve patient safety culture? An extens	2015	This paper presents the effects of lean management activities on t	4. Evaluation of an intervention	Mixed methods	Yes	HSOPS	Yes	Yes				Yes
Simons, P., R. Houben, P. Reijnders, M. Pijls-Johannesma, W. Marneffe, A. Vlayen, J. Hellings and D. Vandijc	2018	we constructed a factorial survey measuring safety awareness and	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Simpson KR, Knox GE, Martin M, George C, Watson SR. Michigan Health & Hospital Association Keystone C	2011	to promote safe care practices during labor and birth using the Co	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Simsekler, M. C. E. (2019). "The link between healthcare risk identification and patient safety culture." Inte	2019	As previous research on risk identification practices, as applied to	2. Testing relationships among varia	Mixed methods	Yes	Simsekler 2019	Yes					Yes
Sine DM, Northcutt N. Interactive qualitative assessment of patient safety culture survey scores. Journal of	2008	To compare the results of a qualitative assessment of safety cultu	3. Methodological focus	Mixed methods	Yes	HSOPS		Yes				
Singer S, Lin S, Falwell A, Gaba D, Baker L. Relationship of safety climate and safety performance in hospita	2009	To examine the relationship between measures of hospital safety	2. Testing relationships among varia	Quantitative	Yes	PSCHO						
Singer SJ, Falwell A, Gaba DM, Baker LC. Patient safety climate in US hospitals: variation by management le	2008	To determine whether frontline workers and supervisors perceive	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Singer SJ, Gaba DM, Falwell A, Lin S, Hayes J, Baker L. Patient safety climate in 92 US hospitals: differences	2009	To understand workers' perceptions of safety climate and ways in	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Singer SJ, Hartmann CW, Hanchate A, et al. Comparing Safety Climate between Two Populations of Hospita	2009	To compare safety climate between diverse U.S. hospitals and Vet	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Singer SJ, Rivard PE, Hayes JE, Shokeen P, Gaba D, Rosen A. Improving Patient Care Through Leadership En	2013	A study was conducted to assess the ability to refine, implement,	4. Evaluation of an intervention	Quantitative	Yes	PSCHO						
Singh HP, Durani P, Dias JJ. Enhanced Morbidity and Mortality Meeting and Patient Safety Education for Sp	2015	to enhance the delivery of patient safety education.	4. Evaluation of an intervention	Quantitative	Yes	afety Attitudes and Climate Questionnaire,						
Sirriyeh R, Lawton R, Armitage G, Gardner P, Ferguson S. Safety subcultures in health-care organizations ar	2012	to explore the perspectives of staff who are tasked in some way w	1. Descriptive or exploratory assess	Qualitative			Yes					
Sklebar I, Habek D, Jurkovic I. The correlation between patient safety culture and regional anesthesia deve	2013	the purpose of detecting differences in patient safety culture amo	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Sklebar I, Mustajbegovic J, Sklebar D, et al. HOW TO IMPROVE PATIENT SAFETY CULTURE IN CROATIAN HO	2016	to assess the state of PSC in Croatian hospitals and compare it wit	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Smeds Alenius L, Tishelman C, Runesdotter S, Lindqvist R. Staffing and resource adequacy strongly related	2014	how RNs assess the safety of patient care at their workplace and h	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Smiley, K., L. Ofori, C. Spangler, R. Acquaah-Arhin, D. Deh, J. Enos, S. Manortey, F. Baiden, S. Finlayson, R. F	2019	we evaluated staff and patient perceptions of safety and quality, a	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Smith DR, Muto T, Sairenchi T, et al. Examining the Dimensions of Hospital Safety Climate and Psychosocia	2011	to examine the dimensions, reliability, and main loading factors as	3. Methodological focus	Quantitative	Yes	HSCS						
Smith DR, Muto T, Sairenchi T, et al. Hospital Safety Climate, Psychosocial Risk Factors and Needlestick Inju	2010	To investigate the interactions between safety climate, psychosoc	2. Testing relationships among varia	Quantitative	Yes	HSCS						
Smith DR, Zhao I, Wang L, Ho A. Dimensions and reliability of a hospital safety climate questionnaire in Chi	2013	the aim of the current study was to examine the dimensions and r	3. Methodological focus	Quantitative	Yes	HSCS						
Smith SA, Yount N, Sorra J. Exploring relationships between hospital patient safety culture and Consumer F	2017	This study examines whether patient safety culture perceptions o	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Smits M, Christiaans-Dingelhoff I, Wagner C, Wal G, Groenewegen PP. The psychometric properties of the	2008	The aim of this study was to examine the underlying dimensions a	3. Methodological focus	Quantitative	Yes	HSOPS						
Smits M, Wagner C, Spreeuwenberg P, Timmermans DR, van der Wal G, Groenewegen PP. The role of patie	2012	the relationship between specialty and patient safety is mediated	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Smits M, Wagner C, Spreeuwenberg P, van der Wal G, Groenewegen PP. Measuring patient safety culture:	2009	To test the claim that the Hospital Survey on Patient Safety Cultur	3. Methodological focus	Quantitative	Yes	HSOPS						
Snijders C, Kollen BJ, van Lingen RA, Fetter WP, Molendijk H. Which aspects of safety culture predict incide	2009	The objective of this study was to examine which aspects of safety	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Soh SE, Barker A, Morello R, Dalton M, Brand C. Measuring safety climate in acute hospitals: Rasch analysis	2016	to extend the psychometric evaluations of the SAQ by using Rasch	3. Methodological focus	Quantitative	Yes	SAQ						
Soh SE, Morello R, Rifat S, Brand C, Barker A. Nurse perceptions of safety climate in Australian acute hospi	2017	The aim of the present study was to explore nurse perceptions of	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Sollid SJ, Strand K, Soreide E. Percutanous dilatational tracheotomy in the ICU: a Norwegian survey focusin	2008	This study was performed to obtain an impression of risk percepti	1. Descriptive or exploratory assess	Qualitative			Yes					
Songur, C., O. Ozer, C. Gun and M. Top (2018). "Patient Safety Culture, Evidence-Based Practice and Perfor	2018	This study aimed to identify the effects of nurses' patient safety c	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Sorensen JL, van der Vleuten C, Rosthoj S, et al. Simulation-based multiprofessional obstetric anaesthesia t	2015	To investigate the effect of in situ simulation (ISS) versus off-site s	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Sorokin R, Riggio JM, Moleski S, Sullivan J. Physicians-in-training attitudes on patient safety: 2003 to 2008.	2011	We assess whether trainees' patient safety attitudes have evolved	1. Descriptive or exploratory assess	Quantitative	Yes	Sorokin et al. 2005						
Sorra J, Khanna K, Dyer N, Mardon R, Famolaro T. Exploring relationships between patient safety culture a	2014	The purpose of this study was to examine relationships among 2 A	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Sorra J, Nieva V, Fastman BR, Kaplan H, Schreiber G, King M. Staff attitudes about event reporting and pati	2008	The goal of this study was to assess transfusion service staff attitu	1. Descriptive or exploratory assess	Quantitative	Yes	sion service error reporting and safety culture survey						
Sorra JS, Dyer N. Multilevel psychometric properties of the AHRQ hospital survey on patient safety culture	2010	to examine the multilevel psychometric properties of the survey.	3. Methodological focus	Quantitative	Yes	HSOPS						
Speroff T, Nwosu S, Greevy R, et al. Organisational culture: variation across hospitals and connection to pa	2010	To determine if an organisational group culture shows better align	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Squires M, Tourangeau A, Laschinger HKS, Doran D. The link between leadership and safety outcomes in h	2010	To test and refine a model examining relationships among leaders	2. Testing relationships among varia	Quantitative	Yes	SCSu						
Sreeramoju, P., L. Dura, M. E. Fernandez, A. Minhajuddin, K. Simacek, T. B. Fomby and B. N. Doebbeling (20	2018	We evaluated the impact of a social change intervention on health	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Staines, A., E. Lecureux, P. Rubin, C. Baralon and A. Farin (2019). "Impact of TeamSTEPPS on patient safety	2019	To assess the impact of implementation of the TeamSTEPPS team	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						

Stark, H. E., L. V. Graudins, T. M. McGuire, C. Y. Y. Lee and M. J. Duguid (2020). "Implementing a sustainabl	2020	To determine the feasibility of implementing the World Health Or	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Stepaniak PS, Heij C, Buise MP, Mannaerts GHH, Smulders JF, Nienhuijs SW. Bariatric Surgery with Operati	2012	the effects on patient outcomes, teamwork and safety climate, an	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Stewart, D., B. Thomas, K. MacLure, A. Pallivalapila, W. El Kassem, A. Awaisu, J. S. McLay, K. Wilbur, K. Wilk	2018	The study aimed to quantify and explain key aspects of patient saf	1. Descriptive or exploratory assess	Mixed methods	Yes	HSOPS		Yes			
Steyrer J, Latzke M, Pils K, Vetter E, Strunk G. Development and validation of a patient safety culture quest	2011	To describe the development of a German-language instrument fo	3. Methodological focus	Quantitative	Yes	y Climate Questionnaire (VSCQ)					
Steyrer J, Schiffinger M, Huber C, Valentin A, Strunk G. Attitude is everything? The impact of workload, saf	2013	this study investigates to which extent production pressure (i.e., ir	2. Testing relationships among varia	Quantitative	Yes	y Climate Questionnaire (VSCQ)					
Stock GN, McFadden KL. Improving service operations: linking safety culture to hospital performance. Jour	2017	to examine the relationship between patient safety culture and ho	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Stolldorf, D. P., J. L. Schnipper, A. S. Mixon, M. Dietrich and S. Kripalani (2019). "Organisational context of h	2019	we evaluated and compared organisational contextual factors and	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Storm, M., J. Schulz and K. Aase (2018). "Patient safety in transitional care of the elderly: effects of a quasi-	2018	The study objective was to assess the effects of an interorganisati	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Stoyanova, R., R. Dimova, M. Tarnovska and T. Boeva (2018). "Linguistic Validation and Cultural Adaptation	2018	The present paper aims to illustrate the process of linguistic valida	3. Methodological focus	Quantitative	Yes	HSOPS					
Stoyanova, R., R. Dimova, M. Tarnovska, T. Boeva, R. Dimov and I. Doykov (2019). "Comparing patient safe	2019	The aim of the study is to assess the status of PSC in Bulgarian hos	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Stringer B, Haines AT, Goldsmith CH, Berguer R, Blythe J. Is use of the hands-free technique during surgery	2009	assessed whether "management support," the most commonly m	2. Testing relationships among varia	Quantitative	Yes	HSCS					
Sturm, H., M. A. Rieger, P. Martus, E. Ueding, A. Wagner, M. Holderried, J. Maschmann, A. Hammer, M. Ho	2019	This study aims to understand the relation between survey-derive	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Sujan MA. A novel tool for organisational learning and its impact on safety culture in a hospital dispensary.	2012	The aim of the work reported in this paper was to prototype a nov	4. Evaluation of an intervention	Qualitative			Yes				
Suliman M, Aljezawi M, AlBashtawy M, Fitzpatrick J, Aloush S, Al-Awamreh K. Exploring Safety Culture in Jo	2017	This study explored Jordanian nurses' perceptions of the culture o	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Sze-Ee, S., R. Morello, S. Rifat, C. Brand and A. Barker (2018). "Nurse perceptions of safety climate in Austr	2018	The aim of the present study was to explore nurse perceptions of	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Tabrizchi N, Sedaghat M. The first study of patient safety culture in Iranian primary health centers. Acta m	2012	To measure the patient safety culture in primary health centers.	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Taher S, Hejaili F, Karkar A, et al. Safety climate in dialysis centers in Saudi Arabia: a multicenter study. J Pa	2014	The aim of this study was to assess the safety climate as perceive	2. Testing relationships among varia	Quantitative	Yes	SCSu					
Taylor JA, Dominici F, Agnew J, Gerwin D, Morlock L, Miller MR. Do nurse and patient injuries share comm	2012	To examine the association of unit-level safety climate and specifi	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Teleş, M. and S. Kaya (2019). "Staff perceptions of patient safety culture in general surgery departments in	2019	The objective of this study was to evaluate patient safety culture a	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Tereanu, C., G. Sampietro, F. Sarnataro, D. Siscanu, R. Palaria, V. Savin, T. Cliscovscaia, V. Pislaru, V. Oglind	2018	This study aims to identify which areas of the patient safety cultur	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Tereanu, C., S. A. Smith, M. S. Ghelase, G. Sampietro, A. Molnar, A. Dragoescu, F. L. Furtunescu, C. Stanesc	2018	To explore psychometrics of the Romanian version of the Hospita	3. Methodological focus	Quantitative	Yes	HSOPS					
Theeb Moeidh A, Abdulaziz Shah F, Mohammed Al-Matari E. The relationship between prosocial voice and	2015	examines the direct linkage between prosocial voice and patient s	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Thomas L, Galla C. Republished: Building a culture of safety through team training and engagement. Postg	2013	Our goal was to build a culture of patient safety within a structure	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Thomas-Hawkins C, Flynn L. Patient safety culture and nurse-reported adverse events in outpatient hemod	2015	to examine relationships between patient safety culture and nurs	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Thomson, H., A. Tourangeau, L. Jeffs and M. Puts (2018). "Factors affecting quality of nurse shift handover	2018	The aim of this study was to explore and test factors hypothesized	2. Testing relationships among varia	Quantitative	Yes	SAQ					
Thorp J, Baqai W, Witters D, et al. Workplace engagement and workers' compensation claims as predictors	2012	Demonstrate the relationship between employee engagement an	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Tiessen B. On the journey to a culture of patient safety. Healthcare quarterly (Toronto, Ont.). 2008;11(4):5	2008	numerous patient safety initiatives were implemented.	4. Evaluation of an intervention	Quantitative	Yes	PSCHO					
Timmel J, Kent PS, Holzmueller CG, Paine L, Schulick RD, Pronovost PJ. Impact of the Comprehensive Unit-l	2010	A culture of teamwork and learning from mistakes are universally	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Tlili, M. A., W. Aouicha, M. Ben Rejeb, J. Sahli, M. Ben Dhiab, S. Chelbi, A. Mtiraoui, H. S. Laatiri, T. Ajmi, C.	2020	This study aimed to assess patient safety culture (PSC) in intensive	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Tomazoni A, Kuerten Rocha P, Batista Ribeiro M, Serapião LS, de Souza S, Figueiredo Manzo B. Perception	2017	To describe patient safety in the perception of nursing and medica	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Tomazoni A, Rocha PK, de Souza S, Anders JC, de Malfussi HF. Patient safety culture at Neonatal Intensive	2014	To verify the assessment of the patient safety culture according to	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Tomazoni A, Rocha PK, Kusahara DM, Souza AIJd, Macedo TR. Evaluation of the patient safety culture in ne	2015	This quantitative, survey type study aimed to analyze the patient s	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Top M, Tekingunduz S. Patient Safety Culture in a Turkish Public Hospital: A Study of Nurses' Perceptions A	2015	The objective of this study was to investigate nurses' perceptions	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Toso GL, Golle L, Magnago TS, et al. Patient safety culture in hospitals within the nursing perspective. Revis	2016	Evaluate the atmosphere regarding patient safety from the perspe	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Tourani S, Hassani M, Ayoubian A, Habibi M, Zaboli R. Analyzing and Prioritizing the Dimensions of Patient	2015	an attempt to analyze the cultural aspects of patient safety in the	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Turkmen E, Baykal U, Intepeler SS, Altuntas S. Nurses' perceptions of and factors promoting patient safety	2013	This descriptive and cross-sectional study	1. Descriptive or exploratory assess	Quantitative	Yes	ulture Scale (Turkmen et al., 2011)					
Turnberg W, Daniell W. Evaluation of a healthcare safety climate measurement tool. Journal of safety rese	2008	evaluated the psychometric properties of the Gershon tool when	3. Methodological focus	Quantitative	Yes	HSCS					
Turunen H, Partanen P, Kvist T, Miettinen M, Vehvilainen-Julkunen K. Patient safety culture in acute care: a	2013	We explored their views on the patient safety culture in four acut	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Ugurluoglu O, Ugurluoglu E, Payziner PD, Ozatkan Y. Patient Safety Culture: Sample of a University Hospita	2012	To assess health personnel perspectives of patient safety culture i	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Ullman AJ, Long DA, Rickard CM. Prevention of central venous catheter infections: A survey of paediatric IC	2014	To assess nursing knowledge of evidence based guidelines to prev	2. Testing relationships among varia	Quantitative	Yes	tream Infection Prevention Practices Survey					
Ulrich B, Kear T. Patient Safety Culture in Nephrology Nurse Practice Settings: Initial Findings. Nephrology (	2014	to assess patient safety cultures in nephrology nurse practice sett	1. Descriptive or exploratory assess	Quantitative	Yes	e Survey on Patient Safety Culture, HSOPS					
Unver, G. T. and A. K. H. Seren (2018). "Defining the patient safety attitudes and influencing factors of heal	2018	To determine patient safety attitudes of midwives, nurses and phy	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ					
Upadhyay, S., R. Weech-Maldonado, C. H. Lemak, A. L. Stephenson and D. G. Smith (2019). "Hospital staffi	2019	The processes by which registered nurse (RN) and hospitalist staffi	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Valentin A, Schiffinger M, Steyrer J, Huber C, Strunk G. Safety climate reduces medication and dislodgemer	2013	To assess the frequency and contributing factors of medication an	2. Testing relationships among varia	Quantitative	Yes	y Climate Questionnaire (VSCQ)					
van der Nelson HA, Siassakos D, Bennett J, et al. Multiprofessional team simulation training, based on an o	2014	This interrupted time-series study evaluated the impact of multipr	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
van Noord I, de Bruijne MC, Twisk JW. The relationship between patient safety culture and the implement	2010	to investigate the association between 11 patient safety culture d	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Velji K, Baker GR, Fancott C, et al. Effectiveness of an Adapted SBAR Communication Tool for a Rehabilitati	2008	evaluated the effectiveness of an adapted SBAR tool for both urge	4. Evaluation of an intervention	Quantitative	Yes	HSOPS					
Verbeek-Van Noord I, Wagner C, Van Dyck C, Twisk JW, De Bruijne MC. Is culture associated with patient s	2014	To describe the patient safety culture of Dutch emergency depart	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Vifladt A, Simonsen BO, Lydersen S, Farup PG. The association between patient safety culture and burnout	2016	To study the associations between registered nurses' (RNs) percep	2. Testing relationships among varia	Quantitative	Yes	HSOPS					
Vlayen A, Hellings J, Claes N, Peleman H, Schrooten W. A nationwide hospital survey on patient safety cult	2012	To measure patient safety culture in Belgian hospitals and to exam	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Vlayen A, Schrooten W, Wami W, et al. Variability of patient safety culture in Belgian acute hospitals. J Pat	2015	The aim of this study was to measure differences in safety culture	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Wagner C, Smits M, Sorra J, Huang CC. Assessing patient safety culture in hospitals across countries. Int J C	2013	to examine similarities and differences in hospital patient safety c	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Wagner, A., A. Hammer, T. Manser, P. Martus, H. Sturm, M. A. Rieger and C. WorkSafeMed (2018). "Do Oc	2018	The aim of this study was to consider and test relevant predictors	2. Testing relationships among varia	Quantitative	Yes	Safety Culture Survey (Wagner et al., 2018)					
Wagner, A., M. A. Rieger, T. Manser, H. Sturm, J. Hardt, P. Martus, C. Lessing and A. Hammer (2019). "Heal	2019	The present study investigated hospital staffs' perceptions of four	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS					
Wagner, A., M. Michaelis, E. Luntz, A. Wittich, M. Schrappe, C. Lessing and M. A. Rieger (2018). "Assessme	2018	The aim of this feasibility study in a German university hospital wa	3. Methodological focus	Quantitative	Yes	Safety Culture Survey (Wagner et al., 2018)					
Wailling, J., B. Robinson and M. Coombs (2019). "Surveillance, anticipation and firefighting: Perspectives o	2019	This study explored how doctors, nurses and managers working in	1. Descriptive or exploratory assess	Qualitative			Yes	Yes			
Wakefield JG, McLaws ML, Whitby M, Patton L. Patient safety culture: factors that influence clinician invol	2010	To develop an understanding of the factors that influence patient	2. Testing relationships among varia	Quantitative	Yes	- Patient safety behavioural intent (PSBI)					
Wallin CJ, Kalman S, Sandelin A, Farnert ML, Dahlstrand U, Jylli L. Creating an environment for patient safe	2015	This pilot project aimed to implement and test whether a new int	4. Evaluation of an intervention	Quantitative	Yes	SAQ					
Walpola RL, Chen TF, Fois RA, Ashcroft DM, Lalor DJ. Evaluation of a survey tool to measure safety climate	2017	To evaluate the construct validity and internal consistency of a su	3. Methodological focus	Quantitative	Yes	Safety Climate Questionnaire					

Walston SL, Al-Omar BA, Al-Mutari FA. Factors affecting the climate of hospital patient safety: a study of h	2010	The purpose of this paper is to describe three organizational dime	2. Testing relationships among varia	Quantitative	Yes	Walston et al. 2010						
Wami SD, Demssie AF, Wassie MM, Ahmed AN. Patient safety culture and associated factors: A quantitativ	2016	the aim of this study was to assess the level of patient safety cultu	1. Descriptive or exploratory assess	Mixed methods	Yes	HSOPS	Yes					
Wang X, Liu K, You LM, et al. The relationship between patient safety culture and adverse events: a questi	2014	to describe nurses' perception of patient safety culture and freque	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Wang, K. Y., C. C. Chou and J. C. Y. Lai (2019). "A structural model of total quality management, work value	2019	This study explores the impact of total quality management on pa	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Wang, Y., H. Han, L. Qiu, C. Liu, Y. Wang and W. Liu (2019). "Development of a patient safety culture scale (	2019	This study aimed to develop a patient safety culture (PSC) scale fo	3. Methodological focus	Quantitative	Yes	and Child Health Institutions (PSCS-MCHI) (Wang et al. 2019)						
Waterson P, Griffiths P, Stride C, Murphy J, Hignett S. Psychometric properties of the Hospital Survey on Pa	2010	This study reports an assessment of the psychometric properties a	3. Methodological focus	Quantitative	Yes	HSOPS						
Watts BV, Percarpio K, West P, Mills PD. Use of the Safety Attitudes Questionnaire as a measure in patient	2010	We report on an attempt to use the Safety Attitude Questionnaire	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Weaver SJ, Rosen MA, DiazGranados D, et al. Does teamwork improve performance in the operating room?	2010	Does teamwork improve performance in the operating room? A m	4. Evaluation of an intervention	Quantitative	Yes	tiple - HSOPS, ORMAQ						
Weaver SJ, Weeks K, Pham JC, Pronovost PJ. On the CUSP: Stop BSI: evaluating the relationship between c	2014	to investigate the relationship between intensive care unit (ICU) p	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Weaver, S. J., S. E. Mossburg, M. Pillari, P. S. Kent and E. L. D. Biddison (2018). "Examining Variation in Mer	2018	This study explored similarities and differences in the views on tea	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Weng R-H, Chen J-C, Pong L-J, Chen L-M, Lin T-C. The Impact of Market Orientation on Patient Safety Clima	2016	explores how market orientation affects the climate of patient saf	2. Testing relationships among varia	Quantitative	Yes	climate scale (Weng et al., 2012)						
Weng RH, Huang CY, Huang JA, Wang MH. The cross-level impact of patient safety climate on nursing inno	2012	To explore the cross-level effects of the four dimensions of patien	2. Testing relationships among varia	Quantitative	Yes	climate scale (Weng et al., 2012)						
Weng SJ, Kim SH, Wu CL. Underlying influence of perception of management leadership on patient safety (	2017	We aim to draw insights on how medical staff's perception of mar	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Wijaya, M. I., A. R. Mohamad and M. Hafizurrachman (2020). "Shift schedule realignment and patient safe	2020	The purpose of this paper is to assess the association between shi	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Wolf, F. A., Way, L. W., & Stewart, L. (2010). The efficacy of medical team training: improved team perform	2010	The efficacy of medical team training: improved team performanc	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Wu JC, Tung TH, Chen PY, Chen YL, Lin YW, Chen FL. Determinants of workplace violence against clinical ph	2015	understanding how organizational factors influence workplace vio	2. Testing relationships among varia	Quantitative	Yes	(1980) safety climate scale						
Wu WT, Wu YL, Hou SM, et al. Examining the effects of an interprofessional crew resource management tr	2016	The study aimed to investigate the effectiveness of this interventi	4. Evaluation of an intervention	Quantitative	Yes	ude Survey (HFAS, Grogan et al., 2004)						
Wu Y, Fujita S, Seto K, et al. The impact of nurse working hours on patient safety culture: a cross-national s	2013	to clarify the impact of long nurse working hours on PSC in Japan,	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Wu, C. F., H. H. Wu, Y. C. Lee and C. H. Huang (2019). "What attributes determine overall satisfaction in pa	2019	This study assessed the attributes of the overall perception of (sat	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Xu, X. P., D. N. Deng, Y. H. Gu, C. S. Ng, X. Cai, J. Xu, X. S. Zhang, D. G. Ke, Q. H. Yu and C. K. Chan (2018). "C	2018	This paper evaluates the current patient safety culture in an exper	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Yang, R., G. A. Pepper, H. Wang, T. Liu, D. Wu and Y. Jiang (2020). "The mediating role of power distance a	2020	(1) Identify nurses' perceptions of patient safety emphasis, face-s	2. Testing relationships among varia	Quantitative	Yes	ate scale (Blegen et al., 2005)						
Yari, S., M. H. Naseri, H. Akbari, S. Shahsavari and H. Akbari (2019). "Interaction of Safety Climate and Safe	2019	The aim of this study isto investigate the interaction between safe	2. Testing relationships among varia	Quantitative	Yes	sment), Zeidi et al. 2012 (Safety Climate Questionnaire)						
Yilmaz Z, Goris S. Determination of the patient safety culture among nurses working at intensive care units	2015	To determine the patient safety culture among nurses working at t	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Yu, B., C.-F. Wen, H.-L. Lo, H.-H. Liao and P.-C. Wang (2020). "Improvements in patient safety culture: a nat	2020	To assess national trends in patient safety culture in Taiwan	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Yu, M., H. Y. Lee, G. Sherwood and E. Kim (2018). "Nurses' handoff and patient safety culture in perinatal c	2018	To examine nursing handoff, identify causes of handoff errors, eva	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Zadow AJ, Dollard MF, McLinton SS, Lawrence P, Tuckey MR. Psychosocial safety climate, emotional exhau	2017	to examine how psychosocial safety climate (PSC) influences the c	2. Testing relationships among varia	Quantitative	Yes	social safety climate (PSC)						
Zadvinskis, I. M., P. J. Salsberry, E. M. Chipps, E. S. Patterson, L. A. Szalacha and K. A. Crea (2018). "An Explo	2018	This descriptive, cross-sectional study used the Safety Attitudes Q	2. Testing relationships among varia	Quantitative	Yes	SAQ						
Zaheer S, Ginsburg L, Chuang YT, Grace SL. Patient safety climate (PSC) perceptions of frontline staff in acu	2015	The broader organizational literature suggests that ease of report	2. Testing relationships among varia	Quantitative	Yes	CAN PSCS						
Zaheer, S., L. R. Ginsburg, H. J. Wong, K. Thomson and L. Bain (2018). "Importance of safety climate, teamv	2018	The impact of staff perceptions of safety climate (ie, senior and su	2. Testing relationships among varia	Quantitative	Yes	CAN PSCS						
Zarei E, Khakzad N, Reniers G, Akbari R. On the relationship between safety climate and occupational burn	2016	to investigate and establish a relationship between safety climate	2. Testing relationships among varia	Quantitative	Yes	Nurses (SCN, Sarasangi et al., 2015)						
Zecevic AA, Li AH, Ngo C, Halligan M, Kothari A. Improving safety culture in hospitals: Facilitators and barri	2017	The purpose of this study was to assess the facilitators and barrier	1. Descriptive or exploratory assess	Mixed methods	Yes	MSI	Yes	Yes				
Zech A, Gross B, Jasper-Birzele C, et al. Evaluation of simparteam - a needs-orientated team training forma	2017	Evaluation of simparteam - a needs-orientated team training form	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Zenere A, Zanolin ME, Negri R, Moretti F, Grassi M, Tardivo S. Assessing safety culture in NICU: psychomet	2016	The aims of the study are to verify the psychometric properties of	3. Methodological focus	Quantitative	Yes	SAQ						
Zhang, D., M. Liao, Y. Zhou and T. Liu (2020). "Quality control circle: a tool for enhancing perceptions of pa	2020	To explore whether quality control circle (QCC) is associated with	4. Evaluation of an intervention	Quantitative	Yes	HSOPS						
Zhang, F., L. Tian, X. Shang, X. Li, R. Xue, S. Cheng and C. Chen (2018). "Exploring relationships between firs	2018	To examine first-line nurse managers' perceptions of safety cultur	1. Descriptive or exploratory assess	Quantitative	Yes	SAQ						
Zhang, X., Q. Li, Y. Guo and S.-Y. Lee (2019). "From organisational support to second victim-related distress	2019	This cross-sectional correlation study aimed to explore the associa	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Zhao, C., Q. Chang, X. Zhang, Q. Wu, N. Wu, J. He and Y. Zhao (2019). "Evaluation of safety attitudes of hos	2019	The objectives of this study are to test the psychometric propertie	3. Methodological focus	Quantitative	Yes	SAQ						
Zhong, X., Y. Song, C. Dennis, D. J. Slovensky, L. Y. Wei, J. Chen and J. Ji (2019). "Patient safety culture in Pe	2019	This study aims to assess the patient safety culture in Peking Univ	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						
Zhou P, Bundorf MK, Gu J, He X, Xue D. Survey on patient safety climate in public hospitals in China. BMC H	2015	This study aimed to describe staff's perceptions of patient safety c	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Zhou, P., F. Bai, H. Q. Tang, J. Bai, M. Q. Li and D. Xue (2018). "Patient safety climate in general public hosp	2018	This study analysed differences in the perceived patient safety clir	1. Descriptive or exploratory assess	Quantitative	Yes	PSCHO						
Zhu J, Li L, Zhao H, Han G, Wu AW, Weingart SN. Development of a patient safety climate survey for Chines	2014	To develop and evaluate a patient safety climate instrument for h	3. Methodological focus	Quantitative	Yes	e Hospital Survey on Patient Safety Climate						
Zhu J, Li L, Zhou Z, Lou Q, Wu AW. Variations in Patient Safety Climate in Chinese Hospitals. J Patient Saf. 2	2016	We examined variations in safety climate by work area and job ca	2. Testing relationships among varia	Quantitative	Yes	HSOPS						
Zhu, J. (2019). "Measurement equivalence of patient safety climate in Chinese hospitals: can we compare a	2019	This study aims to examine the degree of measurement equivalen	3. Methodological focus	Quantitative	Yes	e Hospital Survey on Patient Safety Climate						
Zimmermann N, Kueng K, Sereika SM, Engberg S, Sexton B, Schwendimann R. Assessing the safety attitude	2013	The objective of this study was to assess the psychometric proper	3. Methodological focus	Quantitative	Yes	SAQ						
Zohar D, Werber YT, Marom R, Curlau B, Blondheim O. Modifying head nurse messages during daily conve	2017	The current study adapts an empirically supported strategy develo	4. Evaluation of an intervention	Quantitative	Yes	PSCHO						
Zullo MD, McCarroll ML, Mendise TM, et al. Safety culture in the gynecology robotics operating room. Jou	2014	To measure the safety culture in the robotics surgery operating ro	4. Evaluation of an intervention	Quantitative	Yes	SAQ						
Zwijnenberg NC, Hendriks M, Hoogervorst-Schilp J, Wagner C. Healthcare professionals' views on feedback	2016	to explore healthcare professionals' views on the feedback of a pa	1. Descriptive or exploratory assess	Quantitative	Yes	HSOPS						

Name of survey, tool or approach	New survey developed by study authors?	Quantitative, Qualitative or Mixed methods assessment	Details	As used in	Leadership and their support for safety	Perceptions of safety	Teamwork and collaboration	Safety systems	Prioritisation of safety	Resources and constraints	Reporting and just culture	Openness	Learning and improvement	Awareness of human limits	Wellbeing	Other	Details of other	Number of dimensions covered by tool
SAQ + interviews analysed using framework anaysis method		Mixed meth	Following survey, semi-structured interviews were c	Abdi Z, Delgoshaei B, Ravaghi H, Abbasi M, Heyrani A	1	1	1	1		1	1	1	1	1	1	1	Recommendations for improvement	11
Pharmacy Survey on Patient Safety Culture (PSOPSC) - based on HSC		Quantitative	Originally for community pharmacy it has been adap	Jia PL, Zhang LH, Zhang MM, et al. Safety culture in a	1	1	1	1	1	1	1	1	1	1		1	Documenting mistakes; patient counseling	11
SAQ and Safety Climate Scale + interviews		Mixed meth	Interviews followed survey (which included open-en	Allen S, Chiarella M, Homer CS. Lessons learned from	1	1	1	1		1	1	1	1	1	1			10
Patient Safety Behavioural Intent (PSBI) - developed by W	Yes	Quantitative	constructs based in part on the SAQ: 1. Prior	Wakefield JG, McLaws ML, Whitby M, Patton L. Patie	1	1		1	1	1	1	1	1	1	1	1	Norms and behaviours of colleagues and the individual.	11
HSOPS + analysis of open responses and interviews		Mixed meth	Interviews were based on a thematic guide comprisir	Boussat, B., K. Kamalanavin and P. François (2018). "T	1	1	1	1	1	1	1	1	1		1			10
HSOPS including open responses + interviews		Mixed meth	Interview schedule not focused on dimensions of safe	Campione, J. and T. Famolaro (2018). "Promising Pract	1	1	1	1	1	1	1	1	1					9
Operating Room Management Attitudes Questionnaire (ORMAQ)		Quantitative	Fifty-six item questionnaire covering eight dimensio	Geraghty, A., Reid, S., & McIlhenny, C. (2014). Exami	1		1	1	1	1	1	1		1	1	1	Values	10
new Chinese SAQ - developed by Taiwan Joint Commission		Quantitative	The new Chinese SAQ has 46 items with additional c	Huang, C.-H., H.-H. Wu and Y.-C. Lee (2018). "The pe	1	1	1	1		1	1	1	1	1	1			10
Document analysis + untitled survey - developed by Chapp	Yes	Mixed meth	To assess radiation safety culture, analysis of safety	Chapple CL, Bradley A, Murray M, et al. Radiation safety culture in the UK					1	1								3
Safety Attitudes Questionnaire - developed by Sexton et al. 2006		Quantitative	A 60 item survey of health professionals attitudes to	Sexton JB, Sharek PJ, Thomas EJ, et al. Exposure to Le	1	1	1	1		1	1	1	1	1	1			10
Focus groups + HSOPS		Mixed meth	Focus group interviews first conducted, with the inte	Elder NC, Brungs SM, Nagy M, Kudel I, Render ML. I	1	1	1	1	1	1	1	1	1					9
Patient Safety Climate in Healthcare Organizations - developed by Sin		Quantitative	Developed from the PSCI, this 38 item survey evalu	Singer S, Lin S, Falwell A, Gaba D, Baker L. Relations	1	1	1	1	1	1	1	1	1			1	Flexibility	10
Patient and Occupational Safety Culture Survey - develope	Yes	Quantitative	Based on both the HSOPS and SAQ, this survey asse	Wagner, A., M. Michaelis, E. Luntz, A. Wittich, M. Sch	1	1	1	1	1	1	1	1	1	1				10
untitled survey - developed by Johnston et al. 2018 + inter	Yes	Mixed meth	Survey of junior and senior surgeons included the SA	Johnston, M. J., S. Arora, D. King and A. Darzi (2018). "Improving the Quali	1	1	1	1				1			1	1	Supervision - adequancy of junior-senior supervisory relati	7
SAQ + interviews		Mixed meth	Operating room (OR) version of the SAQ used first,	Khoshbin A, Lingard L, Wright JG. Evaluation of precq	1	1	1	1	1	1	1	1	1	1	1			11
untitled survey - developed by Kirkegaard et al., 2018 + ot	Yes	Mixed meth	Focused on occupational health and safety, and viole	Kirkegaard, M. L., P. Kines, H. B. Nielsen and A. H. G	1	1			1		1					1	Risk-taking behaviour "If I take chances at work, where th	5
Manchester Patient Safety Framework (MaPSaF) + SAQ		Mixed meth	Evaluation of an improvement project using baseline	Lachman P, Linkson L, Evans T, Clausen H, Hothi D. I	1		1	1	1	1	1	1	1	1	1			10
Hospital Survey on Patient Safety Culture - developed by the Agency f		Quantitative	(dimensions): 1. Management support for patient	Sorra JS, Dyer N. Multilevel psychometric properties o	1	1	1	1	1	1	1	1	1					9
SAQ + interviews with SAQ as framework for analysis		Mixed meth	Interviews used to probe topic of safety culture deep	Listyowardojo TA, Yan X, Leyshon S, et al. (2017) A s	1	1	1			1						1	Scope of work	5
HSOPS + interviews and framework for analysis		Mixed meth	Expanding on survey results, three safety domains w	Livorsi D, Knobloch MJ, Blue LA, et al. A rapid assess	1	1	1	1	1	1	1	1	1					9
Interviews, focus groups + Cultural Assessment Survey		Mixed meth	Qualitative data used to inform survey development.	Milne JK, Bendaly N, Bendaly L, Worsley J, Fitzgerald J, Nisker J. A measur	1	1		1				1				1	Valuing individuals - which people are valued, their input,	5
Junior Doctor-Patient Safety Attitudes and Climate Questio	Yes	Quantitative	The Junior Doctors' Patient Safety Attitudes and Cli	Durani P, Dias J, Singh HP, Taub N. Junior doctors and	1	1		1		1	1	1	1	1				8
Safety Climate Survey (SCSu) - based on SAQ - developed by Sexton		Quantitative	A 19-21 item derivative of the SAQ, it assesses safe	Squires M, Tourangeau A, Laschinger HKS, Doran D.	1	1		1	1	1	1	1	1	1				8
Chinese Hospital Survey on Patient Safety Climate - devel	Yes	Quantitative	Based theoretically on Zohar's model of safety cultur	Zhu J, Li L, Zhao H, Han G, Wu AW, Weingart SN. Da	1	1	1	1		1	1	1	1					8
Safety Climate Survey -developed by Ahmed et al. 2018	Yes	Quantitative	Insufficient detail to classify entirely, but contains th	Ahmed, S. M., N. Fekry and F. A. Abed (2018). "Asses	1	1	1	1	1	1					1			7
The Patient Safety and Infection Prevention Assessment - described in		Quantitative	Survey includes Likert and open response questions	Braun BI, Harris AD, Richards CL, et al. Does health c	1	1	1	1		1		1	1					7
Hazardous Drug Handling Questionnaire - developed by Polovich and		Quantitative	This survey has 65 questions that measure variables:	Callahan A, Ames NJ, Manning ML, Touchton-Leonar	1	1	1	1		1		1				1	E.g., "My work area is kept clean"	7
Modified Stanford Patient Safety Culture Survey Instrument (MSI) - d		Quantitative	Available in a number of iterations (2005, 2006), the	Castel ES, Ginsburg LR, Zaheer S, Tamim H. Understa	1	1		1	1		1	1	1					7
MaPSaF + HSOPS + interviews, focus groups + analysis of reports		Mixed meth	Pre-post intervention study with MaPSaF run as worl	Simons PAM, Houben R, Vlayen A, et al. Does lean ma	1	1	1	1	1	1	1	1	1			1	Systems view of safety	10
Hospital Organizational Culture Questionnaire - developed	Yes	Quantitative	A Japanese language survey with 24 items in eight di	Kobuse H, Morishima T, Tanaka M, Murakami G, Hirose M, Imanaka Y. Visualizing variat	1	1	1	1	1	1	1	1	1					7
Manchester Patient Safety Culture Assessment Tool (MaPSCAT) - dev		Quantitative	Based on the MaPSaF, this 24 question survey uses c	Law MP, Zimmerman R, Baker GR, Smith T. Assessm	1		1	1	1	1	1		1					7
untitled survey - developed by Matsubara et al. 2008	Yes	Quantitative	A 50 item, Japanese language safety culture survey a	Matsubara S, Hagihara A, Nobutomo K, Matsubara S, I	1		1	1			1	1	1			1	Patient and family involvement - "Patient opinions are	7
Patient Safety Culture Questionnaire - developed by Veterans Affairs		Quantitative	Similar to the HSOPS, the survey consisted of 68 ite	Mohr, D. C., J. L. Eaton, K. M. McPhaul and M. J. Hod	1	1			1	1	1	1	1					7
Patients' Perceptions of Safety Culture scale (PaPSC) - dev	Yes	Quantitative	Developed from the HSOPS, SAQ and patient experi	Monaca, C., B. Bestmann, M. Kattein, D. Langner, H. M	1	1	1	1	1	1						1	Communication with patients - "I always felt that staff liste	7
Hospital Safety Climate Scale (HSCS) - developed by Gershon et al. 2		Quantitative	A 20 item survey covering the following six dimensi	Stringer B, Haines AT, Goldsmith CH, Berguer R, Blyt	1		1	1	1	1		1				1	E.g., "my work area is kept clean"	7
Pharmacy Safety Climate Questionnaire - developed by Ashcroft and P		Quantitative	Thirty-four item survey covering: investigating and l	Walpola RL, Chen TF, Fois RA, Ashcroft DM, Lalor D	1		1	1	1	1	1		1					7
Safety Climate and Capability (SCC) - developed by Benn	Yes	Quantitative	Based on established safety culture surveys, this 31 i	Benn J, Burnett S, Parand A, Pinto A, Vincent C. Facto	1	1	1	1	1				1					6
Victorian Safety Climate Survey (SCS) - developed by Sexton et al. 20		Quantitative	The 74 survey items were adapted from the SAQ and	Bennett PN, Ockerby C, Stinson J, Willcocks K, Chalm	1			1	1				1		1			6
Leading a Culture of Quality in Infection Prevention (LCQ-IP) - develo		Quantitative	The 19 item survey measures four dimensions of an i	Colet, P. C., J. P. Cruz, G. Cacho, H. Al-Qubeilat, S. S.	1		1		1	1		1	1					6
Patient Safety Culture Assessment Scale (PSCAS) - adapted by Chen a		Quantitative	Based on the SAQ for use in China, the PSCAS cont	Hong S, Li Q. The reasons for Chinese nursing staff to	1	1	1				1			1	1			6
Safety Climate Scale - developed by Zohar and Luria 2005		Quantitative	Originally two 27 item scales assessing organisations	Hughes LC, Chang Y, Mark BA. Quality and strength o	1			1	1	1		1	1					6
Quality and Patient Safety Culture Survey - developed by I	Yes	Quantitative	Based on PSCI/PSCHO and HSOPS, and other instru	Kagan, I., N. Porat and S. Barnoy (2019). "The quality	1		1	1			1	1				1	Perception of quality and implementation of improvement	6
Patient Safety Culture Scale - developed by Calgary Health Region		Quantitative	Management and Leadership (17 items), Employee	Kline TJ, Willness C, Ghali WA. Predicting patient con	1	1		1	1	1	1							6
untitled survey - developed by Liu et al. 2009	Yes	Quantitative	Developed for the Chinese population, this survey in	Liu Y, Kalisch BJ, Zhang L, Xu J. Perception of safety	1		1	1		1	1		1					6
The National Emergency Department Safety Study Survey	Yes	Quantitative	The survey contains 50 questions about systems in th	Magid DJ, Sullivan AF, Cleary PD, et al. The safety of	1	1	1	1		1		1						6
untitled survey - developed by Schwonke et al. 2016	Yes	Quantitative	A Portuguese language survey based on the PSCI, it	Schwonke CRGB, Filho WDL, Lunardi GL, da Silveira	1	1			1	1	1	1		1				6
untitled survey - developed by Walston et al. 2010	Yes	Quantitative	In part based on Zohar (1980), survey is made up of	Walston SL, Al-Omar BA, Al-Mutari FA. Factors affec	1		1	1	1	1			1					6
Patient Safety Culture Survey - Maternal and Child Health	Yes	Quantitative	This survey focuses on safety culture in the context	Wang, Y., H. Han, L. Qiu, C. Liu, Y. Wang and W. Liu	1			1		1	1				1	1	Patient Engagement - "I respect patient rights and willingn	6
Attitudes to Patient Safety Questionnaire III (APSQ III) (Carruthers et		Quantitative	Survey measures patient safety attitudes over nine fa	Abu-El-Noor, N. I., M. K. Abu-El-Noor, Y. Z. Abuowda, M. Alfaqawi and B. Böttcher (20	1					1	1			1		1	Patient involvement in reducing error: "Encouraging patie	5
General Safety Climate Survey - developed by Neal et al. 2000		Quantitative	A 25 item survey of safety culture in general (rather	Amponsah-Tawaih K, Adu MA. Work Pressure and Sa	1			1	1	1		1						5
California-Patient Safety Culture Survey (Cal-PSCS) - dev	Yes	Quantitative	A 22 item survey covering four factors: 1. Assistance	Avramchuk, A. S. and S. J. J. McGuire (2018). "Patient	1		1		1	1	1							5
untitled survey - developed by Birkmeyer et al. 2013	Yes	Quantitative	Survey developed from the HSOPS and SAQ; focus	Birkmeyer NJ, Finks JF, Greenberg CK, et al. Safety cu	1	1	1	1		1								5
Ethnography - interviews, observations, document analysis using Habe		Qualitative	Informal conversations, interview conducted. Docum	Bressers, G., I. Wallenburg, R. Stalmeijer, M. oude Egbrink and K. Lombarts (2020). "Patient safety in medical residency training: Balancing bravery and checklists." Health (United Kingdom).												1	workarounds, independence and medical autonomy	1
untitled survey - developed by Ferrer et al. 2018	Yes	Quantitative	Developed in part from the HSOPS, this 35 item mea	Ferrer, M., R. Santa, A. Medhekar, E. Calvo and L. P. Molina-Sánchez (2018). "Drivers of		1	1	1		1		1						5
untitled survey - developed by Gibbs et al. 2018	Yes	Quantitative	Specifically focused on infection control, this survey	Gibbs, K., S. DeMaria, S. McKinsey, A. Fede, A. Harrington, D. Hutchison, C	1		1			1		1		1				5
Interviews and observation		Qualitative	Interviews and observations used to evaluate training	De Korne DF, Van Wijngaarden JD, Van Dyck C, Hidd	1		1				1		1	1				5
untitled survey - developed by Simsekler 2019 + interview	Yes	Mixed meth	Survey based on some dimensions of the MaPSaF (m	Simsekler, M. C. E. (2019). "The link between healthcare risk identification and patient safety culture." International Jou	1						1		1			1	Dimensions reformulated to assess risk identification pract	4
untitled survey - developed by Kudo et al. 2008	Yes	Quantitative	A Japanese survey with 30 items assessing the follow	Kudo Y, Kido S, Taruzuka Shahzad M, Saegusa Y, Sat	1		1			1	1			1				5
untitled survey - developed by Listyowardojo et al. 2012	Yes	Quantitative	Developed from the SAQ and Jobs Diagnostic Surve	Listyowardojo TA, Nap RE, Johnson A. Variations in h	1		1		1	1					1			5
HSOPS + focus groups with interactive qualitative analysis		Mixed meth	Following administration of the surve, a focus group	Sine DM, Northcutt N. Interactive qualitative assessme	1	1	1	1	1	1	1	1	1					9
HSOPS + focus groups, Theoretical Domains Framework		Mixed meth	HSOPS data used to identify areas to cover in intervi	Stewart, D., B. Thomas, K. MacLure, A. Pallivalapila,	1	1	1	1	1	1	1	1	1					9
Ethnography - Interviews and observation		Qualitative	Interview schedule focused on teamwork and commu	Gillespie BM, Gwinner K, Chaboyer W, Fairweather N. Team communications in surgery -		1						1				1	developing a shared understanding, managing safety in cor	3
Just Culture Assessment Tool (JCAT) - developed by Petsc	Yes	Quantitative	of safety culture (from Reason's theory: "a	Petschonek S, Burlison J, Cross C, et al. Development of the just culture assessment tool: measuring the p				1			1	1	1			1	Trust - The extent to which individuals use the	5
Veterans Health Administration (VHA) Hospital Survey on	Yes	Quantitative	This survey assesses 5 domains of safety climate: 1.	Pringle J, Weber RJ, Rice K, Kirisci L, Sirio C. Examin	1		1	1	1							1	Patient safety as a personal priority - Assuming personal re	5
Hospital Transfusion Service Error Reporting and Safety Culture Surv		Quantitative	transfusion medicine, this survey contains 79 survey	Sorra J, Nieva V, Fastman BR, Kaplan H, Schreiber G,	1	1	1					1	1					5
Vienna Safety Climate Questionnaire (VSCQ) - developed	Yes	Quantitative	A German-language survey based in large part on otl	Steyrer J, Schiffinger M, Huber C, Valentin A, Strunk G	1	1					1	1	1					5
Focus groups and observation		Qualitative	Data collection and data analysis occurred in success	Gimenes FR, Torrieri MC, Gabriel CS, et al. Applying an ecological restorati	1	1	1		1			1						5
Human Factors Attitude Survey (HFAS) - developed by Grogan et al. 2		Quantitative	A 23 item survey based on one in aviation, it assesse	Wu WT, Wu YL, Hou SM, et al. Examining the effects	1		1					1	1	1				5
Safety Climate of Nurses (SCN) - Sarasangi et al. 2015		Quantitative	Focused specifically on nurses, it contains 22 questio	Zarei E, Khakzad N, Reniers G, Akbari R. On the relati	1		1			1	1				1			5
Medical Students' Patient Safety Questionnaire - developed by Flin et a		Quantitative	Originally developed to assess the students' levels of	Ahmed M, Arora S, Baker P, Hayden J, Vincent C, Sevdalis N. Building capacity and capability for patient safety educa	1							1	1	1				4
Survey on Patient Safety Climate - developed by Naveh et al. 2005		Quantitative	The survey covers four dimensions: - managerial safe	Custo, P. T., R. T. Custo and S. Buttigieg (2019). "The	1			1	1	1								4
untitled survey - developed by Hahn & Murphy, 2008	Yes	Quantitative	A 6 item measure of global work safety climate, ques	Hahn SE, Murphy LR. A short scale for measuring safe	1				1		1		1					4
untitled survey - developed by Lowe 2012	Yes	Quantitative	Survey is comprised of the following six items: Erro	Lowe G. How employee engagement matters for hospital performance. Health		1		1			1		1					4
Error Orientation Scale and Zohar Safety Climate Scale - n	Yes	Quantitative	This survey combines two scales, the Error Orientati	Mark BA, Hughes LC, Belyea M, Bacon CT, Chang Y,	1	1	1	1										4
Medical Safety Climate Questionnaire - developed by Madigosky et al.		Quantitative	Questionnaire with 28 items assessing knowledge, sk	McCarthy SE, O'Boyle CA, O'Shaughnessy A, Walsh G. Online patient safety education programme for junior doctors:					1		1		1	1				4
untitled survey - developed by Gimeno et al. 2005		Quantitative	Based in part on the HSCS and developed for Costa	Menendez CC, Gimeno D, Barrientos-Gutiérrez T, Burs	1				1		1	1						4
VA Employee Safety Perception Survey - developed by the National S		Quantitative	The survey consisted of 50 questions that represente	Mohr, D. C., J. L. Eaton, K. M. McPhaul and M. J. Hod	1	1		1	1									4
untitled survey - developed by Paez et al. 2013	Yes	Quantitative	One domain of their survey focused on safety culture	Paez K, Schur C, Zhao L, Lucado J. A national study of	1	1				1			1					4
HSOPS + interviews and framework for analysis		Mixed meth	Following survey, the interviews were aimed at ident	Wami SD, Demssie AF, Wassie MM, Ahmed AN. Patie	1	1	1	1		1	1	1	1			1	Patient factors - Patient perception about the health service	9
Interviews and grounded theory		Qualitative	Interviews conducted in a grounded theory design (it	Groves PS, Finfgeld-Connett D, Wakefield BJ. It's always something: hospita		1								1		1	Managing risk - viewing patients as always at risk,	3
Safety Climate Questionnaire - developed by Zeidi et al. 2012		Quantitative	This 37 item questionnaire includes 8 aspects of man	Yari, S., M. H. Naseri, H. Akbari, S. Shahsavari and H.	1	1	1		1									4
untitled survey - developed by Al Faqeeh et al. 2019	Yes	Quantitative	An English-language survey assessing: safety behavi	Al Faqeeh, F., K. Khalid and A. Osman (2019). "Integr	1	1			1									3
untitled survey - developed by Berman et al. 2018	Yes	Quantitative	A surgery-focused questionnaire, asking surgeons wh	Berman, L., S. Rangel, A. Goldin, D. Skarda, M. Ottose	1	1			1									3
untitled survey - developed by Boyer et al. 2012	Yes	Quantitative	A brief 7 item measure of safety climate with the foll	Boyer KK, Gardner JW, Schweikhart S. Process quality	1	1						1						3

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